

Regularizing Forecast Combination with Random Subsets

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Abstract

How much should be estimated when combining many forecasts? Equal weighting is robust to estimation error but ignores heterogeneity in forecast quality, while estimated optimal weights exploit this heterogeneity at the cost of sampling uncertainty. This paper shows that random subset combination provides a regularization path between these two endpoints. We characterize the resulting approximation–estimation trade-off and show that, because subset regressions are estimated on a shared sample, the optimal subset size is of order $T^{1/4}$ rather than the $T^{1/2}$ rate for an individually evaluated estimated subset. A feasible finite-sample criterion selects the degree of regularization. Using the real-time Survey of Professional Forecasters data, we show that the criterion selects an interior subset size close to the ex-post loss-minimizing choice and substantially stabilizes full-pool adding-up weights. RSM is the most accurate rule in most of the samples. The application supports both the regularization mechanism and the possibility of practical forecast combination gains. **Keywords:** Forecast combination, random subsets, regularization

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1 Introduction

Forecast combination has been extensively studied in economics. However, it is often presented as a choice between two endpoints. Equal weighting estimates nothing with regard to cross-sectional differences in forecast quality and is therefore stable in short samples. When heterogeneity across forecasts is modest, ignoring those differences may also involve little population loss. By contrast, fully estimated unrestricted optimal weights attempt to learn the entire weight vector and can, in theory, exploit persistent heterogeneity. Yet, in doing so, may pay a large sampling-error cost. Consequently, the question that is usually posed is whether weights should be estimated in the first place (forecast combination puzzle). This paper follows a slightly different path and tries to answer the following question: *how much of the optimal weight vector can the available time-series sample support?*

Random subset method (RSM) used for forecast combination turns this question into a regularization problem. To be more specific, this method provides an alternative form of regularization toward the equal weighting Diebold and Pauly (1990). For each subset of k forecasts, the method estimates unrestricted optimal weights and then averages the resulting subset forecasts. At $k = 1$, this procedure produces equal weighting; at $k = m$, when the full pane is used, RSM coincides with the full-pool, unrestricted estimated optimal combination. Hence, intermediate values of k form a transparent regularization path: increasing k lets the estimator learn more heterogeneity in forecast precision, while also increasing the dimension of each estimated weight vector.

Our main theoretical result makes this trade-off explicit. Under the paper’s common-component forecast-error model, the leading excess risk of the shared-sample infinite-bagged¹ estimator can be presented as follows:

$$L_{T,m}^{\text{bag},\infty}(k) - Q^{\text{opt}} \asymp \underbrace{\frac{\mathcal{D}}{mk^2}}_{\text{approximation cost}} + \underbrace{\frac{bk^2}{mT}}_{\text{estimation cost}}, \quad k^* \asymp \left(\frac{\mathcal{D}T}{b}\right)^{1/4}.$$

The approximation term decreases with k as larger subsets move the RSM weights toward the population optimum. The estimation-error term, on the other hand, increases because more coefficients are learned from the same T observations. We find that the balance of these two terms produces the fourth-root rule for the optimal subset size k^* .

In the current setup, the shared sample is essential. RSM averages many subset regressions,

¹Bagging usually randomly selects observations rather than predictors. In the computer science literature, this is usually labeled as "feature bagging" (Munson and Caruana, 2009). That is why, throughout the paper, what is called bagging is actually "feature bagging".

but those regressions reuse the same observations, so their estimation errors remain correlated. Averaging removes subset-draw noise but not the common estimation component. This yields an estimation penalty of order $k^2/(mT)$. By contrast, a criterion that evaluates a single fitted subset and only then averages its loss has a more standard k/T penalty. That leads to a square-root optimizer for the expected loss of an individually estimated random subset. Therefore, the fourth-root and square-root rules characterize different risk criteria.

This paper makes three contributions. First, it establishes that RSM provides a regularization path governing how much forecast heterogeneity should be estimated. Second, it derives the specific shared-sample inflation term, which leads to the k^* fourth-root rule, and a feasible finite- m selector to demonstrate this regularization representation and the corresponding excess-risk decomposition. This makes the relation to the forecast-pool analysis of Elliott and Liao (2026) straightforward: small population gains from unequal weights can be overwhelmed by the cost of estimating them. This trade-off is characterized by k^* : the smaller the value of k^* , the less heterogeneity the available sample can support. Third, the paper validates the theory-based selector using real-time forecasts of quarterly U.S. GDP-price inflation one and four quarters ahead. Only using training information, the selector chooses a median k close to the ex-post loss-minimizing fixed range. The realized loss curve is U-shaped, and loss increases substantially as k approaches the full panel size of m . These patterns directly support the approximation–estimation mechanism.

As for empirical results, RSM achieves the lowest RMSFE among feasible methods in three of the four horizon–sample comparisons. At $h = 1$, it reduces RMSFE relative to the screened survey mean by 0.95 percent in the full sample and by 1.3 percent in the sample when the global financial crisis and COVID-19 disruptions are excluded. At $h = 4$, RSM is essentially tied with screened equal weighting. However, relative to the unscreened all-respondent mean, it lowers RMSFE by 0.93 and 0.92 percent, respectively. RSM consistently outperforms alternative benchmarks, such as partially egalitarian LASSO of Diebold and Shin (2019), sum-to-one non-negative weights (Conflitti et al., 2015) and regularized shrinkage approach (Stock and Watson, 2004). We conclude that some estimation of cross-sectional heterogeneity is useful, and regularization via RSM preserves the stability of the survey consensus while avoiding the severe estimation error.

Related literature. The paper contributes to several strands of literature. First, it is related to the forecast-combination puzzle: equal weighting is often hard to beat because estimated optimal weights introduce finite-sample error (Smith and Wallis, 2009; Claeskens et al., 2016; Timmermann, 2006; Wang et al., 2023). For instance, Elliott and Liao (2026)

use the estimation cost for the optimal weighting². They emphasize that estimated optimal weights face a sampling cost and that, under empirically plausible forecast-error structures, their population gain over equal weighting may itself be small. The present paper treats the amount of the weight vector to estimate as the choice variable. Its RSM-specific shared-sample estimation penalty then yields an explicit optimal subset size.

A second strand studies regularization and subset averaging when the number of forecasts or predictors is large relative to the available time-series sample. So far, complete subset regressions (Elliott et al., 2013, 2015) and existing random subset methods primarily use subsets of observed predictors to construct forecasts (Boot and Nibbering, 2019; Wichitaksorn et al., 2023; Kotchoni et al., 2019; Pick and Carpay, 2022; Kant et al., 2025; Meligkotsidou et al., 2021; Dinh et al., 2024) with little being known about the properties of these methods in the forecast combination framework. Its regularization interpretation is also not well-documented to the best of our knowledge. For instance, Xiao and Cross (2026) extend the traditional RSM framework by introducing regularization into the specification of variable inclusion probabilities and/or regression coefficients. By contrast, we show that RSM per se can serve as a regularization method.

Several regularization methods for forecast combination have been developed. Bayesian shrinkage already made equal and least-squares weights the endpoints of a continuum (Diebold and Pauly, 1990). Similarly, Stock and Watson (2004) introduce a frequentist approach with the same property. Partially egalitarian Lasso combines selection with shrinkage of surviving weights toward equality (Diebold and Shin, 2019). Explicit divergence penalties that stabilize constrained combinations around equal weights are also studied in (Roccazzella et al., 2022). RSM differs from the previous methods in that k controls the dimension of each weight estimation problem. We also isolate a feature not covered by the usual bias–variance account: because all subset regressions reuse the same time-series observations, their estimation errors are correlated, which changes the leading penalty and the optimal rate.

Finally, the empirical design belongs to the literature on combining survey forecasts and evaluating them with real-time data. SPF studies show both how difficult the survey mean is to beat and how weight constraints or shrinkage can mitigate instability (Genre et al., 2013; Conflitti et al., 2015). Because revised macroeconomic data can change both model estimation and forecast rankings, genuine real-time evaluation requires the vintage available at each forecast origin and an explicit choice of realization (Croushore and Stark, 2003; Croushore, 2011). Our results confirm resilience of equal weighting in terms of forecast gains, even though in most

²Similar to a standard one used, for instance, in Kan and Zhou (2007)

cases RSM outperforms equal weighting. However, the overall regularization performance of RSM is competitive with the regularized alternatives considered in the paper.

Section 2 develops the theory and selector. Section 4 validates the distinct theoretical risk objects in Monte Carlo experiments. Section 5 evaluates the selector in real-time SPF data, and Section 6 concludes.

2 Theoretical Model

Let us restate the main theoretical finding: the leading shared-sample risk trade-off derived below is as follows:

$$L_{T,m}^{\text{bag},\infty}(k) - Q^{\text{opt}} \asymp \underbrace{\frac{\mathcal{D}}{mk^2}}_{\text{approximation cost}} + \underbrace{\frac{bk^2}{mT}}_{\text{estimation cost}}. \quad (1)$$

This means, larger subsets move the bagged population weights toward the oracle, but they also require the shared training sample to estimate more within-subset coefficients. Figure 1 gives the resulting regularization path before the formal derivation³.

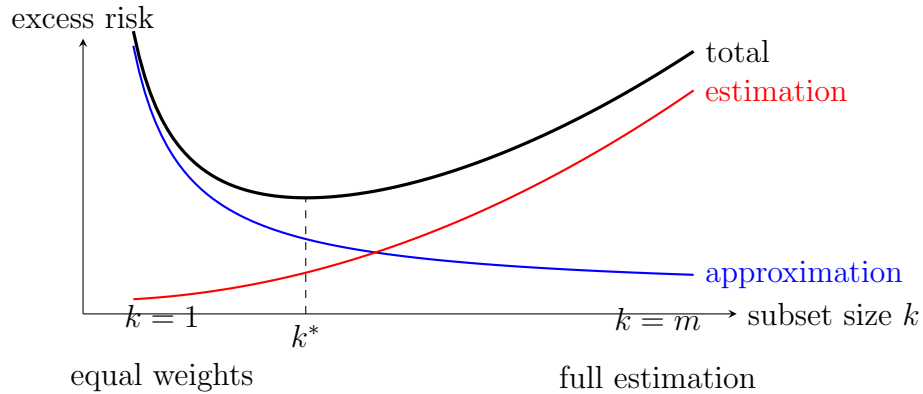


Figure 1: Subset size as a regularization path

More formally, suppose there are m unbiased individual forecasts $f_t = (f_{1t}, \dots, f_{mt})'$ of a scalar target y_t . Let ι_r denote the r -dimensional vector of ones. The realized forecast-error vector is

$$e_t = f_t - y_t \iota_m. \quad (2)$$

It is worth mentioning that expectations below, denoted $\mathbb{E}$, are taken over the unconditional distribution of realized forecast errors across origins. Later, we introduce expectations over random subset selection, that are denoted separately by $\mathbb{E}_S$. We fix the forecast horizon h and suppress its subscript.

³The curves are schematic. Proposition 9 derives their leading orders.

Assumption 1 (Forecast errors). $\mathbb{E}(e_t) = 0$ and $\text{Var}(e_t) = \Omega_{e,m} \succ 0$.

Assumption 2 (Two-component realized forecast errors). *The realized forecast errors satisfy:*

$$e_t = r_t \iota_m + \xi_t, \quad (3)$$

where $\mathbb{E}(r_t) = 0$, $\text{Var}(r_t) = b > 0$, $\mathbb{E}(\xi_t) = 0$, $\text{Var}(\xi_t) = D$, and $\text{Cov}(r_t, \xi_t) = 0$. Moreover,

$$D = \text{diag}(\tau_1^{-1}, \dots, \tau_m^{-1}), \quad 0 < \tau_L \leq \tau_i \leq \tau_U < \infty$$

uniformly in i and m . Denote

$$\bar{\tau}_m = \frac{1}{m} \sum_{i=1}^m \tau_i, \quad v_{\tau,m} = \frac{1}{m} \sum_{i=1}^m (\tau_i - \bar{\tau}_m)^2,$$

assume $\bar{\tau}_m \rightarrow \bar{\tau} > 0$ and $v_{\tau,m} \rightarrow v_{\tau} > 0$. Finally, assume that the precision-profile constant

$$\mathcal{D}_m = \frac{1}{\bar{\tau}_m^6} \frac{1}{m} \sum_{i=1}^m \tau_i \{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)\}^2 \quad (4)$$

converges to a finite constant $\mathcal{D} > 0$.

Assumption 3 (Gaussian shared sample). *The blocks r_t and ξ_t are independent centered Gaussian variables, and $(r_t, \xi_t)'$ is i.i.d. across the T training observations. The evaluation error is an independent draw from the same distribution. Moreover, $T^{1/4} = o(m)$ and $m = o(T)$.*

Assumption 1 is a standard assumption in the forecast combination literature. Assumption 2 is a reduced-form specification for observed realized forecast errors. The common component r_t may combine an unforecastable target innovation with forecast-construction disturbances shared across forecasters. In particular,

$$\Omega_{e,m} = \text{Var}(e_t) = b \iota_m \iota_m' + D. \quad (5)$$

This common-loading structure is related to the forecast-error decompositions used by Davies and Lahiri (1995, 1999); Lahiri and Sheng (2010).⁴ The boundedness and convergence conditions in Assumption 2 keep the precision distribution nondegenerate as m grows. The stronger Gaussian condition in Assumption 3 is imposed only for the exact feasible-loss calculations and the shared-sample expansion, in the spirit of Elliott and Liao (2026) and Kan and Zhou (2007).

⁴Unlike the more general approach, we assume that a forecast-specific bias component is 0, since the forecasts are assumed to be unbiased. Additionally, we assume that matrix D is diagonal. Notice that this specification is equivalent to a factor structure with the common loading. General factor models with heterogeneous loadings and residual cross-sectional dependence are also exploited; see, for instance, Lee and Seregina (2020).

For any adding-up weight vector w satisfying $\iota'_m w = 1$, quadratic forecast risk is as follows⁵:

$$Q(w) = \mathbb{E}(w' e_t e_t' w) = w' \Omega_{e,m} w. \quad (6)$$

The equal-weight and population-optimal combinations have the familiar closed forms

$$w^{\text{ave}} = m^{-1} \iota_m, \quad Q^{\text{ave}} = m^{-2} \iota_m' \Omega_{e,m} \iota_m, \quad (7)$$

$$w^{\text{opt}} = \frac{\Omega_{e,m}^{-1} \iota_m}{\iota_m' \Omega_{e,m}^{-1} \iota_m}, \quad Q^{\text{opt}} = (\iota_m' \Omega_{e,m}^{-1} \iota_m)^{-1}. \quad (8)$$

When $w^{\text{ave}} \neq w^{\text{opt}}$, equal weighting has strictly higher population risk: $Q^{\text{ave}} > Q^{\text{opt}}$. Estimating the optimal weights, however, introduces sampling error that can erase this population gain. This tension is a leading explanation of the forecast-combination puzzle.

2.1 Random Subset Method for Forecast Combination

Before formally introducing RSM, we formulate the subset sampling rule.

Assumption 4 (Subset sampling). *Each subset S is drawn uniformly from the $\binom{m}{k}$ subsets of size k , without replacement within a subset and independently of the data. The subsets $S_1, \dots, S_G$ are independent draws.*

Let P_S be the $m \times k$ selection matrix that embeds a subset weight vector in $\mathbb{R}^m$ by placing its entries in their original coordinates and filling the remaining coordinates with zeros. Notice that the embedding matrix P_S satisfies the following properties: $P_S' P_S = I_k$ and $P_S' \iota_m = \iota_k$. Let $\Omega_{e,k,S}$ denote the $k \times k$ principal submatrix of $\Omega_{e,m}$ indexed by S :

$$\Omega_{e,k,S} := P_S' \Omega_{e,m} P_S = b \iota_k \iota_k' + D_S, \quad D_S = \text{diag}(\tau_i^{-1} : i \in S). \quad (9)$$

Within S , the population-optimal weights and their embedding are

$$w_S = \frac{\Omega_{e,k,S}^{-1} \iota_k}{\iota_k' \Omega_{e,k,S}^{-1} \iota_k}, \quad v_S = P_S w_S, \quad \iota_m' v_S = 1. \quad (10)$$

Three population criteria must be distinguished. First, the average-subset risk evaluates one randomly selected subset and then averages its loss:

$$Q_k^{\text{sub}} = \mathbb{E}_S[v_S' \Omega_{e,m} v_S] = \mathbb{E}_S[Q(S)], \quad Q(S) = (\iota_k' \Omega_{e,k,S}^{-1} \iota_k)^{-1}. \quad (11)$$

⁵The analysis uses symmetric quadratic loss. When w is estimated, the expression is conditional on the training sample and applies to an independent evaluation observation. Asymmetric loss is outside the scope of the paper; see Elliott and Timmermann (2004). Moreover, Aruoba et al. (2012) show that equal weighting is optimal under minimax loss.

Second, finite bagging averages G independently sampled embedded weight vectors before evaluating loss:

$$Q_k^{\text{bag},G} = \mathbb{E}_{S_1, \dots, S_G}(\bar{v}'_{k,G} \Omega_{e,m} \bar{v}_{k,G}), \quad \bar{v}_{k,G} = G^{-1} \sum_{g=1}^G v_{S_g}. \quad (12)$$

Finally, complete, or infinite, bagging evaluates the limit as $G \rightarrow \infty$:

$$Q_k^{\text{bag},\infty} = \bar{v}'_k \Omega_{e,m} \bar{v}_k, \quad \bar{v}_k = \mathbb{E}_S(v_S). \quad (13)$$

The last object is equivalent to averaging over all size- k subsets, as in complete subset regression (Elliott et al., 2013). Moreover, due to convexity of the population loss function in v_S , we can show by applying Jensen's inequality, similar to Boot and Nibbering (2019) that $Q_k^{\text{bag},\infty} \leq Q_k^{\text{sub}}$. Strict convexity of $v' \Omega_{e,m} v$ makes equality possible only when v_S is constant almost surely, which under uniform fixed-size sampling occurs when $k = m$.

Proposition 1 (Finite-bagging bridge). *For every $1 \leq G < \infty$,*

$$Q_k^{\text{bag},G} = Q_k^{\text{bag},\infty} + \frac{1}{G} (Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}). \quad (14)$$

Consequently, $Q_k^{\text{bag},\infty} \leq Q_k^{\text{bag},G} \leq Q_k^{\text{sub}}$.

The proof is in Appendix B. The identity separates variation due to subset sampling from estimation uncertainty due to the common training sample; only the former vanishes as $G \rightarrow \infty$.

Notice the following special cases:

- when $G = 1$, $Q_k^{\text{bag},1} = Q_k^{\text{sub}}$;
- as $G \rightarrow \infty$, $Q_k^{\text{bag},G} \rightarrow Q_k^{\text{bag},\infty}$.

The implementable finite-bag estimator requires a rate condition on the number of subset draws.

Assumption 5 (Number of subset draws G). *For results transferring the infinite-bag risk expansion and optimal-subset rule to the implementable finite-bag estimator, let $G = G_{T,m} \rightarrow \infty$ and assume*

$$\frac{mk}{G} \rightarrow 0.$$

We can analyze how the RSM weights change depending on the subset size k in some general cases. Our analysis closely follows Elliott et al. (2013); however, the interpretation is slightly different.

Proposition 2 (RSM as a unifier). $\bar{v}_1 = w^{\text{ave}}$ and $\bar{v}_m = w^{\text{opt}}$.

The endpoint identities show that RSM contains both standard benchmarks. Thus, in the forecast combination setup, this unification has a stronger interpretation than for pure forecasting, as discussed in Elliott et al. (2013).

3 Population Results

Define the subset and full-panel sums of precisions by

$$A_S = \sum_{i \in S} \tau_i, \quad A_m = \sum_{i=1}^m \tau_i = m\bar{\tau}_m. \quad (15)$$

Proposition 3 (Subset loss under two components). *For every subset S ,*

$$w_{i,S} = \frac{\tau_i}{A_S} \quad (i \in S), \quad Q(S) = v'_S \Omega_{e,m} v_S = b + \frac{1}{A_S}. \quad (16)$$

For the full panel and equal weighting,

$$w_i^{\text{opt}} = \frac{\tau_i}{A_m}, \quad Q^{\text{opt}} = b + \frac{1}{m\bar{\tau}_m}, \quad (17)$$

$$w_i^{\text{ave}} = 1/m, \quad Q^{\text{ave}} = b + \frac{1}{m^2} \sum_{i=1}^m \tau_i^{-1}. \quad (18)$$

Appendix A.2 derives these identities. The subset loss $Q(S)$ is the sum of the irreducible common component b and the inverse of the total precision in the subset. Along any nested sequence, adding a forecast strictly lowers population risk because every τ_i is positive. The minimum is attained at the full panel, i.e. $k = m$. Also, notice that optimal subset weight, unlike the subset loss, does not depend on b : it is only a function of the forecast precision vector τ .

To illustrate the relationship between the two losses, Q^{opt} and Q^{ave} , let $B = \bar{\tau}_m^{-1}$ and $H = (m^{-1} \sum_{i=1}^m \tau_i^{-1})\bar{\tau}_m$, where $\bar{\tau}_m$ is the mean precision and $m^{-1} \sum_{i=1}^m \tau_i^{-1}$ is the mean idiosyncratic variance.

Then, B is the reciprocal of average precision. It illustrates the scale of idiosyncratic noise and sets the overall level of the latter. Intuitively, it depicts how noisy a typical forecast is without taking into account heterogeneity. If forecasts are more precise, it leads to a larger $\bar{\tau}_m$ value. This, in turn, decreases the value of B and allows for achieving lower loss.

The quantity H is the product of mean variance $m^{-1} \sum_{i=1}^m \tau_i^{-1}$ and mean precision $\bar{\tau}_m$. It measures the dispersion of idiosyncratic forecast variances and equals the ratio of their

arithmetic mean to their harmonic mean. Thus, $H \geq 1$, with equality if and only if all τ_i are equal, the case in which equal weights are optimal.

The endpoint risks can therefore be written as follows:

$$Q^{opt} = b + \frac{B}{m}, \quad Q^{ave} = b + \frac{BH}{m}, \quad Q^{ave} - Q^{opt} = \frac{B(H-1)}{m} \geq 0$$

Thus, equal weighting cannot dominate the unrestricted optimum in population. Persistent precision heterogeneity implies $H > 1$ and creates room for an interior RSM combination to improve on equal weighting. Yet, the population risk gap is $B(H-1)/m$ and therefore shrinks with the pool size when B and H remain bounded. This is consistent with Elliott and Liao (2026): as the number of forecasts grows, the population risks of equal and optimal weighting become close even when their weights differ.

Proposition 4 (Average subset loss). *For every k , the exact average-subset risk is defined as:*

$$Q_k^{sub} = b + \mathbb{E}_S(A_S^{-1}). \quad (19)$$

As $m \geq k \rightarrow \infty$,

$$Q_k^{sub} = b + \frac{1}{k\bar{\tau}_m} + \frac{m-k}{k^2(m-1)} \frac{v_{\tau,m}}{\bar{\tau}_m^3} + O(k^{-5/2}), \quad (20)$$

and hence

$$Q_k^{sub} = b + \frac{1}{k\bar{\tau}_m} + O(k^{-2}). \quad (21)$$

The proof is in Appendix C.

For infinite bagging, let us introduce the following notations:

$$g_i(k) = k\mathbb{E}_S(A_S^{-1} \mid i \in S), \quad z_k = \frac{1}{k} - \frac{1}{m}. \quad (22)$$

Proposition 5 (Bagged loss representation). *The infinite-bagged weights and their excess risk satisfy*

$$\bar{v}_{i,k} = \frac{\tau_i}{m} g_i(k), \quad \sum_{i=1}^m \tau_i g_i(k) = m, \quad (23)$$

$$Q_k^{bag,\infty} = Q^{opt} + R_k^{bag}, \quad R_k^{bag} = \sum_{i=1}^m \frac{(\bar{v}_{i,k} - w_i^{opt})^2}{\tau_i} = \frac{1}{m^2} \sum_{i=1}^m \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}_m} \right)^2 \geq 0 \quad (24)$$

Appendix D proves the representation. It shows that R_k^{bag} is the precision-weighted squared distance between the bagged weights $\bar{v}_{i,k}$ and the population-optimal weights w_i^{opt} . At $k = 1$, this is the distance from equal weighting to the optimum; at $k = m$, it is zero. Intermediate subset sizes trace the path between the endpoints, and their population performance depends on how quickly the averaged subset weights approach the optimum.

3.1 Estimation Error Inflation

The population risks above treat weights as known. We now fit the unrestricted optimal weights on the common training sample and evaluate them on an independent error draw. Let $L_{T,m}^{\text{sub}}(k)$ denote the expected test loss when one fitted size- k subset is evaluated, and the result is averaged over S . Let $L_{T,m}^{\text{bag},\infty}(k)$ denote the expected test loss of the literal complete-subset estimator, which averages all fitted subset-weight vectors before evaluation.

Proposition 6 (Exact average subset inflation). *If $T > k$, then*

$$L_{T,m}^{\text{sub}}(k) = Q_k^{\text{sub}} \mathcal{I}_{\text{sub}}(T, k), \quad \mathcal{I}_{\text{sub}}(T, k) = \frac{T-1}{T-k}. \quad (25)$$

Thus, $\mathcal{I}_{\text{sub}}(T, k) = 1 + (k-1)/(T-k)$.

For each fixed S , the adding-up restriction leaves $k-1$ free Gaussian regression coefficients. Proposition 14 proves that the expected independent-test loss is $Q(S)(T-1)/(T-k)$.

For infinite bagging, all subset regressions reuse the same observations. Their estimation errors are correlated, so (25) cannot be applied after averaging the weight vectors.

Proposition 7 (Shared-sample infinite-bag inflation). *Let $k = k_{T,m}$ satisfy $k \rightarrow \infty$, $k/m \rightarrow 0$, and $k < T$. Under Assumptions 1–3 and 4,*

$$L_{T,m}^{\text{bag},\infty}(k) = Q_k^{\text{bag},\infty} + \frac{\Gamma_{m,k}}{T} + o\left(\frac{k^2}{mT}\right), \quad (26)$$

where the complete first-order coefficient $\Gamma_{m,k}$ is given in (66). Equivalently,

$$\mathcal{I}_{\infty}(T, m, k) = 1 + \frac{\Gamma_{m,k}}{TQ_k^{\text{bag},\infty}} + o\left(\frac{k^2}{mT}\right) = 1 + \frac{k^2}{mT}\{1 + o(1)\}. \quad (27)$$

Moreover,

$$\Gamma_{m,k} = b \frac{k^2}{m} \{1 + o(1)\}. \quad (28)$$

Appendix E first derives $\Gamma_{m,k}$ from the covariance between two fitted subsets and then controls the nonlinear estimation remainder. Hence, Equation (27) is a derived first-order factor, not an exact finite- T identity. Its order is smaller than the single-subset penalty because averaging stabilizes the estimated weights, but it does not vanish as if the subset fits were independent. Their common sample is precisely what generates $\Gamma_{m,k}$.

3.2 Optimal subset size

With both inflation factors established, we can solve the two distinct risk-complexity problems. The first result optimizes the leading feasible average subset criterion.

Proposition 8 (Square-root rule for the average-subset criterion). *Combining Equation (21) and Equation (25) gives the following leading continuous criterion:*

$$\tilde{L}_{T,m}^{\text{sub}}(k) = \left(b + \frac{1}{k\bar{\tau}_m}\right) \frac{T-1}{T-k}, \quad 0 < k < T. \quad (29)$$

It has the unique minimizer

$$\tilde{k}_{T,m}^{\text{sub}} = \frac{-1 + \sqrt{1 + b\bar{\tau}_m T}}{b\bar{\tau}_m}. \quad (30)$$

If $T^{1/2} = o(m)$, as stated in Assumption 2, we obtain:

$$\tilde{k}_{T,m}^{\text{sub}} \sim \sqrt{\frac{T}{b\bar{\tau}}}. \quad (31)$$

Proposition 9 (Optimal infinitely bagged subset size). *Let*

$$k_{T,m}^* \in \arg \min_{1 \leq k \leq \min\{m, T-1\}} L_{T,m}^{\text{bag}, \infty}(k).$$

Under Assumptions 1–3 and 4,

$$\frac{k_{T,m}^*}{T^{1/4}} \rightarrow \left(\frac{\mathcal{D}}{b}\right)^{1/4}, \quad k_{T,m}^* \sim \left(\frac{\mathcal{D}T}{b}\right)^{1/4}. \quad (32)$$

The proof is in Appendix E.8. The two leading terms are

$$R_k^{\text{bag}} \sim \frac{\mathcal{D}}{mk^2}, \quad \frac{\Gamma_{m,k}}{T} \sim \frac{bk^2}{mT}.$$

Balancing the two terms gives the fourth-root rate. The common $1/m$ factor cancels, so m affects feasibility and boundary behavior but not the leading interior power of T .

The square-root and fourth-root rates solve different problems. The former evaluates a fitted subset before averaging over subset draws. The latter first averages the correlated fitted weights from the shared sample and then evaluates the resulting combination. This distinction parallels the risk–complexity role of subset size in Elliott et al. (2013) and Boot and Nibbering (2019), but neither paper derives the two inflation factors or optimizer rates obtained here.

Feasible finite- m selector. The asymptotic rule has a direct boundary-corrected implementation. Let $\hat{C} = T^{-1} \sum_{t=1}^T e_t e_t'$ be the training second-moment matrix. Then it is possible to estimate the common component $\hat{b}$ by the mean off-diagonal element of $\hat{C}$, and construct $\hat{\mathcal{D}}_m$ from the implied idiosyncratic variances as in Equation (4). We select:

$$\hat{k} = \arg \min_{1 \leq k \leq \min\{m, T-2\}} \left\{ \hat{\mathcal{D}}_m \left(\frac{1 - k/m}{k} \right)^2 + \hat{b} \frac{k^2}{T} \right\}. \quad (33)$$

The factor $1 - k/m$ restores the exact zero approximation cost at $k = m$. The selector uses only training-sample second moments and is therefore available before the evaluation outcome is observed. Section 5.1 states the numerical safeguard used in the empirical implementation.

Remark 1 (Fixed forecast pool). *If m is fixed while $T \rightarrow \infty$, estimation becomes asymptotically costless and $k = m$ is an asymptotically optimal boundary choice.*

4 Monte Carlo Simulations

Three separate exercises evaluate feasible finite- G performance, literal shared sample infinite bagging, and the average single subset criterion. This separation validates the corresponding results in Section 2.

4.1 Monte Carlo Design

For replication r , the target and the i th candidate forecast are generated as follows:

$$y_t = \mu_t + \varepsilon_t, \tag{34}$$

$$f_{i,t} = \mu_t + c_t + \xi_{i,t}, \quad i = 1, \dots, m, \tag{35}$$

where the predictable component follows

$$\mu_t = 0.6\mu_{t-1} + \nu_t.$$

The innovations are mutually independent over t and across components, with

$$\nu_t \sim N(0, 1), \quad \varepsilon_t \sim N(0, s), \quad c_t \sim N(0, q), \quad \xi_{i,t} \sim N(0, \sigma_i^2).$$

We set $s = q = 1$. The term c_t is common to all candidate forecasts, whereas $\xi_{i,t}$ generates heterogeneity in forecast precision. Because all combination weights are constrained to sum to one, the common predictable component μ_t does not affect the population comparison between alternative combination weights.

We characterize each precision profile by B and H , as discussed previously. To generate a precision profile, we draw a fixed set of centered Gaussian values z_i and set

$$\tau_i \propto \exp(\omega z_i).$$

The parameter ω is chosen numerically so that the finite pool attains the specified value of H exactly. The resulting precisions are then rescaled so that their sample mean equals $1/B$. For a given value of H , the same normalized precision profile is used across different values of B .

The main factorial experiment uses

$$m = 50, \quad T = 240, \quad T_{\text{test}} = 100,$$

and considers

$$B \in \{3, 8\}, \quad H \in \{1.05, 6\}.$$

The case $H = 1.05$ represents an approximately homogeneous forecast pool, whereas $H = 6$ represents substantial heterogeneity.

For every subset size $k = 1, \dots, m$, we draw $G = 800$ subsets independently and uniformly from the collection of all subsets of size k . For each subset S_g , the combination weights are estimated by constrained least squares:

$$\hat{w}_{S_g} = \arg \min_w \sum_{t=1}^T \left(y_t - \sum_{i \in S_g} w_i f_{i,t} \right)^2 \quad \text{subject to} \quad \sum_{i \in S_g} w_i = 1.$$

The finite-bagged RSM forecast is obtained by averaging the resulting subset forecasts,

$$\hat{y}_t^{\text{RSM}}(k) = \frac{1}{G} \sum_{g=1}^G \sum_{i \in S_g} \hat{w}_{i,S_g} f_{i,t}.$$

Its performance is evaluated using the out-of-sample mean squared prediction error

$$\widehat{\text{MSPE}}_{\text{RSM}}(k) = \frac{1}{T_{\text{test}}} \sum_{t=T+1}^{T+T_{\text{test}}} (y_t - \hat{y}_t^{\text{RSM}}(k))^2.$$

We compare bagged RSM with four benchmarks. The first is the equal-weight combination, with $w_i = 1/m$. The second is the full estimated combination, obtained by applying constrained least squares to all m forecasts. The third is the average performance of the individual subset combinations before bagging. Finally, the infeasible oracle uses the population-optimal weights

$$w_i^{\text{opt}} = \frac{\tau_i}{\sum_{j=1}^m \tau_j}.$$

Each design is evaluated using 500 Monte Carlo replications. To avoid evaluating a data-dependent subset size on the same simulated outcomes used to select it, the replications are divided into selection and evaluation samples. The odd-numbered replications are used to select

$$\hat{k}^* = \arg \min_k \widehat{\text{MSPE}}_{\text{RSM}}(k),$$

and the even-numbered replications are used to evaluate the selected subset size. Gains relative to equal weighting and gaps relative to the oracle are calculated from paired evaluation losses. The gain reported in Table 1 is the ratio of paired sample means, and its Monte Carlo standard error uses the paired delta-method covariance.

In addition to the finite-sample experiment, we calculate the population subset and infinite-bag risks using 50,000 random subset draws for each design. The infinite-bag quadratic risk is

estimated with an all-pairs U-statistic, avoiding the upward bias that results from squaring a finite Monte Carlo estimate of the mean weight vector.

Table 1 reports feasible finite- G forecasting performance. Forecast heterogeneity is the principal determinant of the gains from subset combination. When $H = 1.05$, bagged RSM and equal weighting are statistically indistinguishable. When $H = 6$, bagged RSM reduces MSPE by 11.75 percent for $B = 3$ and 25.38 percent for $B = 8$. The selected value of k in the nearly homogeneous designs should not be given a structural interpretation because the corresponding risk curves are essentially flat.

B	H	Selected k	RSM MSPE	Gain (%)	MCSE (pp)
3	1.05	4	2.050	-0.03	0.06
8	1.05	4	2.151	0.05	0.08
3	6.00	8	2.073	11.75	0.34
8	6.00	11	2.201	25.38	0.42

Notes: Each design has $m = 50$, $T = 240$, 500 replications, $G = 800$ bags, and 100 test observations. Odd replications select k and even replications evaluate it. Gain is $100(\bar{L}_{EW} - \bar{L}_{RSM})/\bar{L}_{EW}$. MCSE is the paired delta-method standard error of this ratio in percentage points. This table reports feasible finite- G forecasting performance, not the infinite-bag rate experiment.

Table 1: Exogenous Monte Carlo performance across scale and heterogeneity

4.1.1 Monte Carlo design for the shared-sample theory

The performance experiment establishes that subset combination can pay. It does not establish *why*, nor does it discipline the choice of k . We therefore turn to a second set of experiments designed to identify the objects of Section 3.2 directly: the shared-sample inflation coefficient $\Gamma_{m,k}$ and the subset-size implications it generates.

The simulation is constructed to match the estimand and asymptotic sequence of the theory. Section 2 distinguishes the population criteria in (11)–(13), and Section 3.1 attaches different estimation terms to the single-subset and bagged objects. We therefore evaluate each criterion separately. For the bagged criterion, the object of interest is the expected loss of the *averaged* weight vector, $L_{T,m}^{\text{bag},\infty}(k)$. Because the simulation uses a finite number of subset draws, we recover the infinite-bag loss from G draws using the exact conditional bridge of Proposition 16,

$$\hat{L}_{\infty} = \frac{G\hat{L}_G - \hat{L}_1}{G - 1}.$$

The baseline uses $G = 800$. Appendix F shows that the bridge-corrected risk is stable to further

increases in G and reports the corresponding finite-bag sensitivity checks.

Proposition 9 is a triangular-array result requiring $T^{1/4} = o(m)$ and $m = o(T)$; under a fixed pool it degenerates, as the accompanying remark makes explicit. A design that holds m fixed while growing T therefore cannot test the fourth-root rule. We instead let the pool grow with the sample, setting m to the nearest even integer to $5\sqrt{T}$, so that $m \asymp T^{1/2}$ satisfies both rate conditions along the entire sequence. Even pool sizes keep the balanced two-point profile exactly balanced, so $\bar{\tau}_m = 0.05$ and $\mathcal{D}_m = 4.608$ hold exactly in every row rather than approximately.

The quantity predicted by the shared-sample expansion is a difference between feasible and population risk, and that difference is two to four orders of magnitude smaller than either level. Estimating it by differencing two independent Monte Carlo averages would therefore be unnecessarily noisy. We evaluate the feasible and population weight vectors on the *same* subset draws within each replication, so that their common subset-sampling noise cancels. The resulting control-variate penalty has expectation $\Gamma_{m,k}/T + o(k^2/(mT))$ but a much smaller variance. Test risk is computed analytically as $\hat{v}'C_m\hat{v}$ rather than simulated, removing test-sample noise. The population risk is exact for the balanced two-point profile, by a hypergeometric calculation over subset composition, and is otherwise estimated by independent unbiased U -statistic blocks that also deliver a Monte Carlo standard error. The coefficient $\Gamma_{m,k}$ is integrated conditional on the exact hypergeometric distribution of the overlap $|S \cap R|$, avoiding reliance on rare nonzero overlaps when k/m is small.

The baseline uses $b = 0.2$ and the balanced profile $\tau_i \in \{0.01, 0.09\}$, so $\bar{\tau}_m = 0.05$ and $\mathcal{D}_m = 4.608$, with $T \in \{256, 512, 1024, 2048, 4096\}$ and correspondingly $m \in \{80, 114, 160, 226, 320\}$. To break the two-group structure, the experiment is repeated on two bounded continuous profiles, one symmetric and one right-skewed, recalibrated at every pool size to the same $\bar{\tau}_m$ and the same $\mathcal{D}_m = 4.608$. Because the theory asserts that only b and $\mathcal{D}$ enter the leading constant, these profiles provide a sharp test: they share the two constants the theory says matter while differing substantially in constants it says are irrelevant, with the dispersion index H ranging from 1.13 to 2.78. Appendix F reports the numerical verification of every estimator used here, including agreement of the closed-form Γ_{SR} with a direct matrix evaluation to machine precision and agreement of both the population and the Γ integrators with exact small-pool enumeration.

4.1.2 Validating the shared-sample risk expansion

The first and sharpest result concerns $\Gamma_{m,k}$ itself. If the derivation in Appendix E is right, then the control-variate penalty measured in the simulation should equal $\Gamma_{m,k}/T$ cell by cell, not merely in order of magnitude.

Figure 2 plots the two against each other for every design and every subset size, for the two-point baseline and the smooth continuous profile; the skewed profile behaves identically and is omitted only to keep the figure legible. The points lie on the forty-five degree line across roughly three orders of magnitude. The statistics reported below cover all three profiles. Regressing the empirical penalty on the theoretical penalty through the origin, weighting by the combined Monte Carlo variance, gives slopes of 1.005, 1.018, and 1.014 for the two-point, smooth, and skewed profiles, based on 138, 85, and 85 cells respectively, with no cell excluded. The median ratio of empirical to theoretical penalty is 1.005, 1.017, and 1.010, and the median absolute standardized discrepancy is 0.64, 1.00, and 0.84 — that is, the typical cell sits within one Monte Carlo standard error of the prediction. These magnitudes are insensitive to the signal-to-noise convention used for plotting: raising the cutoff from two to five changes the median ratios in the third decimal and removes at most five of 138 cells. They are mildly sensitive to weighting, as one would expect when the precision of a cell varies by orders of magnitude across the grid: an unweighted regression through the origin gives 1.023, 1.041, and 1.045. The defensible statement is that the measured penalty agrees with $\Gamma_{m,k}/T$ to within roughly five percent under any weighting and to within two percent when cells are weighted by their precision. One scope restriction should be attached to the weighted figure: because the theoretical penalty is nearly zero at small k , where two random subsets rarely overlap, precision weighting concentrates attention on large subsets. Roughly a third of the total weight falls in the band $10 \leq k \leq 20$ that contains every optimizer in this paper, and only a few percent below $k = 8$. The calibration statistic is thus a statement about the grid as a whole rather than a local verification at k^* .

The coefficient $\Gamma_{m,k}$ is not a free parameter fitted to the simulation; it is computed from the precision vector and the subset-pair overlap distribution alone, and it is the quantity that a single-subset calculation cannot deliver, because no one-subset moment formula involves two subsets at once. The evidence supports the central analytical claim of Section 3.1: averaging correlated subset fits estimated on a common training sample carries an estimation cost of order $k^2/(mT)$ with the specific constant derived in (66).

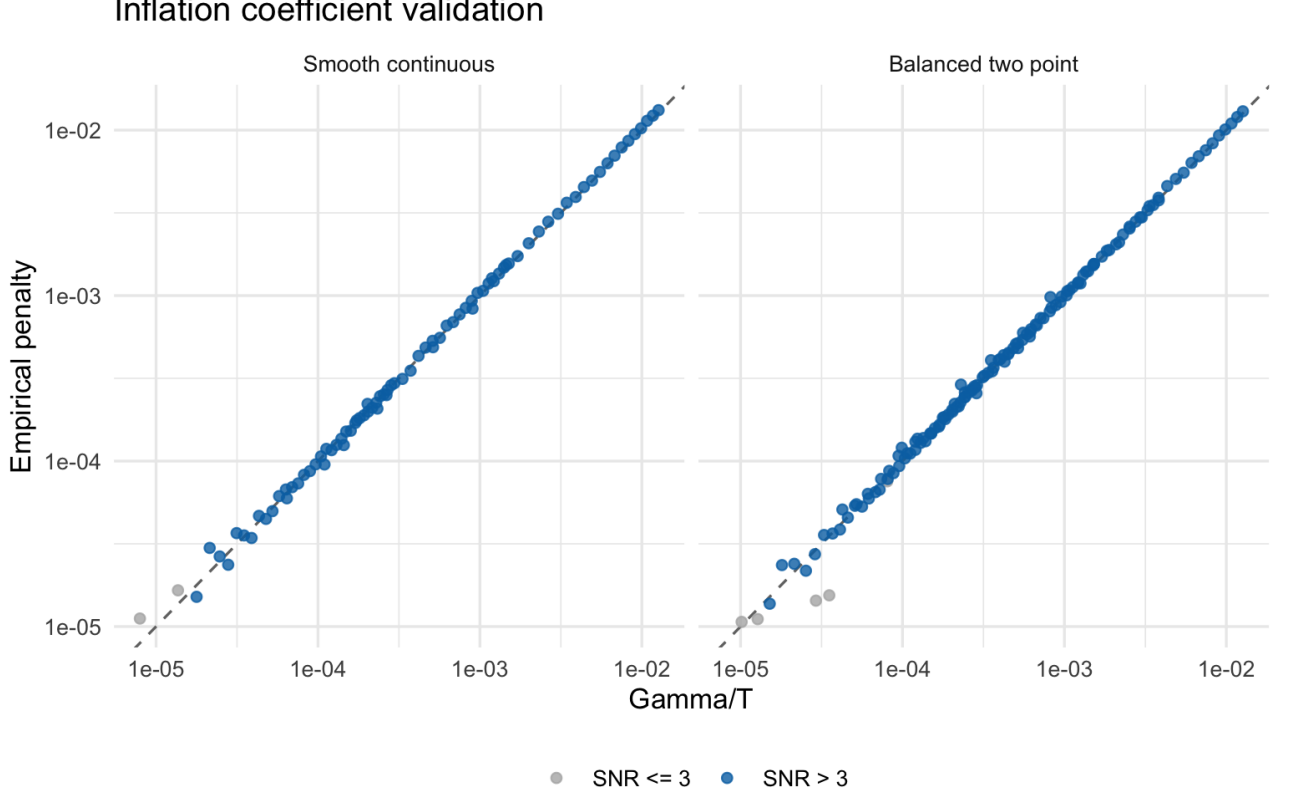


Figure 2: Validation of the shared sample inflation coefficient. Each point is one (T, m, k) cell; the horizontal axis is the theoretical penalty $\Gamma_{m,k}/T$ and the vertical axis is the control-variate empirical penalty. The dashed line is the forty-five degree line. Every cell of the two displayed profiles is plotted without filtering; shading marks cells whose theoretical penalty is small relative to its Monte Carlo standard error. Both axes are logarithmic.

4.1.3 Subset-size implications

Given the coefficient, the optimal subset size follows from balancing the approximation error $R_k^{\text{bag}} \sim \mathcal{D}/(mk^2)$ against the estimation penalty $bk^2/(mT)$. Table 2 compares three objects at every design: the empirical optimizer, the minimizer of the complete criterion $Q_k^{\text{bag},\infty} + \Gamma_{m,k}/T$, and the leading expression $(\mathcal{D}_m T/b)^{1/4}$.

The empirical and complete-theory optimizers agree to within a single subset in all eleven designs, across three precision profiles whose heterogeneity indices differ by a factor of 2.5. This agreement, not the fitted exponent, is the main evidence for the theory: it says that the risk criterion the theory writes down is the criterion the simulation is actually minimizing.

The fitted exponent requires more care, and we state the result precisely rather than favorably. Regressing $\log k^*$ on $\log T$ gives descriptive slopes of 0.29 and 0.27 (integer and locally smoothed optima) for the two-point profile and 0.32 and 0.30 for the continuous profiles, against

an asymptotic target of 0.25. The informative comparison is not against 0.25 but against the slope of the theory’s *own* finite- T optimizer, obtained by applying the identical regression to the minimizer of the complete criterion. That slope is 0.34 for the smooth profile and 0.32 for the skewed profile — at or above the corresponding empirical values. For the continuous profiles the departure from 0.25 is therefore demonstrably a finite- T property of the exact criterion rather than a defect of the simulation. These two profiles happen to produce identical integer optima, so on the empirical side they furnish one observation rather than two; the theoretical optima do differ between them, which is what makes the comparison informative.

The two-point baseline yields a slope of 0.24, marginally below 0.25 and below the empirical 0.27; the two differ by 0.026 against a regression standard error of roughly 0.017, so they are statistically indistinguishable, but the deviations run in opposite directions and the agreement is within noise rather than a shared remainder.

A similar gap appears in the level, though its sign is profile-specific. For the two continuous profiles the leading expression $(\mathcal{D}_m T/b)^{1/4}$ overstates the optimal subset size throughout the grid — at $T = 256$ it returns 8.8 where both the empirical and the complete-theory optima are 7 — and approaches them only as T grows. For the two-point baseline it instead crosses from above to below the two optimizers near $T = 1024$.

Appendix F reports three comparative-static experiments in the same spirit, varying b , $\mathcal{D}_m$, and the precision scale one at a time at $T = 1024$. They point the same way, but they are much weaker evidence than the experiments above and we treat them accordingly: each elasticity is fitted on three to five integer optima, and because those optima are themselves uncertain by one to two units, moving a single cell by one subset can change a fitted elasticity by half its value. We therefore read them only as confirming the sign and rough magnitude of each channel, and not as discriminating between the complete and leading predictions.

The theoretical section also insists that the square-root and fourth-root rules solve different problems rather than competing as answers to one problem. Because both rules are statements about the same DGP and the same estimated weights, differing only in whether losses or weights are averaged first, this distinction is testable within a single experiment.

Repeating the design while evaluating each fitted subset before averaging — the criterion $L_{T,m}^{\text{sub}}(k)$ — reverses the conclusion in exactly the way the theory requires. The empirical optimizer coincides with the minimizer of $Q_k^{\text{sub}}(T-1)/(T-k)$ in all eight (profile, T) cells, taking the values 10, 15, 23, and 30 at $T = 128, 256, 512, 1024$. The locally smoothed log-log slopes are 0.51 and 0.52 for the two profiles against a target of $1/2$; the unsmoothed integer slopes are 0.54 for both. The exact inflation factor $(T-1)/(T-k)$ of Proposition 6 is verified

Profile	T	m	H	$\hat{k}^*$	$\hat{k}_{\text{smooth}}^*$	k_{Γ}^*	$(\mathcal{D}_m T/b)^{1/4}$
Two-point	256	80	2.78	8	8.66	9	8.76
Two-point	512	114	2.78	10	10.63	10	10.42
Two-point	1024	160	2.78	13	12.27	12	12.39
Two-point	2048	226	2.78	15	15.06	15	14.74
Two-point	4096	320	2.78	18	18.43	17	17.53
Smooth	256	80	1.25	7	7.33	7	8.76
Smooth	1024	160	1.25	11	10.81	11	12.39
Smooth	4096	320	1.25	17	17.82	18	17.53
Skewed	256	80	1.14	7	7.20	7	8.76
Skewed	1024	160	1.13	11	10.50	10	12.39
Skewed	4096	320	1.13	17	16.54	17	17.53
Log-log slope on $\log T$, two-point				0.29	0.27	0.24	0.25
smooth / skewed				0.32	0.32 / 0.30	0.34 / 0.32	0.25

Notes: All designs use $b = 0.2$ and $\bar{\tau}_m = 0.05$, and every profile is calibrated so that $\mathcal{D}_m = 4.608$ exactly; the profiles therefore share the two constants that enter the leading rule and differ in the dispersion index $H = \bar{\tau}_m m^{-1} \sum_i \tau_i^{-1}$, which does not. $\hat{k}^*$ is the empirical control-variate optimizer and $\hat{k}_{\text{smooth}}^*$ its five-point local-quadratic counterpart. k_{Γ}^* minimizes the complete criterion $Q_k^{\text{bag}, \infty} + \Gamma_{m,k}/T$, with $\Gamma_{m,k}$ integrated conditional on the exact hypergeometric overlap distribution. The last column is the leading expression of Proposition 9. Empirical risk is the $G = 800$ conditional-bridge estimate of expected infinite-bag risk. The bottom rows report descriptive log-log slopes. They are given for the theoretical columns as well, so that the empirical slope is compared against the criterion the theory implies on this finite grid rather than against its asymptote; these five-point and three-point slopes are descriptive summaries, not formal inference.

Table 2: Optimal subset size for shared-sample infinite bagging under alternative precision profiles

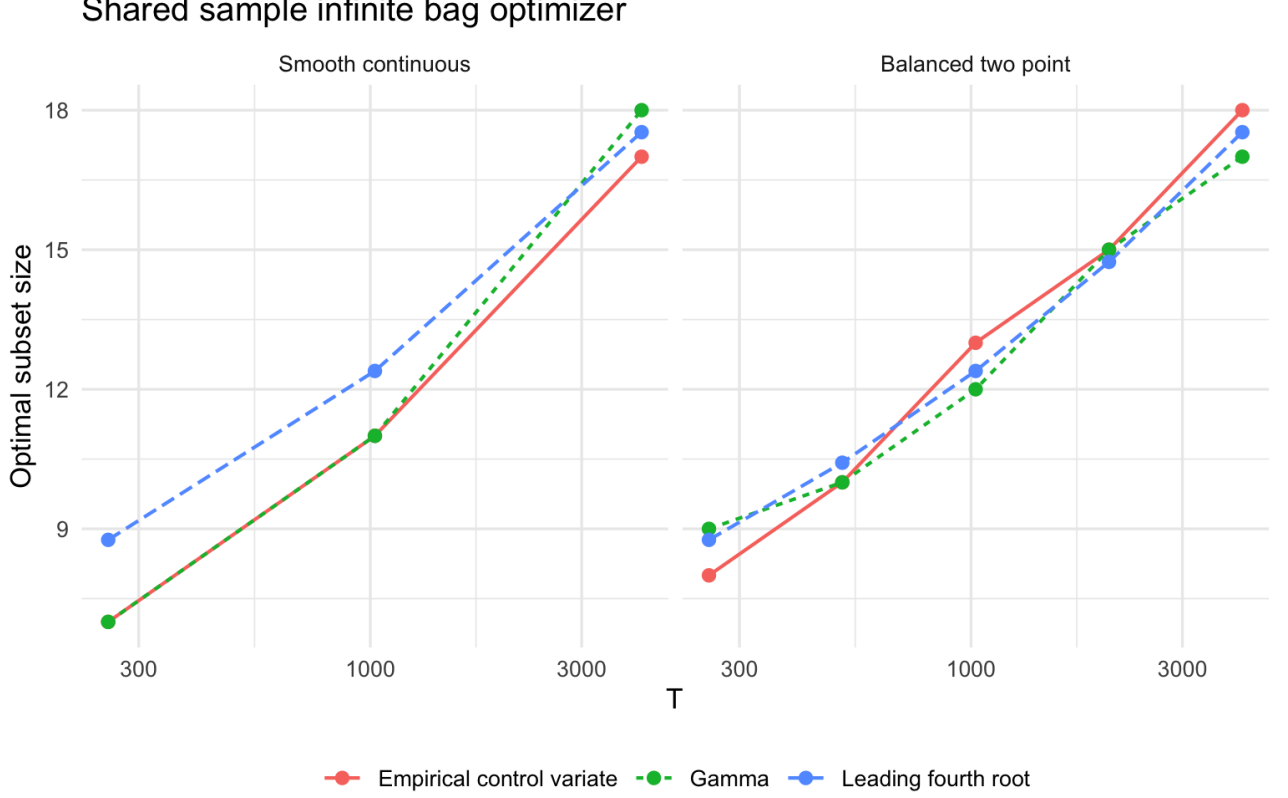


Figure 3: Optimal subset size for shared sample infinite bagging, for the balanced two-point baseline and the smooth continuous profile. The empirical control variate optimizer and the minimizer of the complete criterion $Q_k^{\text{bag}, \infty} + \Gamma_{m,k}/T$ track each other closely at every design. The leading fourth root expression lies above both throughout for the continuous profile and crosses from above to below them near $T = 1024$ for the two-point profile. Figure 8 adds the skewed profile.

separately to within four Monte Carlo standard errors on a design with a general, non-grouped precision vector.

Two qualifications keep this from being read as a sharper measurement than it is. The subset grid is logarithmic and coarse near the optimum — the neighbors of $k = 30$ are 27 and 35 — so “coincides” means agreement within one grid cell, and shifting the largest design by a single grid point would move the fitted slope to 0.47 or 0.63. The exponent is therefore resolved to roughly $[0.47, 0.63]$.

4.1.4 Pool size and the limits of the evidence

Proposition 9 asserts that m affects feasibility and higher-order terms but not the leading power of T , because the factor $1/m$ cancels between the approximation and estimation terms. The claim is about the *location* of the optimum, not about the level of the penalty: over the same

range the coefficient $\Gamma_{m,k}$ at $k = 12$ falls by an order of magnitude, from 1.46 to 0.15, exactly as the $1/m$ factor requires. What should be invariant is the argmin.

Holding $T = 1024$ and varying the pool from $m = 60$ to $m = 400$ — a factor of nearly seven, over which a quarter-power law would move k^* from 10 to 16 — the empirical optimizer takes the values 11, 11, 13, 13, 11 and the complete-theory optimizer 10, 11, 11, 12, 12. Regressing $\log k^*$ on $\log m$ gives an empirical elasticity of 0.04 (standard error 0.07), which is indistinguishable from zero and distinguishable from 0.25. The complete-theory optimizer, however, does drift slightly, with an elasticity of 0.10 that is marginally significant. Finite-pool drift in the argmin is thus a property of the exact criterion rather than an artifact of the simulation, and the evidence supports the absence of a *leading* pool-size effect over the evaluated range without establishing exact invariance in finite pools.

Last, it is worth mentioning that the risk curves are flat near their minima, and this flatness worsens as T grows. From a practical standpoint, the flatness is reassuring rather than concerning: a criterion this flat means that a moderately misspecified k costs little, which is precisely what makes a simple rule of thumb viable in the empirical application.

5 Empirical Application

5.1 Data, real-time design, and feasible selector

We use individual one-quarter-ahead forecasts of the U.S. GDP price index from the Federal Reserve Bank of Philadelphia’s Survey of Professional Forecasters (SPF). For a survey submitted in quarter t , the target is annualized quarter-on-quarter GDP price-index inflation in $t + 1$, constructed from the respondent’s adjacent level forecasts as $100[(PGDP3/PGDP2)^4 - 1]$. Historical forecast errors at each origin use only the GDP price-index vintage available in the middle of that quarter, and combinations are scored against the first official release. This timing convention follows the real-time-data principle that a forecasting procedure should be estimated on the information set actually available at the origin (Croushore and Stark, 2001; Stark, 2010; Croushore, 2011). Scoring against the first release also keeps the evaluation target close to what the forecasters were asked to predict; later releases are examined separately because data revisions can alter measured forecast performance (Stark and Croushore, 2002).

The common evaluation sample contains 146 forecasts between 1985Q2 and 2025Q2. It is not contiguous: 15 quarters are omitted because fewer than eight forecasters satisfy the

eligibility rule; 1995Q4 also lacks a first-release outcome.⁶ Every method is evaluated on the same feasible origins.

At each origin, a forecaster must submit a current forecast and must have responded in at least 20 of the last 48 resolved rounds. Current forecasts are never imputed. Within the training window, an interior missing response is filled using that forecaster’s expanding mean of genuinely observed earlier responses; gaps before entry are backfilled with the first observed response. Entry, exit, and intermittent nonresponse are intrinsic to professional-forecaster panels (Capistrán and Timmermann, 2009; Bürgi, 2023); the expanding, past-only fill used here adapts the panel-balancing logic of Conflitti et al. (2015) without using future responses. The screen is applied before imputation. The median eligible pool contains 22 forecasters, compared with 35 current respondents, and the median imputed share of the training panel is 28 percent.

The finite- m selector is a direct plug-in version of the theoretical trade-off. Let $\hat{C} = T^{-1} \sum_t e_t e_t'$, let $\hat{c}$ be the mean off-diagonal element of $\hat{C}$, and let $\hat{d}_i = \hat{C}_{ii} - \hat{c}$. Nonpositive $\hat{d}_i$ are replaced by five percent of the median positive value before constructing $\hat{\mathcal{D}}_m$ from Equation (4). The selected size is

$$\hat{k}_t = \arg \min_{1 \leq k \leq \min(m_t, T-2)} \left\{ \hat{\mathcal{D}}_{m,t} \left(\frac{1 - k/m_t}{k} \right)^2 + \hat{c}_t \frac{k^2}{T} \right\}. \quad (36)$$

All inputs are computed from the 48-quarter training window; neither the contemporaneous outcome nor evaluation-sample losses enter the rule. The COVID-19 targets 2020Q2 and 2020Q3 remain in the full evaluation sample but are excluded from subsequent weight estimation and selection windows.

The five-percent floor is the baseline because it is scale equivariant and replaces an inadmissible nonpositive idiosyncratic-variance estimate by a small, rather than dominant, fraction of the typical positive estimate. It prevents inversion from assigning effectively infinite precision while leaving every positive estimate unchanged. Table 4 shows that the substantive result does not depend on this convention: all alternatives select only exactly enumerated subset sizes, and the full-sample RMSFE remains between 1.258 and 1.271. The five-percent setting gives the lowest RMSFE among the predeclared values, but it was not selected from evaluation losses.

⁶The omitted quarters are 1986Q1, 1992Q4, 1993Q2–1994Q4, and 1995Q1–1996Q2; 1993Q1 remains in the sample.

5.2 Does the selector choose the right amount of regularization?

Figure 4 is the central external validation. It is rebuilt from the fully real-time object and retains the original diagnostic's two displayed samples: Ex-COVID and Non-crisis (called Calm in the tables). The Full-sample curve is not drawn, preserving the original visual design; its results are reported in the text. To keep the evaluation origins fixed, the connected curves report fixed $k = 1, \dots, 8$, the largest subset size feasible at every origin. Horizontal lines show the equal-weight and simplex benchmarks; large open circles mark the infeasible ex-post minima, and the out-of-range unrestricted endpoints are reported in the legend. The selected headline forecasts are enumerated exactly at every origin. Some unselected middle- k points in the diagnostic curve use 20,000 uniform subset draws when $\binom{m}{k} > 300,000$.

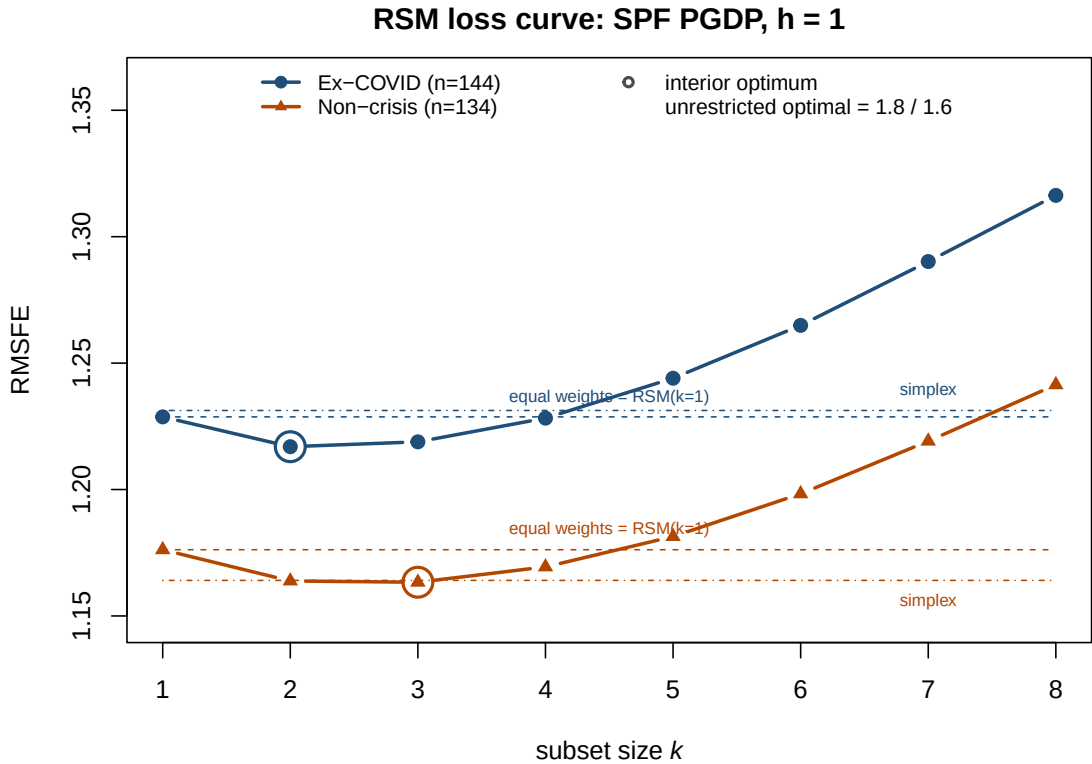


Figure 4: Fully real-time RSM loss curves and benchmarks: Ex-COVID and Non-crisis samples

The displayed loss curves are U-shaped. RMSFE is minimized at $k = 2$ in the Ex-COVID sample and at $k = 3$ in the Non-crisis sample, which additionally excludes 2008Q3–2009Q4 and 2020Q1–2021Q2. The unshown Full-sample calculation is also minimized at $k = 2$. Moving to the full estimated endpoint raises RMSFE sharply relative to the interior minimum.

Appendix Figure 12 overlays this real-time curve with the conventional final-vintage pseudo-out-of-sample curve. The levels are not directly comparable because the outcome definition and feasible-origin counts differ, but the diagnostic conclusion is unchanged: both designs have the

same ex-post minimizing fixed k in every sample, and both deteriorate sharply toward the fully estimated endpoint.

Sample	Origins	Median theory-selected k	Ex-post best fixed k
Full	146	3.0	2
Ex-COVID	144	3.0	2
Calm	134	2.5	3

Notes: Ex-COVID removes 2020Q2–Q3. Calm additionally removes 2008Q3–2009Q4 and 2020Q1–2021Q2. The theory-based selector uses only origin-specific training second moments. The ex-post column minimizes realized RMSFE over fixed $k = 1, \dots, 8$ and is infeasible. Every selected k lies in the exact-enumeration region.

Table 3: Theory-Based and Ex-Post Subset Sizes, Fully Real-Time SPF

Floor (%)	Median k , Full	RMSFE, Full	RMSFE, Ex-COVID	Changed k share
1.0	2	1.271	1.226	0.301
2.5	3	1.262	1.217	0.253
5.0	3	1.258	1.212	0.000
10.0	2	1.260	1.214	0.260
20.0	2	1.260	1.214	0.404

Notes: The floor is a fraction of the median positive estimated idiosyncratic variance. The last column is the full-sample share of origins whose selected k differs from the five-percent baseline. All specifications use 146 Full and 144 Ex-COVID origins, and the selected subset is in the exact-enumeration region at every origin.

Table 4: Sensitivity to the idiosyncratic-variance floor

The agreement concerns the average level of regularization rather than precise quarter-by-quarter timing. Figure 5 shows that the feasible rule remains interior as the eligible panel expands and contracts.

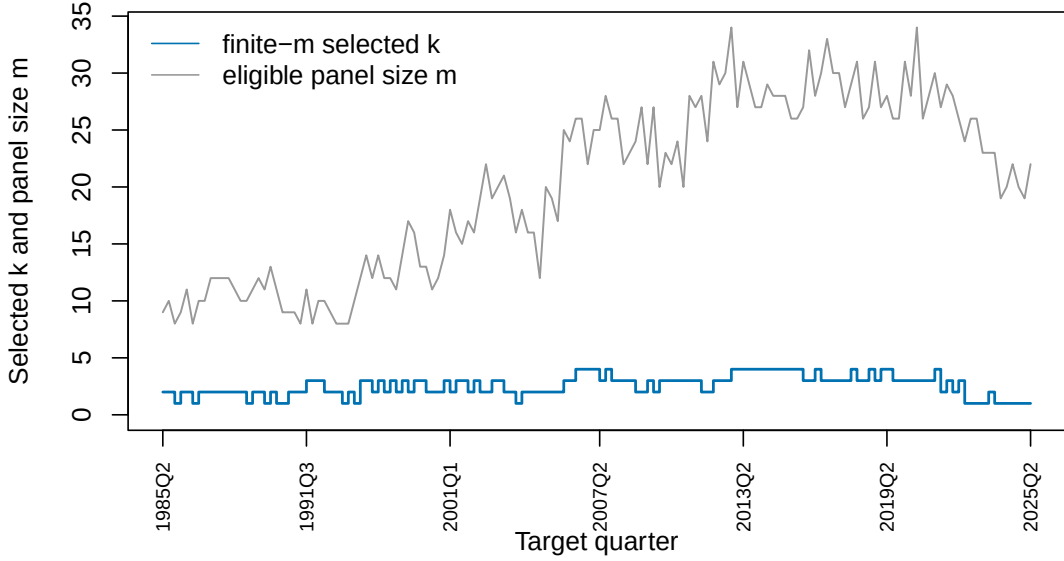


Figure 5: Fully real-time selected subset size and eligible panel size

5.3 Forecast evaluation and shrinkage comparison

We compare RSM with the all-respondent survey mean, equal weighting on the screened estimation panel, nonnegative sum-to-one simplex weights, and unrestricted adding-up weights. We add three distinct regularization benchmarks. The first is the existing conventional shrinkage path,

$$\hat{w}_{\lambda,t} = (1 - \lambda_t)m_t^{-1}l_{m_t} + \lambda_t\hat{w}_{\text{unrestricted},t}, \quad 0 \leq \lambda_t \leq 1.$$

At each origin, λ_t is selected by leave-one-out validation within the 48-observation vintage-correct training panel. In each fold the unrestricted endpoint is re-estimated without the held-out observation, and the squared-error-minimizing intensity is clipped to $[0, 1]$. No evaluation outcome is used. Because ordinary leave-one-out folds retain serially adjacent quarters, this is a transparent training-only comparator rather than a claim of theoretically optimal time-series cross-validation (Tashman, 2000; Bergmeir et al., 2018). The median selected intensity is 0.347, with an interquartile range of 0.103 to 0.581.

The second is the operational two-step partially egalitarian Lasso of Diebold and Shin (2019). At each origin, an unstandardized, no-intercept Lasso of the vintage-correct outcome on the forecast panel selects a support; the surviving current forecasts are then averaged exactly. This is the authors’ “Lasso, Average” implementation, not the discontinuous conceptual one-step objective. We choose $\lambda_t/\lambda_{\max,t}$ from 100 logarithmically spaced values between 0.95 and

0.01 by ordered validation inside the training window: observations 1–24 initialize the fit, observations 25–48 are predicted recursively, and missing forecasts are re-imputed from the available prefix in every fold. The full-sample fit is then recomputed at the selected penalty. The median selected support is 4.5 forecasters (mean 4.86), and the median normalized penalty is 0.034 with interquartile range 0.010–0.114. No selected support is empty.

For comparability with the authors’ empirical search, we also reproduce their common raw grid $\lambda \in \exp\{15, \dots, -15\}$ with 200 equally spaced log points and choose the point minimizing the realized SPF loss among penalties whose second-stage support is nonempty at every origin. This deliberately infeasible exercise selects the interior value $\lambda = 1.0783$ in both evaluation samples. It yields RMSFE 1.263 in the Full sample and 1.222 Ex-COVID, with median support 16 (mean 15.30 in the Full sample). Thus even the ex-post best value on the authors’ grid does not beat RSM, although it improves on the feasible rolling-validation peLASSO. Because the evaluation outcomes select this single common penalty, it is a diagnostic upper bound for that grid, not an implementable forecast rule.

Third, we adapt the shrinkage rule of Stock and Watson (2004) to the same rolling real-time information set. Stock and Watson shrink an otherwise unconstrained no-intercept OLS coefficient vector toward equal weights using

$$\lambda_t(\kappa) = \max \left\{ 0, 1 - \kappa \frac{m_t}{T_t - 1 - m_t} \right\}, \quad \kappa \in \{0.25, 0.5, 1\}.$$

Thus, 0.25, 0.5, and 1 are values of κ , not fixed values of λ . The present application retains its rolling 48-eligible-observation, vintage-correct window and origin-specific panel rather than Stock and Watson’s expanding balanced panel. The resulting median intensities are 0.780, 0.560, and 0.120, respectively; the larger κ therefore moves the combination much closer to equal weighting. The current LOO estimator and this Stock–Watson adaptation share the same affine interpolation form, but trace different paths because their estimated endpoints, windows, and tuning rules differ.

Method	All 146 origins		Ex-COVID, 144 origins	
	MAFE	RMSFE	MAFE	RMSFE
All-respondent mean	0.902	1.259	0.870	1.209
Equal weights, screened panel	0.913	1.275	0.883	1.229
RSM, finite- m theoretical k	0.911	1.258	0.882	1.212
peLASSO, rolling-validation λ	0.917	1.279	0.885	1.229
peLASSO, ex-post common $\lambda^\dagger$	0.906	1.263	0.878	1.222
Stock–Watson, $\kappa = 0.25$	1.161	1.693	1.139	1.667
Stock–Watson, $\kappa = 0.50$	1.014	1.434	0.985	1.394
Stock–Watson, $\kappa = 1.00$	0.899	1.260	0.869	1.213
Shrinkage to equal weights, LOO tuned	0.945	1.354	0.916	1.312
Simplex weights, Conflitti-style	0.908	1.290	0.872	1.231
Unrestricted adding-up weights	1.226	1.825	1.208	1.809

Notes: Forecast errors are annualized percentage points. Historical outcomes used to estimate weights are vintage-correct and forecasts are scored against first releases. Every feasible tuning decision uses only the corresponding origin’s training information; no outer evaluation outcome enters feasible peLASSO, RSM, or LOO-shrinkage tuning. $\dagger$ The common Diebold–Shin-grid penalty is selected using the outer evaluation losses and is infeasible. Relative to screened equal weighting, RSM lowers RMSFE by 1.33 percent in the full sample and 1.36 percent ex-COVID; one-sided unconditional Giacomini–White tests with HAC(4) variance estimates give p -values of 0.148 and 0.165 (Giacomini and White, 2006). Feasible peLASSO raises RMSFE by 0.35 and 0.00 percent, with p -values 0.545 and 0.500. The Stock–Watson $\kappa = 1$ rule lowers RMSFE by 1.15 and 1.25 percent, with p -values 0.273 in both samples. The other two Stock–Watson settings and the LOO line shrinkage retain too much of their unstable estimated endpoints.

Table 5: U.S. SPF GDP Price-Inflation Forecast Evaluation

RSM substantially regularizes the unstable full vector: it lowers RMSFE by 31.09 percent relative to unrestricted weights in the full sample and by 32.99 percent ex-COVID. Feasible peLASSO is competitive with screened equality, especially ex-COVID, but does not improve upon RSM or the all-respondent survey mean; even its infeasible ex-post common penalty remains slightly less accurate than RSM. The Stock–Watson comparison clarifies why shrinkage strength matters. With $\kappa = 0.25$ or 0.5, the rule retains substantial exposure to unstable no-intercept OLS and performs poorly; $\kappa = 1$ shrinks much more strongly and nearly matches RSM. The original LOO path remains weaker than either peLASSO or the strongest Stock–Watson specification.

Figure 6 displays the role of the peLASSO penalty as an ex-post sensitivity diagnostic. Each fixed normalized-penalty rule is itself feasible, but choosing the best path point after inspecting evaluation losses would be infeasible and is not the tuning rule used in Table 5. Moving right weakens regularization and increases the average number of retained forecasters. Over the displayed range, RMSFE falls toward the weakest penalty: aggressive Lasso selection is not rewarded in this U.S. SPF panel. The rolling-validation rule instead varies the penalty by origin using training information alone and attains RMSFE 1.279, slightly below every fixed-ratio point in the figure.

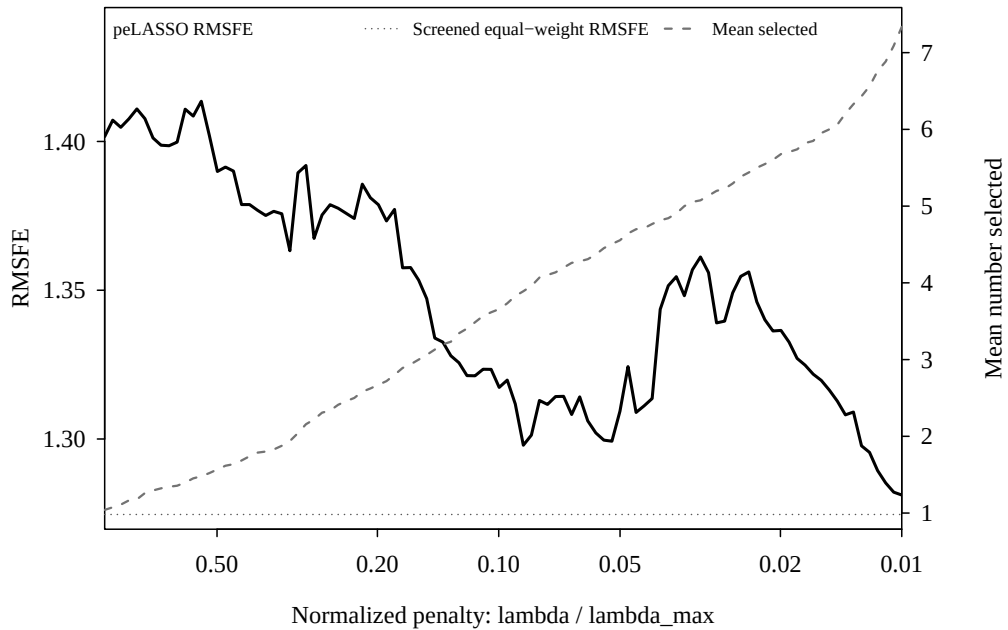


Figure 6: Role of the normalized peLASSO penalty in the fully real-time SPF exercise

The equal-weight benchmark matters. RSM improves by 1.33 percent over the screened mean, but the unscreened all-respondent mean has RMSFE 1.259. RSM's RMSFE is 1.258, an economically negligible difference. The screen needed for estimated-weight methods discards a median of 13 current respondents, so much of the apparent gain over screened equal weighting recovers the cost of constructing an estimable balanced panel.

5.4 Weight profiles and interpretation

Figure 7 compares real-time weights in 2022Q1 using the visual design of the previous paper version: thick orange segments show RSM, blue crosses show simplex weights, and the dashed line is equal weighting. RSM remains dense and close to equality while allowing moderate cross-sectional heterogeneity; simplex weights concentrate on a few forecasters. Unrestricted weights

are deliberately omitted from the plot because their range, -1.29 to 1.50 , would compress the economically relevant comparison; their instability remains visible in Table 5.

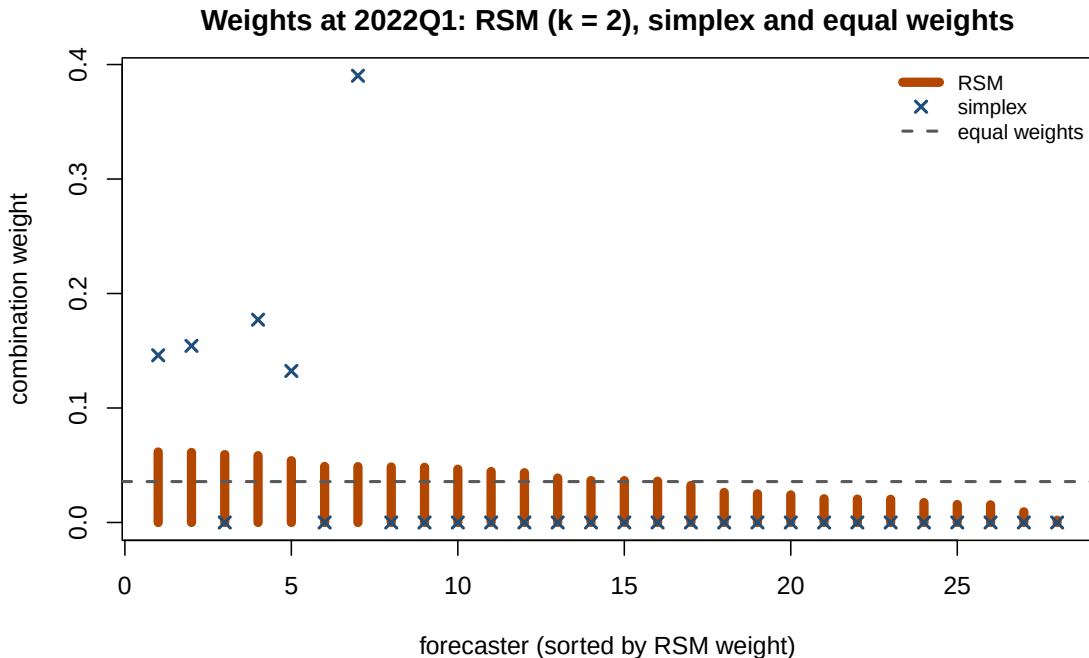


Figure 7: Fully real-time combination-weight profiles, 2022Q1

Three conclusions follow. First, the fully real-time selector chooses an interior k close to the ex-post fixed- k optimum, and the loss curve is U-shaped. Second, the new competitors reinforce the importance of regularization strength: peLASSO is approximately tied with screened equality, while Stock–Watson performs well only at the strongest published shrinkage setting. Third, RSM’s accuracy gain over the unscreened survey consensus remains negligible. The empirical evidence therefore validates the amount-of-regularization mechanism, not a broad claim of forecast dominance.

6 Conclusion

This paper reframes forecast combination as a question of how much of the optimal combination-weight vector the available sample can support. Equal weighting and full-pool estimated optimal weights form the endpoints of a regularization path indexed by the random-subset size k . Increasing k allows the estimator to exploit more heterogeneity in forecast quality, but it also raises the cost of estimating combination weights from a finite time-series sample.

The theoretical results characterize this approximation–estimation trade-off. Under the paper’s common-component forecast-error model, the leading excess risk of shared-sample infinite-

bag RSM balances an approximation cost of order $1/(mk^2)$ against an estimation cost of order $k^2/(mT)$. Because the subset regressions are fitted on the same observations, their estimation errors remain correlated after bagging. Balancing the two terms therefore yields the fourth-root rule $k^* \asymp T^{1/4}$. By contrast, the expected loss of an individually estimated random subset defines a different risk criterion and produces a square-root rule. A feasible finite- m criterion translates the shared-sample result into a data-based choice of the degree of regularization.

The real-time SPF application provides external support for this mechanism. Using training information only, the theoretical selector chooses a median subset size close to the ex-post loss-minimizing fixed range. Realized loss is U-shaped in k , and performance deteriorates as the estimator approaches the unrestricted full-pool endpoint. Conventional shrinkage toward equality can also perform similarly when sufficiently strong. Thus, the empirical contribution is a selector derived from the theoretical mechanism identifies approximately the amount of heterogeneity the available sample can support.

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Appendices

A Proofs for the Population Results

A.1 A common component identity

Every weight vector used in the paper satisfies $\iota'_m w = 1$. Hence the difference $\delta = w_1 - w_2$ between any two such vectors satisfies $\iota'_m \delta = 0$, and therefore

$$\delta'(a\iota_m\iota'_m + D)\delta = a(\iota'_m\delta)^2 + \delta'D\delta = \delta'D\delta \quad (37)$$

for every scalar a . Thus the common realized-error component remains in risk levels but cancels from comparisons between adding-up weight vectors.

A.2 Subset algebra

Lemma 1 (Subset algebra). *Fix a subset S of size k and define $C_S(a) = a\iota_k\iota'_k + D_S$ for $a \geq 0$. Then*

$$C_S(a)^{-1} = D_S^{-1} - \frac{aD_S^{-1}\iota_k\iota'_kD_S^{-1}}{1 + aA_S}, \quad (38)$$

$$C_S(a)^{-1}\iota_k = \frac{D_S^{-1}\iota_k}{1 + aA_S}, \quad \iota'_k C_S(a)^{-1}\iota_k = \frac{A_S}{1 + aA_S}. \quad (39)$$

The minimizer of $w'C_S(a)w$ subject to $\iota'_k w = 1$ and its minimum are

$$w_S = \frac{D_S^{-1}\iota_k}{A_S}, \quad \min_{\iota'_k w = 1} w'C_S(a)w = a + \frac{1}{A_S}. \quad (40)$$

Finally,

$$C_S(a)^{-1} - \frac{C_S(a)^{-1}\iota_k\iota'_k C_S(a)^{-1}}{\iota'_k C_S(a)^{-1}\iota_k} = D_S^{-1} - \frac{D_S^{-1}\iota_k\iota'_k D_S^{-1}}{A_S}. \quad (41)$$

Proof. Let $x = C_S(a)^{-1}y$. The equation $C_S(a)x = y$ yields:

$$C_S(a)x = a\iota_k(\iota'_k x) + D_S x = y,$$

moving the first term across gives $D_S x = y - a\iota_k(\iota'_k x)$, that is,

$$x = D_S^{-1}y - a(\iota'_k x)D_S^{-1}\iota_k. \quad (42)$$

Premultiplying by ι'_k and using $\iota'_k D_S^{-1}\iota_k = A_S$ yields

$$(1 + aA_S)\iota'_k x = \iota'_k D_S^{-1}y, \quad \iota'_k x = \frac{\iota'_k D_S^{-1}y}{1 + aA_S}.$$

Putting this value of $\iota'_k x$ back into the right-hand side of Equation (42) gives:

$$x = \left(D_S^{-1} - \frac{a D_S^{-1} \iota_k \iota'_k D_S^{-1}}{1 + a A_S} \right) y.$$

Because this holds for every y , the bracket is $C_S(a)^{-1}$, since $x = C_S(a)^{-1} y$. It proves Equation (38).

Setting $y = \iota_k$ gives the first identity in Equation (39). Namely,

$$C_S(a)^{-1} \iota_k = D_S^{-1} \iota_k \left(1 - \frac{a A_S}{1 + a A_S} \right) = \frac{D_S^{-1} \iota_k}{1 + a A_S},$$

Premultiplying by ι'_k gives the second statement in Equation (39).

For (40), first note that $C_S(a)$ is positive definite for every $a \geq 0$: $D_S \succ 0$ because $0 < \tau_i < \infty$, and $a \iota_k \iota'_k \succeq 0$, so $w' C_S(a) w \geq w' D_S w > 0$ for every $w \neq 0$. The map $w \mapsto w' C_S(a) w$ is therefore strictly convex, and its minimizer on the affine set $\{w : \iota'_k w = 1\}$ is characterized by the first-order condition of the Lagrangian $w' C_S(a) w - 2\lambda(\iota'_k w - 1)$, namely $C_S(a) w = \lambda \iota_k$ for a scalar multiplier λ . Hence $w = \lambda C_S(a)^{-1} \iota_k$, and the constraint $\iota'_k w = 1$ pins down $\lambda = 1/(\iota'_k C_S(a)^{-1} \iota_k)$, so

$$w_S = \frac{C_S(a)^{-1} \iota_k}{\iota'_k C_S(a)^{-1} \iota_k}, \quad w'_S C_S(a) w_S = \lambda^2 \iota'_k C_S(a)^{-1} \iota_k = \frac{1}{\iota'_k C_S(a)^{-1} \iota_k}.$$

Strict convexity makes this stationary point the unique global constrained minimizer. Substituting (39),

$$w_S = \frac{D_S^{-1} \iota_k / (1 + a A_S)}{A_S / (1 + a A_S)} = \frac{D_S^{-1} \iota_k}{A_S}, \quad \min_{\iota'_k w = 1} w' C_S(a) w = \frac{1 + a A_S}{A_S} = a + \frac{1}{A_S},$$

which proves both statements in (40). In particular, since the diagonal entries of D_S^{-1} are the precisions, $w_{i,S} = \tau_i / A_S$.

For (41), equations (38)–(39) give

$$\frac{C_S(a)^{-1} \iota_k \iota'_k C_S(a)^{-1}}{\iota'_k C_S(a)^{-1} \iota_k} = \frac{D_S^{-1} \iota_k \iota'_k D_S^{-1}}{A_S (1 + a A_S)}.$$

Subtracting this expression from (38), the coefficient on $D_S^{-1} \iota_k \iota'_k D_S^{-1}$ is

$$-\frac{a}{1 + a A_S} - \frac{1}{A_S (1 + a A_S)} = -\frac{1}{A_S},$$

which proves the result. $\square$

Proof of Proposition 3. Apply Lemma 1 with $a = b$, so that $C_S(b) = b \iota_k \iota'_k + D_S = \Omega_{e,k,S}$ by (9). Equation (40) then gives $w_{i,S} = \tau_i / A_S$ directly. For the risk, the embedding $v_S = P_S w_S$ satisfies

$$v'_S \Omega_{e,m} v_S = w'_S P'_S \Omega_{e,m} P_S w_S = w'_S \Omega_{e,k,S} w_S = b + \frac{1}{A_S},$$

using $\Omega_{e,k,S} = P'_S \Omega_{e,m} P_S$ from (9) and the minimum value in (40); the same identity shows that the two definitions of $Q(S)$ in (11) coincide. Componentwise, $(v_S)_i = w_{i,S} = \tau_i/A_S$ for $i \in S$ and $(v_S)_i = 0$ otherwise, since P_S places the coordinates of w_S in their original positions.

The full-panel case (17) is the lemma applied to $S = \{1, \dots, m\}$, for which $P_S = I_m$, $A_S = A_m = m\bar{\tau}_m$: it yields $w_i^{\text{opt}} = \tau_i/A_m$ and $Q^{\text{opt}} = b + 1/(m\bar{\tau}_m)$.

Finally, (18) follows by direct evaluation of the quadratic form at $w^{\text{ave}} = m^{-1}\iota_m$:

$$Q^{\text{ave}} = m^{-2} \iota'_m \Omega_{e,m} \iota_m = m^{-2} \iota'_m (b \iota_m \iota'_m + D) \iota_m = m^{-2} \left(b m^2 + \sum_{i=1}^m \tau_i^{-1} \right) = b + \frac{1}{m^2} \sum_{i=1}^m \tau_i^{-1}.$$

□

A.3 Moments of the subset precision sum

Lemma 2 (Moments of A_S). *Under Assumptions 2 and 4:*

(1) $k\tau_L \leq A_S \leq k\tau_U$.

(2) For $m \geq 2$,

$$\mathbb{E}_S(A_S) = k\bar{\tau}_m, \quad \text{Var}_S(A_S) = \frac{k(m-k)}{m-1} v_{\tau,m}. \quad (43)$$

(3) For $m \geq 3$, $2 \leq k \leq m$, and every i ,

$$\mathbb{E}_S(A_S \mid i \in S) = k\bar{\tau}_m + \frac{m-k}{m-1} (\tau_i - \bar{\tau}_m), \quad (44)$$

$$\text{Var}_S(A_S \mid i \in S) = \frac{(k-1)(m-k)}{m-2} \left\{ \frac{m}{m-1} v_{\tau,m} - \frac{m}{(m-1)^2} (\tau_i - \bar{\tau}_m)^2 \right\}. \quad (45)$$

(4) The constant $C = (4^{7/4} + 4)(\tau_U - \tau_L)^3 < \infty$, which depends only on $\tau_U - \tau_L$, satisfies, uniformly in i, m, k ,

$$\mathbb{E}_S |A_S - k\bar{\tau}_m|^3 \leq Ck^{3/2}, \quad (46)$$

$$\mathbb{E}_S (|A_S - k\bar{\tau}_m|^3 \mid i \in S) \leq Ck^{3/2}. \quad (47)$$

Proof. Statement (1) holds because A_S is a sum of exactly k terms, each of which lies in $[\tau_L, \tau_U]$ by Assumption 2.

For statement (2), write $A_S = \sum_{i=1}^m \tau_i \mathbf{1}\{i \in S\}$. Under Assumption 4, $\Pr(i \in S) = k/m$ for every i , so $\mathbb{E}_S(A_S) = \frac{k}{m} \sum_i \tau_i = \frac{k}{m} \cdot m\bar{\tau}_m = k\bar{\tau}_m$.

For the variance, we use the finite-population sampling formula (Cochran, 1977, Theorem 2.2), restated here because it is applied twice with different population sizes and divisors: if a simple random sample of size n is drawn without replacement from a finite population

$\{x_1, \dots, x_N\}$ with mean $\bar{x}$ and variance $\sigma_N^2 = N^{-1} \sum_{j=1}^N (x_j - \bar{x})^2$ computed with divisor N , then

$$\text{Var}\left(\sum_{j \in \text{sample}} x_j\right) = \frac{n(N-n)}{N-1} \sigma_N^2. \quad (48)$$

Applying (48) with $(N, n) = (m, k)$ and $\sigma_N^2 = v_{\tau, m}$ (which is computed with divisor m , exactly as defined in Assumption 2) proves (43).

Conditional on $i \in S$, the remaining $k-1$ indices form a simple random sample from the other $m-1$ precisions. Their mean is as follows:

$$\bar{\tau}_{-i} = \frac{m\bar{\tau}_m - \tau_i}{m-1} = \frac{(m-1)\bar{\tau}_m + \bar{\tau}_m - \tau_i}{m-1} = \bar{\tau}_m - \frac{\tau_i - \bar{\tau}_m}{m-1}$$

Hence:

$$\mathbb{E}_S(A_S \mid i \in S) = \tau_i + (k-1) \left(\bar{\tau}_m - \frac{\tau_i - \bar{\tau}_m}{m-1} \right) = k\bar{\tau}_m + (\tau_i - \bar{\tau}_m) \left(\frac{m-k}{m-1} \right),$$

which simplifies to (44).

The variance of the remaining population, with divisor $m-1$, equals:

$$\begin{aligned} \frac{1}{m-1} \sum_{j \neq i} \left(\tau_j - \bar{\tau}_m + \frac{\tau_i - \bar{\tau}_m}{m-1} \right)^2 &= \frac{1}{m-1} \left\{ mv_{\tau, m} - (\tau_i - \bar{\tau}_m)^2 - \frac{(\tau_i - \bar{\tau}_m)^2}{m-1} \right\} \\ &= \frac{mv_{\tau, m}}{m-1} - \frac{m(\tau_i - \bar{\tau}_m)^2}{(m-1)^2}. \end{aligned}$$

Applying (48) with population size $N = m-1$, sample size $n = k-1$, and σ_N^2 equal to the divisor- $(m-1)$ variance just computed proves (45).

For the third moments, put $d_i = \tau_i - \bar{\tau}_m$. Then $\sum_i d_i = 0$, $|d_i| \leq \tau_U - \tau_L$ (because both τ_i and $\bar{\tau}_m$ lie in $[\tau_L, \tau_U]$), and $A_S - k\bar{\tau}_m = \sum_{i \in S} d_i$. We use Hoeffding's comparison theorem for sampling without replacement (Hoeffding, 1963, Theorem 4); see also Bardenet and Maillard (2015) for a review. It states: if $X_1, \dots, X_k$ are drawn without replacement and $Y_1, \dots, Y_k$ are drawn with replacement from the same finite population, then $\mathbb{E}f(\sum_j X_j) \leq \mathbb{E}f(\sum_j Y_j)$ for every continuous convex function f . Applied to the population $\{d_1, \dots, d_m\}$ with the continuous convex function $f(x) = |x|^3$, it bounds the third absolute moment of $\sum_{i \in S} d_i$ by that of $\sum_{j=1}^k d_{J_j}$, where the J_j are i.i.d. uniform indices on $\{1, \dots, m\}$. By Lyapunov's inequality ($\|X\|_3 \leq \|X\|_4$),

$$\mathbb{E} \left| \sum_{j=1}^k d_{J_j} \right|^3 \leq \left\{ \mathbb{E} \left(\sum_{j=1}^k d_{J_j} \right)^4 \right\}^{3/4}.$$

To compute the fourth moment, expand $(\sum_{j=1}^k d_{J_j})^4$ into the multinomial sum over index quadruples (j_1, j_2, j_3, j_4) . Because the d_{J_j} are i.i.d. with $\mathbb{E}(d_{J_1}) = m^{-1} \sum_i d_i = 0$, every term in

which some index appears exactly once factorizes with a factor $\mathbb{E}(d_{J_1}) = 0$ and vanishes; this includes, in particular, the $\mathbb{E}(d^3)\mathbb{E}(d)$ cross terms. Only the k all-equal quadruples and the $3k(k-1)$ quadruples forming two distinct pairs survive, giving

$$\begin{aligned}\mathbb{E}\left(\sum_{j=1}^k d_{J_j}\right)^4 &= k \mathbb{E}(d_{J_1}^4) + 3k(k-1)\{\mathbb{E}(d_{J_1}^2)\}^2 = \frac{k}{m} \sum_{i=1}^m (\tau_i - \bar{\tau}_m)^4 + 3k(k-1)v_{\tau,m}^2 \\ &\leq k(\tau_U - \tau_L)^4 + 3k(k-1)(\tau_U - \tau_L)^4 \\ &\leq 4(\tau_U - \tau_L)^4 k^2,\end{aligned}$$

where the last step uses $k + 3k(k-1) = 3k^2 - 2k \leq 4k^2$. This proves (46) with the constant $4^{3/4}(\tau_U - \tau_L)^3$, which is smaller than the stated $C = (4^{7/4} + 4)(\tau_U - \tau_L)^3$.

For (47), write $\mu_i = \mathbb{E}_S(A_S \mid i \in S)$ and $\bar{\tau}_{-i} = (m\bar{\tau}_m - \tau_i)/(m-1)$. Recall that $S \setminus \{i\}$ consists of $k-1$ draws without replacement from $\{1, \dots, m\} \setminus \{i\}$, a population with mean $\bar{\tau}_{-i}$; hence, writing $d_j^{(i)} = \tau_j - \bar{\tau}_{-i}$ for $j \neq i$,

$$A_S - \mu_i = \sum_{j \in S \setminus \{i\}} d_j^{(i)}, \quad \frac{1}{m-1} \sum_{j \neq i} d_j^{(i)} = 0, \quad |d_j^{(i)}| \leq \tau_U - \tau_L.$$

The unconditional argument above — Hoeffding's comparison, Lyapunov, and the fourth-moment computation, whose hypotheses (a centered finite population with entries bounded by $\tau_U - \tau_L$) have just been checked for $\{d_j^{(i)}\}_{j \neq i}$ — therefore applies with (m, k) replaced by $(m-1, k-1)$, which gives

$$\mathbb{E}_S(|A_S - \mu_i|^3 \mid i \in S) \leq 4^{3/4}(\tau_U - \tau_L)^3(k-1)^{3/2} \leq 4^{3/4}(\tau_U - \tau_L)^3 k^{3/2}.$$

Notice also

$$|\mu_i - k\bar{\tau}_m| = \frac{m-k}{m-1} |\tau_i - \bar{\tau}_m| \leq \tau_U - \tau_L.$$

By convexity of $x \mapsto x^3$ on $[0, \infty)$, $(|a| + |b|)^3 \leq 4(|a|^3 + |b|^3)$ for all real a, b ; applying this with $a = A_S - \mu_i$ and $b = \mu_i - k\bar{\tau}_m$ gives $|A_S - k\bar{\tau}_m|^3 \leq 4(|A_S - \mu_i|^3 + |\mu_i - k\bar{\tau}_m|^3)$. Combining this with the two previous displays, and using $(\tau_U - \tau_L)^3 \leq (\tau_U - \tau_L)^3 k^{3/2}$ since $k \geq 1$, yields the second bound:

$$\begin{aligned}\mathbb{E}_S(|A_S - k\bar{\tau}_m|^3 \mid i \in S) &\leq 4\{4^{3/4}(\tau_U - \tau_L)^3 k^{3/2} + (\tau_U - \tau_L)^3\} \\ &\leq (4^{7/4} + 4)(\tau_U - \tau_L)^3 k^{3/2}.\end{aligned}$$

□

A.4 Reciprocal identities

Lemma 3 (Exact reciprocal expansions). *For $a > 0$, $\mu > 0$,*

$$\frac{1}{a} = \frac{1}{\mu} - \frac{a - \mu}{\mu^2} + \frac{(a - \mu)^2}{a\mu^2}, \quad (49)$$

$$\frac{1}{a} = \frac{1}{\mu} - \frac{a - \mu}{\mu^2} + \frac{(a - \mu)^2}{\mu^3} - \frac{(a - \mu)^3}{a\mu^3}. \quad (50)$$

Proof. Multiplying either right-hand side by $a\mu^2$ in (49), or by $a\mu^3$ in (50), reduces both identities to the corresponding binomial expansion of $a = \mu + (a - \mu)$. Division by the positive denominators proves the claims. $\square$

B Proof of the Finite Bagging Bridge

Proof of Proposition 1. Write $V_g = v_{S_g}$ and $\bar{v}_k = \mathbb{E}_S(V_g)$. Expanding the quadratic form in (12) gives

$$Q_k^{\text{bag}, G} = \frac{1}{G^2} \sum_{g=1}^G \sum_{r=1}^G \mathbb{E}(V_g' \Omega_{e,m} V_r).$$

There are G diagonal terms, each equal to Q_k^{sub} . For $g \neq r$, independence of the two subset draws implies

$$\mathbb{E}(V_g' \Omega_{e,m} V_r) = (\mathbb{E}V_g)' \Omega_{e,m} (\mathbb{E}V_r) = \bar{v}_k' \Omega_{e,m} \bar{v}_k = Q_k^{\text{bag}, \infty}.$$

There are $G(G - 1)$ ordered off-diagonal pairs. Consequently,

$$\begin{aligned} Q_k^{\text{bag}, G} &= \frac{G}{G^2} Q_k^{\text{sub}} + \frac{G(G - 1)}{G^2} Q_k^{\text{bag}, \infty} \\ &= Q_k^{\text{bag}, \infty} + \frac{1}{G} (Q_k^{\text{sub}} - Q_k^{\text{bag}, \infty}), \end{aligned}$$

which is (14). The inequalities follow because $Q_k^{\text{sub}} - Q_k^{\text{bag}, \infty} = \mathbb{E}_S[(v_S - \bar{v}_k)' \Omega_{e,m} (v_S - \bar{v}_k)] \geq 0$. $\square$

C Proof of the Average Subset Expansion

Proof of Proposition 4. Equation (16) gives the exact identity $Q_k^{\text{sub}} = b + \mathbb{E}_S(A_S^{-1})$. Apply (50) with $a = A_S$ and $\mu = k\bar{\tau}_m$, then take $\mathbb{E}_S$:

$$\mathbb{E}_S(A_S^{-1}) = \frac{1}{k\bar{\tau}_m} + \frac{\text{Var}_S(A_S)}{k^3\bar{\tau}_m^3} - \frac{1}{k^3\bar{\tau}_m^3} \mathbb{E}_S \left\{ \frac{(A_S - k\bar{\tau}_m)^3}{A_S} \right\}. \quad (51)$$

The linear term vanishes by (43). Substituting the variance from the same equation yields

$$\frac{\text{Var}_S(A_S)}{k^3\bar{\tau}_m^3} = \frac{m - k}{k^2(m - 1)} \frac{v_{\tau, m}}{\bar{\tau}_m^3}.$$

For the remainder, Lemma 2(1) and (46) give

$$\left| \mathbb{E}_S \left\{ \frac{(A_S - k\bar{\tau}_m)^3}{A_S} \right\} \right| \leq \frac{Ck^{3/2}}{k\tau_L} = O(k^{1/2}).$$

After division by $k^3\bar{\tau}_m^3$, this is $O(k^{-5/2})$ uniformly because $\bar{\tau}_m \geq \tau_L$. Substitution into (51) proves (20). The middle term is $O(k^{-2})$ under the uniform precision bounds, which gives (21). $\square$

D Proof of the Infinite-Bagged Risk Representation

Proof of Proposition 5. Put $\delta = \bar{v}_k - w^{\text{opt}}$. Both vectors add to one, hence $\iota'_m \delta = 0$. The first-order condition for the full-panel optimum implies $\Omega_{e,m} w^{\text{opt}} = Q^{\text{opt}} \iota_m$, so

$$(w^{\text{opt}})' \Omega_{e,m} \delta = Q^{\text{opt}} \iota'_m \delta = 0.$$

Expanding around w^{opt} and applying (37),

$$\begin{aligned} Q_k^{\text{bag},\infty} - Q^{\text{opt}} &= 2(w^{\text{opt}})' \Omega_{e,m} \delta + \delta' \Omega_{e,m} \delta \\ &= \delta' D \delta = \sum_{i=1}^m \frac{(\bar{v}_{i,k} - w_i^{\text{opt}})^2}{\tau_i}. \end{aligned}$$

This proves (24).

Next, (16) and the componentwise form of the embedding $v_S = P_S w_S$ established in the proof of Proposition 3 give $v_{S,i} = \mathbf{1}\{i \in S\} \tau_i / A_S$. Since $\Pr(i \in S) = k/m$,

$$\bar{v}_{i,k} = \mathbb{E}_S(v_{S,i}) = \frac{k}{m} \tau_i \mathbb{E}_S(A_S^{-1} \mid i \in S) = \frac{\tau_i}{m} g_i(k).$$

Using $w_i^{\text{opt}} = \tau_i / (m\bar{\tau}_m)$ in the preceding squared distance gives the score representation in (24).

Finally,

$$\sum_{i=1}^m \tau_i g_i(k) = m \sum_{i=1}^m \bar{v}_{i,k} = m,$$

because every v_S , and hence $\bar{v}_k$, adds to one. This completes (23). $\square$

D.1 The endpoint identities

Proof of Proposition 2. Both identities follow from the representation $\bar{v}_{i,k} = \tau_i g_i(k) / m$ in (23), which was just established and whose proof does not use the present proposition.

Take $k = 1$. Conditional on $i \in S$ the subset of size one is necessarily $S = \{i\}$, so $A_S = \tau_i$ with probability one and

$$g_i(1) = 1 \cdot \mathbb{E}_S(A_S^{-1} \mid i \in S) = \frac{1}{\tau_i}, \quad \text{hence} \quad \bar{v}_{i,1} = \frac{\tau_i}{m} \cdot \frac{1}{\tau_i} = \frac{1}{m}$$

for every i . Therefore $\bar{v}_1 = m^{-1}\iota_m = w^{\text{ave}}$ by (7).

Take $k = m$. There is exactly one subset of size m , namely $S = \{1, \dots, m\}$, so under Assumption 4 it is drawn with probability one and $\mathbb{E}_S$ is the identity operator. Then $A_S = A_m = m\bar{\tau}_m$ deterministically, so

$$g_i(m) = m \cdot \frac{1}{m\bar{\tau}_m} = \frac{1}{\bar{\tau}_m}, \quad \text{hence} \quad \bar{v}_{i,m} = \frac{\tau_i}{m} \cdot \frac{1}{\bar{\tau}_m} = \frac{\tau_i}{m\bar{\tau}_m} = w_i^{\text{opt}},$$

the last equality by (17) together with $A_m = m\bar{\tau}_m$. Therefore $\bar{v}_m = w^{\text{opt}}$. □

E Estimation-Error Inflation for Shared-Sample Infinite Bagging

This section proves the two main results: the estimation-error coefficient of the literal infinite-bag estimator, whose subset regressions all share one training sample, and the resulting fourth-root rule for k . The route is: (a) pin down how fast the *population* approximation error R_k^{bag} vanishes (E.1–E.2); (b) linearize the estimated subset weights exactly (E.3); (c) compute the variance of the average of those linearizations, which is where the shared sample enters (E.4); (d) control the nonlinear remainder (E.6); and (e) balance the approximation and estimation terms to obtain k^* (E.8).

Throughout, $\mathcal{D}_m$ is the precision-profile constant

$$\mathcal{D}_m = \frac{1}{\bar{\tau}_m^6} \cdot \frac{1}{m} \sum_{i=1}^m \tau_i \{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)\}^2 \longrightarrow \mathcal{D} > 0. \quad (52)$$

E.1 Local form of the population approximation error

Proposition 10 (Local form of R_k^{bag}). *Under Assumptions 2 and 4, as $k \rightarrow \infty$ with $k/m \rightarrow 0$,*

$$g_i(k) - \frac{1}{\bar{\tau}_m} = z_k \left\{ \frac{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)}{\bar{\tau}_m^3} + o(1) \right\} \quad \text{uniformly in } i, \quad (53)$$

and consequently

$$R_k^{bag} = \frac{\mathcal{D}_m}{m} z_k^2 \{1 + o(1)\}, \quad \mathcal{D}_m \geq \frac{\tau_L}{\bar{\tau}_m^6} (v_{\tau,m}^2 + \bar{\tau}_m^2 v_{\tau,m}), \quad (54)$$

and the lower bound is strictly positive for all sufficiently large m because $v_{\tau,m} \rightarrow v_\tau > 0$ under Assumption 2.

Proof. Apply the exact reciprocal identity (50) with $a = A_S$ and $\mu = k\bar{\tau}_m$, multiply both sides by k , and then take $\mathbb{E}_S(\cdot \mid i \in S)$. Since $g_i(k) = k\mathbb{E}_S(A_S^{-1} \mid i \in S)$, this gives:

$$\begin{aligned} g_i(k) - \frac{1}{\bar{\tau}_m} &= -\frac{\mathbb{E}_S(A_S - k\bar{\tau}_m \mid i \in S)}{k\bar{\tau}_m^2} + \frac{\mathbb{E}_S\{(A_S - k\bar{\tau}_m)^2 \mid i \in S\}}{k^2\bar{\tau}_m^3} \\ &\quad - \frac{\mathbb{E}_S\{(A_S - k\bar{\tau}_m)^3/A_S \mid i \in S\}}{k^2\bar{\tau}_m^3}. \end{aligned} \quad (55)$$

We treat the three terms in turn. It is convenient to note once that $z_k = (m - k)/(km)$, so that $z_k \sim 1/k$ whenever $k/m \rightarrow 0$.

The linear term. By Lemma 2(3),

$$\begin{aligned} -\frac{\mathbb{E}_S(A_S - k\bar{\tau}_m \mid i \in S)}{k\bar{\tau}_m^2} &= -\frac{(m - k)(\tau_i - \bar{\tau}_m)}{(m - 1)k\bar{\tau}_m^2} = z_k \left\{ -\frac{m}{m - 1} \cdot \frac{\tau_i - \bar{\tau}_m}{\bar{\tau}_m^2} \right\} \\ &= z_k \left\{ -\frac{\tau_i - \bar{\tau}_m}{\bar{\tau}_m^2} + O(m^{-1}) \right\}, \end{aligned}$$

uniformly in i , using $(m-k)/\{(m-1)k\} = z_k m/(m-1)$ and $m/(m-1) = 1 + O(m^{-1})$ together with the uniform precision bounds.

The quadratic term. Write $\mathbb{E}_S\{(A_S - k\bar{\tau}_m)^2 \mid i \in S\} = \text{Var}_S(A_S \mid i \in S) + \{\mathbb{E}_S(A_S - k\bar{\tau}_m \mid i \in S)\}^2$ and divide each piece by $k^2\bar{\tau}_m^3$, factoring z_k out of both. Using Lemma 2(3),

$$\begin{aligned} \frac{\text{Var}_S(A_S \mid i \in S)}{k^2\bar{\tau}_m^3} &= z_k \left\{ \frac{m(k-1)}{k(m-2)} \cdot \frac{\frac{m}{m-1}v_{\tau,m} - \frac{m}{(m-1)^2}(\tau_i - \bar{\tau}_m)^2}{\bar{\tau}_m^3} \right\}, \\ \frac{\{\mathbb{E}_S(A_S - k\bar{\tau}_m \mid i \in S)\}^2}{k^2\bar{\tau}_m^3} &= z_k \left\{ \frac{m(m-k)}{k(m-1)^2} \cdot \frac{(\tau_i - \bar{\tau}_m)^2}{\bar{\tau}_m^3} \right\}. \end{aligned}$$

In the first line $m(k-1)/(k(m-2)) = (1 - \frac{1}{k})(1 + \frac{2}{m-2}) = 1 + O(k^{-1} + m^{-1})$, and the numerator inside the braces is $v_{\tau,m} + O(m^{-1})$ uniformly in i because $\frac{m}{m-1}v_{\tau,m} = v_{\tau,m} + \frac{v_{\tau,m}}{m-1}$ with $v_{\tau,m} \leq (\tau_U - \tau_L)^2$ bounded, and the subtracted term is $O(m^{-1})$ since $\frac{m}{(m-1)^2} = O(m^{-1})$ and $(\tau_i - \bar{\tau}_m)^2 \leq (\tau_U - \tau_L)^2$. Combining the prefactor with the numerator yields:

$$\begin{aligned} \frac{\text{Var}_S(A_S \mid i \in S)}{k^2\bar{\tau}_m^3} &= z_k \{1 + O(k^{-1} + m^{-1})\} \frac{v_{\tau,m} + O(m^{-1})}{\bar{\tau}_m^3} \\ &= z_k \left\{ \frac{v_{\tau,m}}{\bar{\tau}_m^3} + O(k^{-1} + m^{-1}) \right\}. \end{aligned}$$

In the second line $m(m-k)/\{k(m-1)^2\} = O(k^{-1})$, so that whole term $\frac{\{\mathbb{E}_S(A_S - k\bar{\tau}_m \mid i \in S)\}^2}{k^2\bar{\tau}_m^3}$ is absorbed into the $O(k^{-1} + m^{-1})$. Hence

$$\begin{aligned} \frac{\mathbb{E}_S\{(A_S - k\bar{\tau}_m)^2 \mid i \in S\}}{k^2\bar{\tau}_m^3} &= z_k \left\{ \frac{v_{\tau,m}}{\bar{\tau}_m^3} + O(k^{-1} + m^{-1}) \right\} \\ &= z_k \left\{ \frac{v_{\tau,m}}{\bar{\tau}_m^3} + o(1) \right\}, \quad \text{uniformly in } i. \end{aligned}$$

The cubic remainder. Recall $A_S \geq k\tau_L$ and apply Lemma 2(4):

$$\left| \frac{\mathbb{E}_S\{(A_S - k\bar{\tau}_m)^3 \mid i \in S\}}{k^2\bar{\tau}_m^3} \right| \leq \frac{\mathbb{E}_S(|A_S - k\bar{\tau}_m|^3 \mid i \in S)}{k^3\tau_L\bar{\tau}_m^3} \leq \frac{Ck^{3/2}}{k^3\tau_L\bar{\tau}_m^3} = O(k^{-3/2}),$$

uniformly in i . Since $k/m \rightarrow 0$, we have $z_k = (1 - k/m)/k \geq 1/(2k)$ for all sufficiently large (k, m) , so $k^{-3/2} \leq 2z_k k^{-1/2}$; the cubic term is therefore $z_k \cdot O(k^{-1/2}) = z_k \cdot o(1)$, uniformly in i .

After adding up all the terms and using $(\tau_i - \bar{\tau}_m)/\bar{\tau}_m^2 = \bar{\tau}_m(\tau_i - \bar{\tau}_m)/\bar{\tau}_m^3$, we obtain:

$$g_i(k) - \frac{1}{\bar{\tau}_m} = z_k \left\{ -\frac{\tau_i - \bar{\tau}_m}{\bar{\tau}_m^2} + \frac{v_{\tau,m}}{\bar{\tau}_m^3} + O(m^{-1}) + o(1) + o(1) \right\} = z_k \left\{ \frac{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)}{\bar{\tau}_m^3} + o(1) \right\}$$

which gives Equation (53). Substitute it into the score form of R_k^{bag} in (24):

$$R_k^{bag} = \frac{z_k^2}{m} \cdot \frac{1}{m} \sum_{i=1}^m \tau_i \left\{ \frac{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)}{\bar{\tau}_m^3} + o(1) \right\}^2.$$

To expand the square, write $g_i(k) - 1/\bar{\tau}_m = z_k \{F_i + \varepsilon_i\}$ with $F_i = \{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)\}/\bar{\tau}_m^3$ and $\max_i |\varepsilon_i| = o(1)$, so that

$$R_k^{bag} = \frac{z_k^2}{m} \left\{ \mathcal{D}_m + \frac{2}{m} \sum_{i=1}^m \tau_i F_i \varepsilon_i + \frac{1}{m} \sum_{i=1}^m \tau_i \varepsilon_i^2 \right\},$$

where the leading piece is $\mathcal{D}_m$ by (52). The factor F_i is bounded uniformly: $|F_i| \leq \{v_{\tau,m} + \tau_U(\tau_U - \tau_L)\}/\tau_L^3$ by the precision bounds. Hence

$$\left| \frac{2}{m} \sum_{i=1}^m \tau_i F_i \varepsilon_i \right| \leq 2\tau_U \left(\max_i |F_i| \right) \left(\max_i |\varepsilon_i| \right) = o(1), \quad \frac{1}{m} \sum_{i=1}^m \tau_i \varepsilon_i^2 \leq \tau_U \max_i \varepsilon_i^2 = o(1),$$

and since $\mathcal{D}_m$ is bounded away from zero for large m (shown below), the braced factor is $\mathcal{D}_m \{1 + o(1)\}$, which gives (54).

It remains to check that $\mathcal{D}_m$ stays away from zero. Bounding $\tau_i \geq \tau_L$ and computing the unweighted average,

$$\frac{1}{m} \sum_{i=1}^m \{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)\}^2 = v_{\tau,m}^2 - 2\bar{\tau}_m v_{\tau,m} \cdot \frac{1}{m} \sum_i (\tau_i - \bar{\tau}_m) + \bar{\tau}_m^2 \cdot \frac{1}{m} \sum_i (\tau_i - \bar{\tau}_m)^2 = v_{\tau,m}^2 + \bar{\tau}_m^2 v_{\tau,m},$$

since $\sum_i (\tau_i - \bar{\tau}_m) = 0$. Dividing by $\bar{\tau}_m^6$ gives the stated lower bound, which is bounded away from zero because $v_{\tau,m} \rightarrow v_\tau > 0$. $\square$

E.2 A global lower bound on the approximation error

Proposition 10 describes R_k^{bag} on the interior range $k \rightarrow \infty$, $k/m \rightarrow 0$. To prove that the optimizer actually *lies* in that range we also need a bound that holds for every k , with no asymptotics. That is the only purpose of the following lemma.

Lemma 4 (Small- k localization). *Under Assumptions 2 and 4, for all sufficiently large m and every $1 \leq k < m$,*

$$R_k^{bag} \geq \frac{\tau_L v_\tau^2}{4\tau_U^6 m} z_k^2. \quad (56)$$

Proof. Step 1: a lower bound on the excess conditional harmonic mean. Apply the exact reciprocal identity (49) with $a = A_S$ and $\mu = k\bar{\tau}_m$:

$$\frac{1}{A_S} = \frac{1}{k\bar{\tau}_m} - \frac{A_S - k\bar{\tau}_m}{(k\bar{\tau}_m)^2} + \frac{(A_S - k\bar{\tau}_m)^2}{A_S(k\bar{\tau}_m)^2}.$$

Multiply by k and take $\mathbb{E}_S$. Since $\mathbb{E}_S(A_S) = k\bar{\tau}_m$ by Lemma 2(2), the linear term has expectation zero, leaving the exact identity

$$k \mathbb{E}_S(A_S^{-1}) - \frac{1}{\bar{\tau}_m} = \mathbb{E}_S \left\{ \frac{(A_S - k\bar{\tau}_m)^2}{A_S k \bar{\tau}_m^2} \right\}.$$

Using $A_S \leq k\tau_U$ and $\text{Var}_S(A_S) = k(m-k)v_{\tau,m}/(m-1)$ from Lemma 2(2),

$$k \mathbb{E}_S(A_S^{-1}) - \frac{1}{\bar{\tau}_m} \geq \frac{\text{Var}_S(A_S)}{k^2 \tau_U \bar{\tau}_m^2} = \frac{m}{m-1} \cdot \frac{v_{\tau,m}}{\tau_U \bar{\tau}_m^2} z_k \geq \frac{v_\tau}{2\tau_U^3} z_k,$$

the last step using $m/(m-1) \geq 1$, $\bar{\tau}_m \leq \tau_U$, and $v_{\tau,m} \geq v_\tau/2$ for all sufficiently large m .

Step 2: an aggregation identity for the scores. Additionally, because $\Pr(i \in S) = k/m$,

$$\sum_{i=1}^m g_i(k) = k \sum_{i=1}^m \mathbb{E}_S(A_S^{-1} \mid i \in S) = m \mathbb{E}_S\left(\frac{\sum_i \mathbf{1}\{i \in S\}}{A_S}\right) = mk \mathbb{E}_S(A_S^{-1}),$$

so that

$$\sum_{i=1}^m \left(g_i(k) - \frac{1}{\bar{\tau}_m}\right) = m \left\{k \mathbb{E}_S(A_S^{-1}) - \frac{1}{\bar{\tau}_m}\right\}.$$

Step 3: Cauchy–Schwarz. Now split each summand as $\sqrt{\tau_i}(g_i(k) - 1/\bar{\tau}_m) \cdot \tau_i^{-1/2}$ and apply Cauchy–Schwarz, then use the score form (24) and $\sum_i \tau_i^{-1} \leq m/\tau_L$:

$$m^2 \left\{k \mathbb{E}_S(A_S^{-1}) - \frac{1}{\bar{\tau}_m}\right\}^2 \leq \left\{\sum_i \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}_m}\right)^2\right\} \left(\sum_i \frac{1}{\tau_i}\right) \leq m^2 R_k^{\text{bag}} \cdot \frac{m}{\tau_L}.$$

Rearranging gives $R_k^{\text{bag}} \geq \tau_L \{k \mathbb{E}_S(A_S^{-1}) - 1/\bar{\tau}_m\}^2/m$, and inserting the lower bound of Step 1 gives (56). $\square$

E.3 Exact linearization of the estimated subset weights

We now turn to the feasible estimator. Under Assumptions 2 and 4,

$$e_t = h_t \iota_m + \xi_t, \quad C_m = \mathbb{E}(e_t e_t') = b \iota_m \iota_m' + D,$$

so $C_S = P_S' C_m P_S = C_S(b)$ in the notation of Lemma 1. The feasible subset weights minimize the uncentered sum of squared realized combination errors, without an intercept and with the zero mean treated as known:

$$\hat{w}_S = \arg \min_{\iota_k' w = 1} \sum_{t=1}^T (w' P_S' e_t)^2 = \frac{\hat{C}_S^{-1} \iota_k}{\iota_k' \hat{C}_S^{-1} \iota_k}, \quad \hat{C}_S = \frac{1}{T} \sum_{t=1}^T P_S' e_t e_t' P_S,$$

and $\hat{v}_S = P_S \hat{w}_S$, $\hat{v}_k = \mathbb{E}_S(\hat{v}_S)$. Under Assumption 3 and $T \geq k$ the minimizer is almost surely well defined. Indeed, the criterion $\sum_t (w' P_S' e_t)^2 = w'(T \hat{C}_S)w$ is minimized over the affine set $\{\iota_k' w = 1\}$, and by Lemma 7(ii)–(iii) below — whose proof of these two parts uses only Lemma 1 and Assumption 3, not the present subsection — the criterion equals $\sum_t (\eta_t - \beta' x_t)^2$ in contrast coordinates, with Gram matrix $X'X \sim W_{k-1}(I_{k-1}, T)$. A Wishart matrix with $T \geq k-1$ degrees of freedom and nonsingular scale is almost surely nonsingular, so the criterion is almost surely strictly convex in β and $\hat{w}_S$ exists and is unique. All statements below hold on this

almost-sure event. By (40) of Lemma 1 applied with $a = b$, the population counterpart of $\widehat{w}_S$ is exactly w_S : the feasible regression targets the population subset weights, with no bias from the common component. Define

$$K_S = P_S \left[D_S^{-1} - \frac{D_S^{-1} \iota_k \iota_k' D_S^{-1}}{A_S} \right] P_S'. \quad (57)$$

By (41) of Lemma 1, K_S is the embedded “contrast inverse” of C_S , and it does not depend on b .

Lemma 5 (Properties of K_S). *K_S is symmetric and*

$$K_S \iota_m = 0, \quad K_S D K_S = K_S, \quad K_S D v_S = 0, \quad \text{tr}(D K_S) = k - 1. \quad (58)$$

Moreover $D^{1/2} K_S D^{1/2}$ is an orthogonal projection of rank $k - 1$.

Proof. Write M_S for the bracket in (57), so $K_S = P_S M_S P_S'$. The matrix M_S is symmetric: D_S^{-1} is diagonal, hence symmetric, and setting $x = D_S^{-1} \iota_k$ the second term is xx'/A_S with $(xx')' = xx'$, so

$$M_S' = (D_S^{-1})' - \frac{(D_S^{-1} \iota_k \iota_k' D_S^{-1})'}{A_S} = D_S^{-1} - \frac{D_S^{-1} \iota_k \iota_k' D_S^{-1}}{A_S} = M_S.$$

Consequently $K_S' = (P_S M_S P_S')' = P_S M_S' P_S' = K_S$, so K_S is symmetric as well. Since $\iota_k' D_S^{-1} \iota_k = A_S$,

$$M_S \iota_k = D_S^{-1} \iota_k - D_S^{-1} \iota_k \frac{A_S}{A_S} = 0,$$

and $K_S \iota_m = P_S M_S P_S' \iota_m = P_S M_S \iota_k = 0$, using $P_S' \iota_m = \iota_k$.

Because D is diagonal and P_S selects coordinates, $D^{1/2} P_S = P_S D_S^{1/2}$, so

$$D^{1/2} K_S D^{1/2} = P_S \left[I_k - \frac{(D_S^{-1/2} \iota_k)(D_S^{-1/2} \iota_k)'}{A_S} \right] P_S'.$$

As $\|D_S^{-1/2} \iota_k\|^2 = \iota_k' D_S^{-1} \iota_k = A_S$, the bracket is $\Pi := I_k - uu'/\|u\|^2$ with $u = D_S^{-1/2} \iota_k$, the orthogonal projection of $\mathbb{R}^k$ onto the orthogonal complement of u ; it is therefore symmetric idempotent of rank $k - 1$. Its embedding inherits all three properties: using $P_S' P_S = I_k$,

$$(P_S \Pi P_S')' = P_S \Pi' P_S' = P_S \Pi P_S', \quad (P_S \Pi P_S')^2 = P_S \Pi (P_S' P_S) \Pi P_S' = P_S \Pi^2 P_S' = P_S \Pi P_S',$$

and $\text{rank}(P_S \Pi P_S') = \text{rank}(\Pi) = k - 1$ because P_S has orthonormal columns. Idempotence $(D^{1/2} K_S D^{1/2})^2 = D^{1/2} K_S D^{1/2}$, written out as $D^{1/2} K_S D K_S D^{1/2} = D^{1/2} K_S D^{1/2}$, becomes $K_S D K_S = K_S$ after pre- and post-multiplying by the invertible $D^{-1/2}$; and the trace of a projection is its rank, giving $\text{tr}(D K_S) = \text{tr}(D^{1/2} K_S D^{1/2}) = k - 1$.

Finally, $(D v_S)_i = \tau_i^{-1} \tau_i / A_S = 1/A_S$ for $i \in S$ and 0 otherwise, so $P_S' D v_S = \iota_k / A_S$ and $K_S D v_S = P_S M_S \iota_k / A_S = 0$. □

Proposition 11 (Exact linearization). *Suppose Assumptions 2, 3, and 4 hold and $T \geq k$, so that $\widehat{w}_S$ is almost surely well defined. Let $\widehat{\Delta} = T^{-1} \sum_{t=1}^T (e_t e_t' - C_m)$. Then, almost surely, exactly and with no approximation,*

$$\widehat{v}_S - v_S = -K_S \widehat{\Delta} \widehat{v}_S, \quad (59)$$

and therefore

$$\widehat{v}_S - v_S = \frac{1}{T} \sum_{t=1}^T \phi_S(t) + \rho_S, \quad \phi_S(t) = -K_S \xi_t \{h_t + v_S' \xi_t\}, \quad (60)$$

with $\rho_S = -K_S \widehat{\Delta} (\widehat{v}_S - v_S)$ a product of two estimation errors. Moreover $\mathbb{E}\{\phi_S(t)\} = 0$.

The identity (59) supplies the first-order decomposition algebraically, without differentiating the map $C \mapsto C^{-1} \iota / (\iota' C^{-1} \iota)$. More importantly, it exhibits the remainder as the explicit product $-K_S \widehat{\Delta} (\widehat{v}_S - v_S)$, which can be bounded directly.

Proof. The two constrained minimisation problems have first-order conditions $\widehat{C}_S \widehat{w}_S = \widehat{\mu} \iota_k$ and $C_S w_S = \mu \iota_k$ for scalar multipliers $\widehat{\mu}, \mu$. Writing $\widehat{C}_S = C_S + P_S' \widehat{\Delta} P_S$ and subtracting,

$$C_S (\widehat{w}_S - w_S) + P_S' \widehat{\Delta} P_S \widehat{w}_S = (\widehat{\mu} - \mu) \iota_k.$$

Premultiply by M_S . By (41) of Lemma 1, $M_S = C_S^{-1} - C_S^{-1} \iota_k \iota_k' C_S^{-1} / (\iota_k' C_S^{-1} \iota_k)$, from which two facts follow: $M_S \iota_k = 0$, which kills the right-hand side; and $M_S C_S x = x$ for every x with $\iota_k' x = 0$, since $M_S C_S x = x - C_S^{-1} \iota_k (\iota_k' x) / (\iota_k' C_S^{-1} \iota_k)$. As both $\widehat{w}_S$ and w_S sum to one, $x = \widehat{w}_S - w_S$ qualifies, and

$$\widehat{w}_S - w_S + M_S P_S' \widehat{\Delta} P_S \widehat{w}_S = 0.$$

Premultiplying by P_S and using $K_S = P_S M_S P_S'$ together with $P_S P_S' \widehat{v}_S = \widehat{v}_S$ (as $\widehat{v}_S$ is supported on S) gives (59).

Now write $\widehat{v}_S = v_S + (\widehat{v}_S - v_S)$ on the right of (59). The first piece is $-K_S \widehat{\Delta} v_S = T^{-1} \sum_t \{-K_S (e_t e_t' - C_m) v_S\}$, and the second is ρ_S . To identify the summand, note that $K_S \iota_m = 0$ gives $K_S e_t = K_S \xi_t$; that $\iota_m' v_S = 1$ gives $e_t' v_S = h_t + v_S' \xi_t$; and that

$$K_S C_m v_S = b K_S \iota_m (\iota_m' v_S) + K_S D v_S = 0$$

by (58). Hence $-K_S (e_t e_t' - C_m) v_S = -K_S e_t (e_t' v_S) = \phi_S(t)$, which is (60).

Finally $\mathbb{E}\{\phi_S(t)\} = -K_S \mathbb{E}(\xi_t h_t) - K_S \mathbb{E}(\xi_t \xi_t') v_S = 0 - K_S D v_S = 0$, using $\mathbb{E}(\xi_t h_t) = \text{Cov}(h_t, \xi_t) = 0$ from Assumption 2 and (58). $\square$

E.4 The cross-subset coefficient

Infinite bagging averages $\widehat{v}_S$ over *all* subsets, fitted on *one* training sample. Before stating the result, we derive the object it computes, rather than positing it.

Let R be an independent copy of S , and write $\mathbb{E}_R$ for expectation over this second subset draw. Both subset draws are independent of the training sample.

Averaging the exact linearization of Proposition 11 over subsets gives

$$\widehat{v}_k - \bar{v}_k = \frac{1}{T} \sum_{t=1}^T \mathbb{E}_S \phi_S(t) + \mathbb{E}_S(\rho_S).$$

The next step needs a rule converting this error *vector* into a contribution to *risk*. Because the loss is quadratic, writing $\widehat{v}_k = \bar{v}_k + \delta$ in $L_{T,m}^{\text{bag},\infty}(k) = \mathbb{E}(\widehat{v}_k' C_m \widehat{v}_k)$ gives the exact three-term identity

$$\mathbb{E}(\widehat{v}_k' C_m \widehat{v}_k) = \underbrace{\bar{v}_k' C_m \bar{v}_k}_{Q_k^{\text{bag},\infty}} + \underbrace{2 [\mathbb{E}(\delta)]' C_m \bar{v}_k}_{=0 \text{ since } \mathbb{E}(\widehat{v}_S) = \bar{v}_S} + \underbrace{\mathbb{E}(\delta' D \delta)}_{\text{the estimation contribution}},$$

in which the estimation error enters only through a quadratic form, and in the metric D rather than C_m because $\iota_m' \delta = 0$ and (37) imply $\delta' C_m \delta = \delta' D \delta$. The unbiasedness used in the middle term is established directly in Lemma 7(iii), proved in Section E.6 below, and then used formally in Proposition 7. Step 1 of Proposition 7 records this identity formally; here we only need its last term. Set the remainder $\mathbb{E}_S(\rho_S)$ aside for the moment, so that δ is replaced by its linear part; the squared remainder and the cross term it generates are shown to be $o\{k^2/(mT)\}$ in Step 3 of Proposition 7, using Corollary 1 and Cauchy–Schwarz. Expanding that quadratic form:

$$\begin{aligned} & \mathbb{E} \left[\left\{ \frac{1}{T} \sum_{t=1}^T \mathbb{E}_S \phi_S(t) \right\}' D \left\{ \frac{1}{T} \sum_{u=1}^T \mathbb{E}_R \phi_R(u) \right\} \right] \\ &= \frac{1}{T^2} \sum_{t=1}^T \sum_{u=1}^T \mathbb{E} \left[\{ \mathbb{E}_S \phi_S(t) \}' D \{ \mathbb{E}_R \phi_R(u) \} \right] \end{aligned} \quad (61)$$

$$= \frac{1}{T^2} \sum_{t=1}^T \mathbb{E} \left[\{ \mathbb{E}_S \phi_S(t) \}' D \{ \mathbb{E}_R \phi_R(t) \} \right] \quad (62)$$

$$= \frac{1}{T} \mathbb{E} \left[\{ \mathbb{E}_S \phi_S(t) \}' D \{ \mathbb{E}_R \phi_R(t) \} \right] \quad (63)$$

$$= \frac{1}{T} \mathbb{E}_{S,R} \mathbb{E} [\phi_S(t)' D \phi_R(t)] = \frac{1}{T} \mathbb{E}_{S,R} \Gamma_{SR}, \quad (64)$$

where

$$\Gamma_{SR} := \mathbb{E} \{ \phi_S(t)' D \phi_R(t) \}. \quad (65)$$

Each step uses one fact. (61): the product of two averages is a double sum over two *different* indices, so both factors $1/T$ come out as $1/T^2$ and $\mathbb{E}$ passes through the finite sum by linearity.

(62): the $T(T-1)$ terms with $t \neq u$ vanish, because $\mathbb{E}_S \phi_S(t)$ and $\mathbb{E}_R \phi_R(u)$ are then independent (Assumption 3) and each has mean zero (Proposition 11), so the expectation factorises into $\mathbb{E}[\cdot]' D \mathbb{E}[\cdot] = 0$. (63): the T surviving terms are identical because the observations are identically distributed, so the sum equals T times one of them and $T/T^2 = 1/T$. *This is where the rate $1/T$ comes from:* not from cancelling one of the two averages, but from T of the T^2 pairs surviving. (64): $x'Dy$ is bilinear and the two subset draws are independent of each other and of the data, so $\mathbb{E}_S$ and $\mathbb{E}_R$ move outside.

The second index is written u and the second subset R for the same reason: expanding a product of two averages requires an independent copy of each summation variable.

The contrast between (62) and (64) is the whole point: squaring an average over *observations* leaves only the diagonal, whereas squaring an average over *subsets* keeps every pair (S, R) . So the quantity that must be evaluated is the D -inner product of the influence functions of *two different* subsets at the *same* observation. It is a covariance — both factors have mean zero by Proposition 11 — and it is nonzero only because the two subset regressions are fitted on the same training data. Had each subset had its own independent sample, every off-diagonal term would vanish and bagging would drive the estimation error to zero.

Its average over independent subset draws, $\Gamma_{m,k} = \mathbb{E}_{S,R}(\Gamma_{SR})$, is therefore the coefficient multiplying $1/T$ in the loss expansion of Proposition 7. No single-portfolio moment formula supplies it, because no such formula involves two subsets at once.

Proposition 12 (Complete first-order coefficient). *Let S and R be independent uniform subsets of size k , drawn according to Assumption 4, and let Γ_{SR} be as in (65). Under Assumptions 2, 3, and 4,*

$$\Gamma_{m,k} := \mathbb{E}_{S,R}(\Gamma_{SR}) = \mathbb{E}_{S,R} \left[\{b + v'_S D v_R\} \text{tr}(D K_S D K_R) + v'_R D K_S D K_R D v_S \right]. \quad (66)$$

Setting $R = S$ gives, exactly,

$$\Gamma_{SS} = \{Q(S)\}(k-1). \quad (67)$$

Proof. By Assumptions 2 and 3, h_t and ξ_t are independent centred Gaussians with variances b and D . From (60), $\phi_S(t) = -K_S \xi_t (e'_t v_S)$ with $e'_t v_S = h_t + v'_S \xi_t$, and K_S is symmetric, so

$$\Gamma_{SR} = \mathbb{E} \left[(h_t + v'_S \xi_t)(h_t + v'_R \xi_t) \xi'_t K_S D K_R \xi_t \right].$$

Expanding the first product gives three kinds of terms.

(a) *The h_t^2 term.* By independence,

$$\mathbb{E}(h_t^2) \mathbb{E}(\xi'_t K_S D K_R \xi_t) = b \text{tr}(D K_S D K_R),$$

using $\mathbb{E}(\xi' A \xi) = \text{tr}(AD)$ for centred ξ with variance D .

(b) *The two terms with one factor h_t .* Each equals $\mathbb{E}(h_t)$ times a moment of ξ_t , hence zero.

(c) *The fourth-moment term.* For a centred Gaussian ξ with variance D and fixed a, r, A , Isserlis' theorem (pairing the four factors in the three possible ways) gives

$$\begin{aligned} \mathbb{E}\{(a'\xi)(r'\xi)(\xi' A \xi)\} &= \sum_{i,j,p,q} a_i r_j A_{pq} \mathbb{E}(\xi_i \xi_j \xi_p \xi_q) \\ &= \sum_{i,j,p,q} a_i r_j A_{pq} \{D_{ij} D_{pq} + D_{ip} D_{jq} + D_{iq} D_{jp}\}. \end{aligned}$$

The three covariance pairings give $(a' D r) \text{tr}(AD)$, $a' D A D r$, and $a' D A' D r$, respectively. Therefore

$$\mathbb{E}[(a'\xi)(r'\xi)(\xi' A \xi)] = (a' D r) \text{tr}(AD) + a' D (A + A') D r.$$

With $a = v_S$, $r = v_R$, $A = K_S D K_R$ — which need not be symmetric, so both orders survive —

$$\mathbb{E}[(v'_S \xi_t)(v'_R \xi_t) \xi'_t K_S D K_R \xi_t] = (v'_S D v_R) \text{tr}(D K_S D K_R) + v'_S D K_S D K_R D v_R + v'_R D K_S D K_R D v_S.$$

The middle term vanishes, because K_S is symmetric and $K_S D v_S = 0$ by (58):

$$v'_S D K_S D K_R D v_R = (K_S D v_S)' D K_R D v_R = 0.$$

Adding (a) and (c) and averaging over the independent draws of S and R gives (66).

For (67), put $R = S$. The identity $K_S D K_S = K_S$ gives $\text{tr}(D K_S D K_S) = \text{tr}(D K_S) = k - 1$.

The surviving pairing vanishes because

$$v'_S D K_S D K_S D v_S = (K_S D v_S)' D (K_S D v_S) = 0.$$

Finally,

$$v'_S D v_S = \sum_{i \in S} \left(\frac{\tau_i}{A_S} \right)^2 \tau_i^{-1} = \frac{\sum_{i \in S} \tau_i}{A_S^2} = \frac{1}{A_S},$$

so $b + v'_S D v_S = Q(S)$ by Proposition 3. □

Remark 2 (Consistency check). *Expanding the exact single-subset factor of Proposition 14 when $k/T \rightarrow 0$,*

$$\frac{T-1}{T-k} = 1 + \frac{k-1}{T} + O\left(\frac{k^2}{T^2}\right),$$

whose first-order term is Γ_{SS}/T divided by $Q(S)$. So (67) reproduces the exact one-subset answer, as it must.

E.5 The order of the coefficient

Lemma 6 (Two subset-weight pairings). *Let $J = |S \cap R|$. Under bounded precisions there is a constant C depending only on τ_L, τ_U such that, pointwise in (S, R) ,*

$$|(v'_S D v_R) \operatorname{tr}(D K_S D K_R)| \leq C \frac{J}{k}, \quad |v'_R D K_S D K_R D v_S| \leq C \frac{J}{k^2},$$

and therefore, since $\mathbb{E}J = k^2/m$,

$$\mathbb{E}_{S,R} |(v'_S D v_R) \operatorname{tr}(D K_S D K_R)| \leq C \frac{k}{m}, \quad \mathbb{E}_{S,R} |v'_R D K_S D K_R D v_S| \leq \frac{C}{m}. \quad (68)$$

Both are $o(k^2/m)$ as $k \rightarrow \infty$.

Proof. Throughout, $k\tau_L \leq A_S, A_R \leq k\tau_U$ by Lemma 2(1).

First pairing. Only shared indices contribute to $v'_S D v_R$, and

$$v'_S D v_R = \sum_{i \in S \cap R} \frac{\tau_i}{A_S A_R}, \quad \text{so} \quad |v'_S D v_R| \leq \frac{J\tau_U}{k^2 \tau_L^2}.$$

For the trace, $D^{1/2} K_S D^{1/2}$ and $D^{1/2} K_R D^{1/2}$ are orthogonal projections of rank $k-1$ (Lemma 5), so each has Frobenius norm $\sqrt{k-1}$ and Cauchy–Schwarz for the Frobenius inner product gives $|\operatorname{tr}(D K_S D K_R)| \leq k-1$. Multiplying the two bounds gives CJ/k .

Second pairing. Recall $(D v_S)_i = \mathbf{1}\{i \in S\}/A_S$, i.e. $D v_S = \mathbf{1}_S/A_S$ where $\mathbf{1}_S$ is the indicator vector of S ; likewise $D v_R = \mathbf{1}_R/A_R$. Hence

$$v'_R D K_S D K_R D v_S = \frac{1}{A_S A_R} (K_S \mathbf{1}_R)' D (K_R \mathbf{1}_S).$$

Now compute $K_S \mathbf{1}_R$ from (57): for $a \in S$,

$$(K_S \mathbf{1}_R)_a = \tau_a \left\{ \mathbf{1}\{a \in R\} - \frac{A_{S \cap R}}{A_S} \right\}, \quad A_{S \cap R} = \sum_{i \in S \cap R} \tau_i,$$

and $(K_S \mathbf{1}_R)_a = 0$ for $a \notin S$; symmetrically for $K_R \mathbf{1}_S$, which is supported on R . The two vectors are therefore simultaneously nonzero only on $S \cap R$, and there

$$(K_S \mathbf{1}_R)' D (K_R \mathbf{1}_S) = \sum_{a \in S \cap R} \tau_a \left\{ 1 - \frac{A_{S \cap R}}{A_S} \right\} \left\{ 1 - \frac{A_{S \cap R}}{A_R} \right\}.$$

Both braces lie in $[0, 1]$ because $0 \leq A_{S \cap R} \leq \min(A_S, A_R)$, so the sum is at most $J\tau_U$ in absolute value, and dividing by $A_S A_R \geq k^2 \tau_L^2$ gives CJ/k^2 .

Finally $J = \sum_i \mathbf{1}\{i \in S\} \mathbf{1}\{i \in R\}$ and S, R are independent with $\Pr(i \in S) = k/m$, so $\mathbb{E}J = m(k/m)^2 = k^2/m$. Taking expectations gives (68). Relative to k^2/m , the two bounds have ratios $(Ck/m)/(k^2/m) = C/k$ and $(C/m)/(k^2/m) = C/k^2$, respectively. Both ratios converge to zero as $k \rightarrow \infty$. $\square$

Proposition 13 (Order of $\Gamma_{m,k}$). *Let $\bar{K} = \mathbb{E}_S(K_S)$. As $m \geq k \rightarrow \infty$,*

$$\text{tr}(D\bar{K}D\bar{K}) = \frac{k^2}{m} \{1 + O(k^{-1} + m^{-1})\}, \quad (69)$$

and consequently

$$\Gamma_{m,k} = b \frac{k^2}{m} \{1 + o(1)\}. \quad (70)$$

Moreover, for every k and m , without asymptotics,

$$\text{tr}(D\bar{K}D\bar{K}) \geq \frac{(k-1)^2}{m-1}, \quad (71)$$

with equality when all precisions are equal.

Where the $1/m$ comes from. A single subset fit has effective dimension $k-1$. Averaging over subsets does *not* average $k-1$ independent errors: the same m coordinates are re-used, and $\bar{K}$ has entries of order k/m rather than order one. Squaring and summing m diagonal entries turns $(k/m)^2 \cdot m$ into k^2/m . In the orthonormal complete-subset calculation of Elliott et al. (2013, Sec. 2.2), the shrinkage factor is k/K , so the associated variance term has the same k^2/K order. Boot and Nibbering (2019, Thm. 1 and Lemma 1) provide the broader random-subspace bias-variance comparison, but their displayed variance bound is proportional to k , not the k^2/m coefficient here. The content of Proposition 13 is that the complete-subset order survives the adding-up restriction and heterogeneous precisions.

Proof. Step 1: entries of $\bar{K}$. From (57), on the selected coordinates K_S has diagonal entry $\tau_i - \tau_i^2/A_S$ and off-diagonal entry $-\tau_i\tau_j/A_S$. Since $\Pr(i \in S) = k/m$ and $\Pr(i, j \in S) = k(k-1)/\{m(m-1)\}$ for $i \neq j$, taking expectations gives exactly

$$\bar{K}_{ii} = \frac{k}{m} \tau_i \left\{ 1 - \tau_i \mathbb{E}_S(A_S^{-1} \mid i \in S) \right\}, \quad \bar{K}_{ij} = -\frac{k(k-1)}{m(m-1)} \tau_i \tau_j \mathbb{E}_S(A_S^{-1} \mid i, j \in S) \quad (i \neq j).$$

By Lemma 2(1), $A_S^{-1} \leq 1/(k\tau_L)$ holds pointwise on every subset S , hence under any conditioning event; both conditional expectations are therefore $O(k^{-1})$ uniformly in the conditioning indices.

Step 2: the trace. Since $\bar{K}$ is symmetric and $D = \text{diag}(\tau_i^{-1})$,

$$\text{tr}(D\bar{K}D\bar{K}) = \sum_{i,j} \frac{\bar{K}_{ij}^2}{\tau_i \tau_j} = \sum_i \frac{\bar{K}_{ii}^2}{\tau_i^2} + \sum_{i \neq j} \frac{\bar{K}_{ij}^2}{\tau_i \tau_j}.$$

For the diagonal part, $\tau_i \mathbb{E}_S(A_S^{-1} \mid i \in S) = O(k^{-1})$ uniformly, so

$$\sum_i \frac{\bar{K}_{ii}^2}{\tau_i^2} = \frac{k^2}{m^2} \sum_{i=1}^m \{1 + O(k^{-1})\}^2 = \frac{k^2}{m} \{1 + O(k^{-1})\}.$$

For the off-diagonal part, $\bar{K}_{ij} = O(k/m^2)$ uniformly. Hence each summand $\bar{K}_{ij}^2/(\tau_i \tau_j)$ is $O(k^2/m^4)$, because the precisions are bounded above and away from zero. Summing at most

$m(m-1)$ such terms gives $O(k^2/m^2)$, which is $O(m^{-1})$ relative to k^2/m . Adding the two gives (69).

Step 3: from the trace to $\Gamma_{m,k}$. In (66), the term generated by the common component is $b\mathbb{E}_{S,R}\text{tr}(DK_S DK_R)$; because S and R are independent, the expectation passes inside the trace and $\mathbb{E}_{S,R}(K_S DK_R) = \bar{K} D \bar{K}$, giving $b \text{tr}(D \bar{K} D \bar{K})$. The two remaining terms are $o(k^2/m)$ by Lemma 6. Combining with (69) gives (70).

Step 4: the non-asymptotic lower bound. The matrix $D^{1/2} \bar{K} D^{1/2} = \mathbb{E}_S(D^{1/2} K_S D^{1/2})$ is an average of orthogonal projections (Lemma 5), hence positive semidefinite. Its rank is at most $m-1$: since $\bar{K} \iota_m = \mathbb{E}_S(K_S \iota_m) = 0$, the nonzero vector $D^{-1/2} \iota_m$ satisfies

$$D^{1/2} \bar{K} D^{1/2} (D^{-1/2} \iota_m) = D^{1/2} \bar{K} \iota_m = 0.$$

Thus, the kernel is nontrivial. Its trace is $\mathbb{E}_S \text{tr}(DK_S) = k-1$ by (58). Moreover, by cyclicity of the trace,

$$\text{tr}(D \bar{K} D \bar{K}) = \text{tr}\{(D^{1/2} \bar{K} D^{1/2})^2\} = \sum_{j=1}^r \lambda_j^2,$$

where $\lambda_1, \dots, \lambda_r$ are the positive eigenvalues of $D^{1/2} \bar{K} D^{1/2}$ and $r \leq m-1$. Cauchy-Schwarz gives $(\sum_{j=1}^r \lambda_j)^2 \leq r \sum_{j=1}^r \lambda_j^2$, and therefore

$$\text{tr}(D \bar{K} D \bar{K}) = \sum_{j=1}^r \lambda_j^2 \geq \frac{(\sum_{j=1}^r \lambda_j)^2}{r} \geq \frac{(k-1)^2}{m-1},$$

which is (71). Under homogeneity ($\tau_i \equiv \tau$, so $A_S = k\tau$ and the conditional expectations in Step 1 equal $1/(k\tau)$), the entrywise formulas give $\bar{K}_{ii} = \tau(k-1)/m$ and $\bar{K}_{ij} = -\tau(k-1)/\{m(m-1)\}$. Summing the m diagonal and the $m(m-1)$ off-diagonal squared entries,

$$\text{tr}(D \bar{K} D \bar{K}) = m \cdot \frac{(k-1)^2}{m^2} + m(m-1) \cdot \frac{(k-1)^2}{m^2(m-1)^2} = \frac{(k-1)^2}{m} + \frac{(k-1)^2}{m(m-1)} = \frac{(k-1)^2}{m-1},$$

so (71) holds with equality. $\square$

E.6 One Gaussian regression, two consequences

One whitened-contrast calculation yields both the bound on the nonlinear remainder ρ_S and the exact single-subset factor $(T-1)/(T-k)$. We therefore establish the regression representation once and then derive both consequences from it.

Lemma 7 (The subset regression in contrast coordinates). *Fix S and choose a $k \times (k-1)$ matrix H_S whose columns span $\{a \in \mathbb{R}^k : \iota'_k a = 0\}$ and satisfy $H'_S D_S H_S = I_{k-1}$. Put*

$$\eta_t = v'_S e_t, \quad x_t = -H'_S P'_S e_t \in \mathbb{R}^{k-1},$$

and let X be the $T \times (k-1)$ matrix with rows x'_t . Then, under Assumption 3:

(i) $M_S = H_S H'_S$, where M_S is the bracket in (57);

(ii) the restricted least squares problem defining $\widehat{w}_S$ is exactly the unrestricted regression of η on X : writing $\widehat{\beta} = (X'X)^{-1}X'\eta$, we have $\widehat{w}_S = w_S + H_S \widehat{\beta}$, and any sum-to-one $w = w_S + H_S \beta$ satisfies $(w - w_S)' D_S (w - w_S) = \|\beta\|^2$;

(iii) $x_t \sim N(0, I_{k-1})$ i.i.d. and $X'X \sim W_{k-1}(I_{k-1}, T)$. Moreover, η_t is independent of x_t with variance $Q(S)$;

(iv) the influence-function term of (60) corresponds to $\widetilde{\beta} = T^{-1}X'\eta$, i.e. $T^{-1} \sum_t \phi_S(t) = P_S H_S \widetilde{\beta}$ and $\rho_S = P_S H_S (\widehat{\beta} - \widetilde{\beta})$, so that $\rho'_S D \rho_S = \|\widehat{\beta} - \widetilde{\beta}\|^2$.

Proof. Such a matrix exists: if B_S is any full-column-rank basis of $\iota_k^\perp$, then $B'_S D_S B_S$ is positive definite and $H_S = B_S (B'_S D_S B_S)^{-1/2}$ has the required two properties.

(i) Both M_S and $H_S H'_S$ annihilate ι_k : for the first this was shown in Lemma 5, and for the second $H'_S \iota_k = 0$ by construction, so the two matrices agree on $\text{span}(\iota_k)$. They also agree on $D_S(\iota_k^\perp)$, mapping $D_S H_S \beta$ to $H_S \beta$: for $\beta \in \mathbb{R}^{k-1}$,

$$M_S D_S H_S \beta = H_S \beta - \frac{D_S^{-1} \iota_k (\iota'_k H_S \beta)}{A_S} = H_S \beta, \quad H_S H'_S D_S H_S \beta = H_S \beta,$$

using $\iota'_k H_S = 0$ in the first display and $H'_S D_S H_S = I_{k-1}$ in the second. (Since the columns of H_S span $\iota_k^\perp$, every element of $D_S(\iota_k^\perp)$ has the form $D_S H_S \beta$.) The two subspaces together span $\mathbb{R}^k$: D_S is invertible, so $\dim D_S(\iota_k^\perp) = k - 1$, and $\iota_k \notin D_S(\iota_k^\perp)$ — otherwise $\iota_k = D_S a$ for some $a \in \iota_k^\perp$, i.e. $a = D_S^{-1} \iota_k$, and then $0 = \iota'_k a = \iota'_k D_S^{-1} \iota_k = A_S$, contradicting $A_S \geq k\tau_L > 0$. Hence $\mathbb{R}^k = \text{span}(\iota_k) \oplus D_S(\iota_k^\perp)$, and two linear maps that agree on a direct-sum decomposition of the whole space agree everywhere.

(ii) Every sum-to-one w is $w_S + H_S \beta$ for a unique β , and $(H_S \beta)' D_S (H_S \beta) = \|\beta\|^2$ by the normalisation. Substituting into the training criterion,

$$\sum_t (w' P'_S e_t)^2 = \sum_t (\eta_t - \beta' x_t)^2,$$

which is minimised at $\widehat{\beta} = (X'X)^{-1}X'\eta$.

(iii) e_t is Gaussian, so $(\eta_t, x'_t)'$ is Gaussian. Because $\iota'_k H_S = 0$ kills the rank-one part of C_S ,

$$\mathbb{E}(x_t x'_t) = H'_S C_S H_S = H'_S D_S H_S = I_{k-1}.$$

By Lemma 1, $C_S w_S = \mu \iota_k$ with $\mu = 1/(\iota'_k C_S^{-1} \iota_k) = Q(S)$, so

$$\mathbb{E}(\eta_t x_t) = -H'_S C_S w_S = -\mu H'_S \iota_k = 0, \quad \text{Var}(\eta_t) = w'_S C_S w_S = \mu = Q(S).$$

Jointly Gaussian and uncorrelated means independent, and independence over t gives $X'X \sim W_{k-1}(I_{k-1}, T)$.

(iv) By Proposition 11 and (i), $\phi_S(t) = -K_S(e_t e_t' - C_m)v_S$ equals, in $\mathbb{R}^k$, $-M_S(P_S' e_t)\eta_t + M_S C_S w_S = -H_S H_S' P_S' e_t \eta_t = H_S x_t \eta_t$, using $M_S \iota_k = 0$. Averaging over t gives $H_S(T^{-1}X'\eta) = H_S \tilde{\beta}$, and embedding gives the claim. Since $\hat{v}_S - v_S = P_S H_S \hat{\beta}$, the remainder is $P_S H_S(\hat{\beta} - \tilde{\beta})$, and its D -quadratic form is $\|\hat{\beta} - \tilde{\beta}\|^2$ by the normalisation in (ii). $\square$

Proposition 14 (Exact single-subset factor). *Write $L(v) = v' C_m v$ for the independent-test loss of a sum-to-one weight vector v . For $T > k$, the expected independent-test loss of one fitted subset is, exactly,*

$$\mathbb{E} L(\hat{v}_S) = \{Q(S)\} \frac{T-1}{T-k}.$$

For $T > m$, the same argument applied to the full set $S = \{1, \dots, m\}$ (with $P_S = I_m$) gives the factor $(T-1)/(T-m)$ for the full estimated combination.

Proof. Since $\iota_m' \hat{v}_S = 1$, the test loss is $L(\hat{v}_S) = \hat{v}_S' C_m \hat{v}_S$. Writing $\hat{w}_S = w_S + H_S \hat{\beta}$ and expanding,

$$\hat{v}_S' C_m \hat{v}_S = w_S' C_S w_S + 2\hat{\beta}' H_S' C_S w_S + \hat{\beta}' H_S' C_S H_S \hat{\beta} = \{Q(S)\} + 0 + \|\hat{\beta}\|^2,$$

using the facts established in Lemma 7(iii). Because $T > k$ and the Wishart dimension is $p = k-1$, $X'X$ is invertible almost surely and the first inverse moment used below exists. Now $\mathbb{E}(\|\hat{\beta}\|^2 \mid X) = \{Q(S)\} \text{tr}\{(X'X)^{-1}\}$, and for $W \sim W_p(I_p, T)$ with $p = k-1$ one has $\mathbb{E}(W^{-1}) = I_p/(T-p-1) = I_{k-1}/(T-k)$. Hence

$$\mathbb{E}\|\hat{\beta}\|^2 = \{Q(S)\} \frac{k-1}{T-k},$$

and $1 + (k-1)/(T-k) = (T-1)/(T-k)$. $\square$

Corollary 1 (Nonlinear remainder). *For $T > k$,*

$$\mathbb{E}(\rho_S' D \rho_S) = \{Q(S)\} \frac{k(k-1)}{T(T-k)} = O\left(\frac{k^2}{T}\right) \quad \text{when } k \leq m = o(T). \quad (72)$$

Proof. By Lemma 7(iv), $\rho_S' D \rho_S = \|\hat{\beta} - \tilde{\beta}\|^2$ with $\hat{\beta} - \tilde{\beta} = \{(X'X)^{-1} - T^{-1}I\}X'\eta$. Conditioning on X and using $\mathbb{E}(\eta\eta' \mid X) = \{Q(S)\}I_T$, set $W = X'X$ and $B_X = (W^{-1} - T^{-1}I)X'$. Then

$$B_X B_X' = W^{-1} - \frac{2}{T}I_{k-1} + \frac{1}{T^2}W,$$

so

$$\mathbb{E}(\|\hat{\beta} - \tilde{\beta}\|^2 \mid X) = \frac{Q(S)}{T} \text{tr}\left[\left(\frac{X'X}{T}\right)^{-1} - 2I_{k-1} + \frac{X'X}{T}\right].$$

Taking expectations with $\mathbb{E}(X'X/T) = I_{k-1}$ and $\mathbb{E}\{(X'X/T)^{-1}\} = TI_{k-1}/(T-k)$ leaves

$$\frac{Q(S)}{T}(k-1) \left\{ \frac{T}{T-k} - 1 \right\} = \{Q(S)\} \frac{k(k-1)}{T(T-k)}.$$

When $m = o(T)$ and $k \leq m$ we have $T-k \asymp T$, giving $O(k^2/T^2)$. $\square$

E.7 The shared-sample inflation

Proposition (Sharp shared-sample inflation; restatement of Proposition 7). *Let $k = k_{T,m}$ satisfy $k \rightarrow \infty$, $k/m \rightarrow 0$ and $k < T$. Under Assumptions 2, 3, and 4, the expected independent-test loss of the literal shared-sample infinite-bag estimator satisfies*

$$L_{T,m}^{\text{bag},\infty}(k) = Q_k^{\text{bag},\infty} + \frac{\Gamma_{m,k}}{T} + o\left(\frac{k^2}{mT}\right), \quad (73)$$

with $\Gamma_{m,k}$ given by (66). Equivalently, in multiplicative form,

$$\mathcal{I}_\infty(T, m, k) = 1 + \frac{\Gamma_{m,k}}{T\{Q_k^{\text{bag},\infty}\}} + o\left(\frac{k^2}{mT}\right) = 1 + \frac{k^2}{mT}\{1 + o(1)\}.$$

Proof of Proposition 7. Step 1: the estimation error is measured in D and is unbiased.

Both $\widehat{v}_k$ and $\bar{v}_k$ sum to one, so $\iota'_m(\widehat{v}_k - \bar{v}_k) = 0$ and, by the convention of (37), its C_m -quadratic form equals its D -quadratic form. For unbiasedness, fix S and work in the contrast coordinates of Lemma 7. By part (iii), $(\eta_t, x'_t)'$ is Gaussian with η_t independent of x_t for each t , and the pairs are i.i.d. across t ; the joint law therefore factorizes, so the whole vector η is independent of the whole matrix X . Hence $\mathbb{E}(\widehat{\beta} \mid X) = (X'X)^{-1}X'\mathbb{E}(\eta \mid X) = 0$; the conditional expectation is integrable because $\mathbb{E}\|\widehat{\beta}\|^2 = Q(S)(k-1)/(T-k) < \infty$ under $k < T$ (Proposition 14). Thus $\mathbb{E}(\widehat{v}_S) = v_S + P_S H_S \mathbb{E}(\widehat{\beta}) = v_S$ for every S , and averaging over the finitely many subsets gives $\mathbb{E}(\widehat{v}_k) = \bar{v}_k$. Expanding,

$$L_{T,m}^{\text{bag},\infty}(k) = \mathbb{E}(\widehat{v}'_k C_m \widehat{v}_k) = \bar{v}'_k C_m \bar{v}_k + \mathbb{E}[(\widehat{v}_k - \bar{v}_k)' D (\widehat{v}_k - \bar{v}_k)],$$

the cross term vanishing by unbiasedness. The first term is $Q_k^{\text{bag},\infty}$.

Step 2: split the error. By Proposition 11, $\widehat{v}_k - \bar{v}_k = \mathcal{L}_T + \mathcal{R}_T$ with

$$\mathcal{L}_T = \frac{1}{T} \sum_{t=1}^T \mathbb{E}_S \phi_S(t), \quad \mathcal{R}_T = \mathbb{E}_S(\rho_S).$$

The leading term is exactly the one computed at the start of Section E.4, where Γ_{SR} was derived:

$$\mathbb{E}(\mathcal{L}'_T D \mathcal{L}_T) = \frac{\Gamma_{m,k}}{T},$$

which is $\asymp k^2/(mT)$ by Proposition 13.

Step 3: the remainder is negligible. By convexity of the D -quadratic form and Corollary 1,

$$\mathbb{E}(\mathcal{R}'_T D \mathcal{R}_T) \leq \mathbb{E}[\mathbb{E}_S(\rho'_S D \rho_S)] = O\left(\frac{k^2}{T^2}\right),$$

whose ratio to $k^2/(mT)$ is $O(m/T) = o(1)$ by Assumption 3. For the cross term, Cauchy-Schwarz in the D -inner product gives

$$|\mathbb{E}(\mathcal{L}'_T D \mathcal{R}_T)| \leq \left\{ \mathbb{E}(\mathcal{L}'_T D \mathcal{L}_T) \right\}^{1/2} \left\{ \mathbb{E}(\mathcal{R}'_T D \mathcal{R}_T) \right\}^{1/2} = O\left(\frac{k}{\sqrt{mT}} \cdot \frac{k}{T}\right),$$

whose ratio to $k^2/(mT)$ is $O(\sqrt{m/T}) = o(1)$. Adding Steps 2 and 3 gives (73).

Step 4: multiplicative form. By (23), (24), and Lemma 2(1), $g_i(k)$ is bounded above and below, so $\bar{v}_{i,k} = O(m^{-1})$ uniformly and $Q_k^{\text{bag},\infty} = b + \sum_i \bar{v}_{i,k}^2/\tau_i = b + O(m^{-1})$. Hence $Q_k^{\text{bag},\infty} = b + O(m^{-1}) \rightarrow b > 0$; in particular, it is bounded away from zero. Therefore, dividing (73) by it and using $\Gamma_{m,k} = bk^2m^{-1}\{1 + o(1)\}$ from (70) gives both stated forms. $\square$

E.8 The fourth-root rule

Proposition (Optimal infinitely bagged subset size; restatement of Proposition 9). *Let*

$$k_{T,m}^* \in \arg \min_{1 \leq k \leq \min\{m, T-1\}} L_{T,m}^{\text{bag},\infty}(k).$$

Under Assumptions 2, 3, and 4,

$$\frac{k_{T,m}^*}{T^{1/4}} \longrightarrow \left(\frac{\mathcal{D}}{b}\right)^{1/4}, \quad i.e. \quad k_{T,m}^* \sim \left(\frac{\mathcal{D}T}{b}\right)^{1/4}. \quad (74)$$

No power of m appears, because the approximation error and the estimation penalty both carry the same factor $1/m$.

Intuition, and why the exponent differs from 1/2. For the average-single-subset criterion the approximation error falls like $1/k$ and the penalty grows like k/T , so balancing gives $\sqrt{T}$ (Proposition 8). Bagging changes *both* sides: averaging the weight vectors squares the approximation error, so it falls like $1/k^2$; and the shared sample makes the penalty grow like k^2/T rather than k/T . Balancing k^{-2} against k^2/T gives $T^{1/4}$. The factors $1/m$ on both sides cancel, which is why the pool size does not enter the rate.

Elliott et al. (2013, Sec. 2.3) and Boot and Nibbering (2019, Sec. 3.1) likewise use subset dimension to trade approximation bias against estimation variance. Neither paper derives the present $T^{1/4}$ rule: that exponent comes from averaging adding-up-constrained forecast weights and from fitting all subsets on the same sample.

Proof of Proposition 9. Step 1: solve the leading minimisation problem directly. Since $Q_k^{\text{bag},\infty} = Q^{\text{opt}} + R_k^{\text{bag}}$ by Proposition 5, the part of the leading excess-loss criterion that depends on k is

$$H_{T,m}(k) := \frac{\mathcal{D}_m}{k^2} + b\frac{k^2}{T}, \quad k > 0. \quad (75)$$

Treating k temporarily as a positive real number,

$$H'_{T,m}(k) = -\frac{2\mathcal{D}_m}{k^3} + \frac{2bk}{T}, \quad H''_{T,m}(k) = \frac{6\mathcal{D}_m}{k^4} + \frac{2b}{T} > 0.$$

Thus $H_{T,m}$ is strictly convex and its unique minimiser is

$$\tilde{k}_{T,m} = \left(\frac{\mathcal{D}_m T}{b} \right)^{1/4}, \quad (76)$$

obtained from $H'_{T,m}(k) = 0$, or equivalently $\tilde{k}_{T,m}^4 = \mathcal{D}_m T/b$. Because $\mathcal{D}_m \rightarrow \mathcal{D} > 0$, Assumption 3 implies $\tilde{k}_{T,m} \rightarrow \infty$ and $\tilde{k}_{T,m}/m \rightarrow 0$. Let $k_{T,m}^0$ be a feasible integer nearest to $\tilde{k}_{T,m}$; feasibility, that is $1 \leq k_{T,m}^0 \leq \min\{m, T-1\}$, holds for all large (T, m) because $\tilde{k}_{T,m} \asymp T^{1/4} = o(\min\{m, T\})$. By Taylor's theorem around $\tilde{k}_{T,m}$, the first-order condition $H'_{T,m}(\tilde{k}_{T,m}) = 0$, $|k_{T,m}^0 - \tilde{k}_{T,m}| \leq 1$, and the bound $H''_{T,m}(\xi) = 6\mathcal{D}_m/\xi^4 + 2b/T = O(T^{-1})$ for $\xi \in [\tilde{k}_{T,m} - 1, \tilde{k}_{T,m} + 1]$ (as $\xi \asymp T^{1/4}$ implies $\xi^4 \asymp T$),

$$H_{T,m}(k_{T,m}^0) - H_{T,m}(\tilde{k}_{T,m}) = O(T^{-1}) = o(T^{-1/2}). \quad (77)$$

Moreover, Propositions 10 and 7 apply at $k_{T,m}^0$, since $k_{T,m}^0 \asymp T^{1/4} \rightarrow \infty$ and $k_{T,m}^0/m \rightarrow 0$ by $T^{1/4} = o(m)$. Writing $k^0 = k_{T,m}^0$, they give

$$mR_{k^0}^{bag} = \mathcal{D}_m z_{k^0}^2 \{1 + o(1)\} = \frac{\mathcal{D}_m}{(k^0)^2} \left(1 - \frac{k^0}{m}\right)^2 \{1 + o(1)\} = \frac{\mathcal{D}_m}{(k^0)^2} + o(T^{-1/2}),$$

because $(k^0)^{-2} \asymp T^{-1/2}$ while $(1 - k^0/m)^2 = 1 + o(1)$, and

$$\begin{aligned} m\{L_{T,m}^{bag,\infty}(k^0) - Q_{k^0}^{bag,\infty}\} &= \frac{m\Gamma_{m,k^0}}{T} + m \cdot o\left(\frac{(k^0)^2}{mT}\right) \\ &= \frac{b(k^0)^2}{T} \{1 + o(1)\} + o(T^{-1/2}) \\ &= \frac{b(k^0)^2}{T} + o(T^{-1/2}). \end{aligned}$$

using (70) and $(k^0)^2/T \asymp T^{-1/2}$. Since $Q_{k^0}^{bag,\infty} - Q^{\text{opt}} = R_{k^0}^{bag}$ (Proposition 5), adding the two displays yields

$$m\{L_{T,m}^{bag,\infty}(k_{T,m}^0) - Q^{\text{opt}}\} = H_{T,m}(k_{T,m}^0) + o(T^{-1/2}) = O(T^{-1/2}). \quad (78)$$

Step 2: localise the exact global minimiser. This step verifies that the direct solution in (76) lies in the region where the leading expansion describes the exact criterion. The non-asymptotic approximation bound in Lemma 4, valid for every $1 \leq k < m$, gives

$$mR_k^{bag} \geq c_A z_k^2, \quad c_A := \frac{\tau_L v_\tau^2}{4\tau_U^6} > 0, \quad (79)$$

and (79) holds trivially at $k = m$, where both sides vanish because $z_m = 0$ and $R_m^{bag} = 0$. For the estimation component, decompose the estimation error as $\mathcal{L}_T + \mathcal{R}_T$ according to Proposition 11,

equation (60). To bound $\mathbb{E}(\mathcal{L}'_T D \mathcal{L}_T)$ from below, split the averaged influence function into its common-residual and idiosyncratic parts,

$$\mathbb{E}_S \phi_S(t) = -\bar{K} \xi_t h_t - \psi(\xi_t), \quad \psi(\xi) := \mathbb{E}_S [K_S \xi (v'_S \xi)].$$

Both parts have mean zero ($\mathbb{E}\{\psi(\xi_t)\} = \mathbb{E}_S(K_S D v_S) = 0$ by (58)) and are independent across t , so only same- t pairs survive in $\mathbb{E}(\mathcal{L}'_T D \mathcal{L}_T)$:

$$\mathbb{E}(\mathcal{L}'_T D \mathcal{L}_T) = \frac{1}{T} \left\{ \mathbb{E}[h_t^2 \xi_t' \bar{K} D \bar{K} \xi_t] + 2 \mathbb{E}[h_t \xi_t' \bar{K} D \psi(\xi_t)] + \mathbb{E}[\psi(\xi_t)' D \psi(\xi_t)] \right\}.$$

The middle term vanishes because h_t is independent of ξ_t with $\mathbb{E}(h_t) = 0$; the first term equals $(b/T) \text{tr}(D \bar{K} D \bar{K})$ by $\mathbb{E}(\xi_t' A \xi_t) = \text{tr}(AD)$; and the third term is nonnegative. Hence, using (71),

$$\mathbb{E}(\mathcal{L}'_T D \mathcal{L}_T) \geq \frac{b}{T} \text{tr}(D \bar{K} D \bar{K}) \geq \frac{b(k-1)^2}{T(m-1)}.$$

Corollary 1 and $m/T \rightarrow 0$ imply, uniformly over $2 \leq k \leq m$ (using $Q(S) \leq b + 1/(2\tau_L)$, $T - k \geq T - m$, and $k^2/(k-1)^2 \leq 4$ for $k \geq 2$),

$$\frac{\mathbb{E}(\mathcal{R}'_T D \mathcal{R}_T)}{b(k-1)^2/\{T(m-1)\}} = O\left(\frac{m}{T}\right) = o(1).$$

The reverse triangle inequality in the norm $\|X\| := \{\mathbb{E}(X' D X)\}^{1/2}$ therefore yields, uniformly over $2 \leq k \leq m$,

$$\mathbb{E}\{(\mathcal{L}_T + \mathcal{R}_T)' D (\mathcal{L}_T + \mathcal{R}_T)\} \geq \frac{b(k-1)^2}{T(m-1)} \{1 - o(1)\}. \quad (80)$$

At $k = 1$ the bound (80) is vacuous, but $k = 1$ is excluded from the optimum directly by (79): $z_1^2 = (1 - 1/m)^2 \geq 1/4$ is not $O(T^{-1/2})$.

Optimality of $k_{T,m}^*$ and the benchmark bound (78), together with the two nonnegative components (79) and (80), imply

$$z_{k_{T,m}^*}^2 = O(T^{-1/2}), \quad \frac{(k_{T,m}^* - 1)^2}{T} = O(T^{-1/2}).$$

The second relation gives $k_{T,m}^* = O(T^{1/4}) = o(m)$. Consequently, $z_{k_{T,m}^*} = (k_{T,m}^*)^{-1} \{1 - k_{T,m}^*/m\} \sim (k_{T,m}^*)^{-1}$, and the first relation then gives $(k_{T,m}^*)^{-1} = O(T^{-1/4})$. Hence there are fixed constants $0 < c < C < \infty$ such that, eventually,

$$cT^{1/4} \leq k_{T,m}^* \leq CT^{1/4}. \quad (81)$$

Step 3: compare with the exact criterion. Because $\tilde{k}_{T,m}/T^{1/4} \rightarrow (\mathcal{D}/b)^{1/4}$, the constants c and C in (81) may be enlarged, if necessary, so that $\tilde{k}_{T,m}$ and $k_{T,m}^0$ also belong to the

band. Uniformly over this band, $k \rightarrow \infty$, $k/m \rightarrow 0$, and $z_k \sim k^{-1}$. Equations (54), (70), and (73) therefore give

$$m\{L_{T,m}^{\text{bag},\infty}(k) - Q^{\text{opt}}\} = H_{T,m}(k) + r_{T,m}(k), \quad \sup_{cT^{1/4} \leq k \leq CT^{1/4}} |r_{T,m}(k)| = o(T^{-1/2}). \quad (82)$$

The uniform statement follows from the sequential criterion. For each (T, m) the band contains finitely many integers, so there exists $k_{T,m}$ in the band attaining $\sup_{cT^{1/4} \leq k \leq CT^{1/4}} |r_{T,m}(k)|$. Any such extremizing sequence satisfies $k_{T,m} \asymp T^{1/4} \rightarrow \infty$ and $k_{T,m}/m \rightarrow 0$ (by $T^{1/4} = o(m)$), so Propositions 10 and 7 apply along it and force $T^{1/2}|r_{T,m}(k_{T,m})| \rightarrow 0$, which is exactly the stated uniform bound.

By optimality of $k_{T,m}^*$, equations (77) and (82) give

$$H_{T,m}(k_{T,m}^*) - H_{T,m}(\tilde{k}_{T,m}) = o(T^{-1/2}). \quad (83)$$

This gap can be evaluated exactly. Setting $q_{T,m} := k_{T,m}^*/\tilde{k}_{T,m}$ and using $\mathcal{D}_m/\tilde{k}_{T,m}^2 = b\tilde{k}_{T,m}^2/T = \sqrt{\mathcal{D}_m b/T}$ (both equalities from $\tilde{k}_{T,m}^4 = \mathcal{D}_m T/b$), we obtain

$$\begin{aligned} H_{T,m}(k_{T,m}^*) - H_{T,m}(\tilde{k}_{T,m}) &= \sqrt{\frac{\mathcal{D}_m b}{T}} (q_{T,m}^{-2} + q_{T,m}^2 - 2) \\ &= \sqrt{\frac{\mathcal{D}_m b}{T}} (q_{T,m} - q_{T,m}^{-1})^2. \end{aligned}$$

Combining this identity with (83) and $\mathcal{D}_m \rightarrow \mathcal{D} > 0$ gives $(q_{T,m} - q_{T,m}^{-1})^2 \rightarrow 0$. On the band, $q_{T,m}$ is bounded above and below by positive constants (both $k_{T,m}^*$ and $\tilde{k}_{T,m}$ are of exact order $T^{1/4}$), so $q_{T,m}^2 - 1 = q_{T,m}(q_{T,m} - q_{T,m}^{-1}) \rightarrow 0$ and hence $q_{T,m} \rightarrow 1$. Finally, (76) yields

$$\frac{k_{T,m}^*}{T^{1/4}} = \frac{k_{T,m}^*}{\tilde{k}_{T,m}} \left(\frac{\mathcal{D}_m}{b} \right)^{1/4} \rightarrow \left(\frac{\mathcal{D}}{b} \right)^{1/4},$$

which is (74). □

Remark 3 (Fixed m). *If m is fixed while $T \rightarrow \infty$, the triangular-array conditions fail: $k/m \rightarrow 0$ is impossible for $k \rightarrow \infty$. Estimation is then asymptotically costless, and $k = m$ is an asymptotically optimal boundary choice, eventually unique whenever $R_k^{\text{bag}} > 0$ for every $k < m$. Lemma 4 plays no role in this case.*

E.9 The finite- G coefficient

Proposition 15 (Second moment of the finite- G averaged influence function). *Let $S_1, \dots, S_G$ be independent subset draws and define the averaged influence function*

$$\ell_G = \frac{1}{T} \sum_{t=1}^T \frac{1}{G} \sum_{g=1}^G \phi_{S_g}(t).$$

Then

$$\mathbb{E}(\ell'_G D \ell_G) = \frac{\Gamma_{m,k}^G}{T}, \quad \Gamma_{m,k}^G = \frac{1}{G} \mathbb{E}_S \Gamma_{SS} + \left(1 - \frac{1}{G}\right) \Gamma_{m,k}, \quad \Gamma_{SS} = \{Q(S)\}(k-1). \quad (84)$$

Thus $\Gamma_{m,k}^G$ plays for the finite- G estimator the role that $\Gamma_{m,k}$ plays for the infinite bag in Proposition 7: it is the second moment of the linear term of the estimation error. (A complete finite- G loss expansion is not needed in what follows; Proposition 16 compares the complete feasible losses exactly.)

Proof. Expanding the squared D -norm of ℓ_G , the $t \neq u$ pairs vanish exactly as in (62), since each $G^{-1} \sum_g \phi_{S_g}(t)$ is a mean-zero function of (h_t, ξ_t) and the subset draws alone, i.i.d. across t . For the surviving same- t terms, the double sum over draws produces G diagonal pairs, each with expectation $\mathbb{E}_S \Gamma_{SS}$, and $G(G-1)$ off-diagonal pairs, each with expectation $\Gamma_{m,k}$ by independence of the draws — exactly the pair count of Proposition 1. Dividing by G^2 and by the factor T from the time average, as in (63), gives (84). The expression for Γ_{SS} is (67). $\square$

The coefficient calculation does not by itself compare the complete finite- G and infinite-bag losses, because finite subset sampling also creates a population-level Monte Carlo term. The following exact identity accounts for both effects.

Proposition 16 (Exact feasible finite-bag bridge). *Let*

$$\widehat{v}_{k,G} = \frac{1}{G} \sum_{g=1}^G \widehat{v}_{S_g}, \quad L_{T,m}^{\text{bag},G}(k) = \mathbb{E}(\widehat{v}'_{k,G} C_m \widehat{v}_{k,G}).$$

For every finite G ,

$$L_{T,m}^{\text{bag},G}(k) = L_{T,m}^{\text{bag},\infty}(k) + \frac{1}{G} \left\{ L_{T,m}^{\text{sub}}(k) - L_{T,m}^{\text{bag},\infty}(k) \right\}. \quad (85)$$

If $k \rightarrow \infty$, $k/m \rightarrow 0$, and $k^2/T \rightarrow 0$, then

$$L_{T,m}^{\text{bag},G}(k) - L_{T,m}^{\text{bag},\infty}(k) = \frac{1}{Gk\bar{\tau}_m} \{1 + o(1)\}. \quad (86)$$

Proof. Condition on the training sample and abbreviate $\widehat{v}_k = \mathbb{E}_S(\widehat{v}_S)$. Expanding the quadratic form produces G diagonal and $G(G-1)$ ordered off-diagonal pairs. Define their conditional expectations by

$$\begin{aligned} d_T &= \mathbb{E}_S(\widehat{v}'_S C_m \widehat{v}_S \mid \text{training sample}), \\ o_T &= \mathbb{E}_{S,R}(\widehat{v}'_S C_m \widehat{v}_R \mid \text{training sample}). \end{aligned}$$

The pair count then gives

$$\begin{aligned}\mathbb{E}_{S_1, \dots, S_G}(\widehat{v}'_{k,G} C_m \widehat{v}_{k,G} \mid \text{training sample}) &= \frac{G}{G^2} d_T + \frac{G(G-1)}{G^2} o_T \\ &= \frac{1}{G} d_T + \left(1 - \frac{1}{G}\right) o_T.\end{aligned}$$

For an off-diagonal pair, conditional independence of S_g and S_r allows the two subset expectations to factor, giving $o_T = \widehat{v}'_k C_m \widehat{v}_k$. Taking expectation over the training sample proves (85).

For the order calculation, Proposition 14 and Proposition 4 give

$$L_{T,m}^{\text{sub}}(k) = Q_k^{\text{sub}} \frac{T-1}{T-k} = Q_k^{\text{sub}} + O\left(\frac{k}{T}\right),$$

because $Q_k^{\text{sub}} = b + O(k^{-1}) = O(1)$. Also, Propositions 5 and 10 imply

$$Q_k^{\text{sub}} - Q_k^{\text{bag},\infty} = \frac{1}{k\bar{\tau}_m} - \frac{1}{m\bar{\tau}_m} + O(k^{-2}) + O\left(\frac{1}{mk^2}\right) = \frac{1}{k\bar{\tau}_m} \{1 + o(1)\}.$$

Finally, Proposition 7 gives $L_{T,m}^{\text{bag},\infty}(k) = Q_k^{\text{bag},\infty} + O(k^2/(mT))$. The conditions $k^2/T \rightarrow 0$ and $k/m \rightarrow 0$ ensure that both k/T and $k^2/(mT)$ are $o(k^{-1})$. Consequently,

$$L_{T,m}^{\text{sub}}(k) - L_{T,m}^{\text{bag},\infty}(k) = \frac{1}{k\bar{\tau}_m} \{1 + o(1)\},$$

and substitution into (85) proves (86). □

Corollary 2 (Finite-bag fourth-root rule). *Suppose $G = G_{T,m}$ satisfies*

$$\frac{mT^{1/4}}{G} \longrightarrow 0, \tag{87}$$

which is Assumption 5 evaluated on the fourth-root range $k \asymp T^{1/4}$. Then every minimizer

$$k_{T,m}^{*,G} \in \arg \min_{1 \leq k \leq \min\{m, T-1\}} L_{T,m}^{\text{bag},G}(k)$$

satisfies

$$\frac{k_{T,m}^{*,G}}{T^{1/4}} \longrightarrow \left(\frac{\mathcal{D}}{b}\right)^{1/4}.$$

Proof. Let $k_{T,m}^0$ be the integer benchmark defined in the proof of Proposition 9. The gap in the exact bridge (85) can be bounded non-asymptotically. First, $0 \leq Q_k^{\text{sub}} - Q_k^{\text{bag},\infty} \leq Q_k^{\text{sub}} - Q^{\text{opt}} \leq \mathbb{E}_S(A_S^{-1}) \leq 1/(k\tau_L)$ by (19) and Lemma 2(1). Second, $0 \leq L_{T,m}^{\text{sub}}(k) - Q_k^{\text{sub}} = Q_k^{\text{sub}}(k-1)/(T-k)$ by Proposition 6, and $0 \leq L_{T,m}^{\text{bag},\infty}(k) - Q_k^{\text{bag},\infty} \leq \mathbb{E}_S\{Q(S)\}(k-1)/(T-k)$ by conditional Jensen's inequality and Proposition 14. Since Q_k^{sub} and $\mathbb{E}_S\{Q(S)\}$ are bounded and $T-k \asymp T$ on every

band $cT^{1/4} \leq k \leq CT^{1/4}$, combining the three bounds inside (85) gives, uniformly over the band,

$$0 \leq L_{T,m}^{\text{bag},G}(k) - L_{T,m}^{\text{bag},\infty}(k) = \frac{1}{G} \left\{ L_{T,m}^{\text{sub}}(k) - L_{T,m}^{\text{bag},\infty}(k) \right\} = O\left(\frac{1}{Gk} + \frac{k}{GT}\right).$$

After multiplication by the fourth-root normalization $m\sqrt[4]{T}$, the two terms are respectively $O(mT^{1/4}/G)$ and $O\{m/(GT^{1/4})\}$, both $o(1)$ under (87). Equivalently, uniformly on such a band,

$$m\{L_{T,m}^{\text{bag},G}(k) - L_{T,m}^{\text{bag},\infty}(k)\} = o(T^{-1/2}). \quad (88)$$

Moreover, (85) and conditional Jensen's inequality show that the finite- G loss is never below the infinite-bag loss. At the benchmark $k_{T,m}^0$, (78) and (88) give

$$m\{L_{T,m}^{\text{bag},G}(k_{T,m}^0) - Q^{\text{opt}}\} = O(T^{-1/2}).$$

Because the finite- G loss dominates the infinite-bag loss pointwise in k , optimality of $k_{T,m}^{*,G}$ gives the chain

$$m\{L_{T,m}^{\text{bag},\infty}(k_{T,m}^{*,G}) - Q^{\text{opt}}\} \leq m\{L_{T,m}^{\text{bag},G}(k_{T,m}^{*,G}) - Q^{\text{opt}}\} \leq m\{L_{T,m}^{\text{bag},G}(k_{T,m}^0) - Q^{\text{opt}}\} = O(T^{-1/2}).$$

The global lower bounds (79) and (80), applied to $L_{T,m}^{\text{bag},\infty}$ at $k_{T,m}^{*,G}$, therefore force $z_{k_{T,m}^{*,G}}^2 = O(T^{-1/2})$ and $(k_{T,m}^{*,G} - 1)^2/T = O(T^{-1/2})$ — exactly the two inequalities used in Step 2 of Proposition 9 — which localise every finite- G minimiser in a band $cT^{1/4} \leq k_{T,m}^{*,G} \leq CT^{1/4}$.

On that band, combining (82) and (88) yields

$$m\{L_{T,m}^{\text{bag},G}(k) - Q^{\text{opt}}\} = H_{T,m}(k) + o(T^{-1/2})$$

uniformly in k . Optimality of $k_{T,m}^{*,G}$ relative to $k_{T,m}^0$ and (77) consequently imply

$$H_{T,m}(k_{T,m}^{*,G}) - H_{T,m}(\tilde{k}_{T,m}) = o(T^{-1/2}).$$

Applying the exact squared-gap identity from Step 3 of Proposition 9, now with $q_{T,m} = k_{T,m}^{*,G}/\tilde{k}_{T,m}$, gives $q_{T,m} \rightarrow 1$. Equation (76) then proves the stated limit. $\square$

Remark 4 (The two effects of G are different). *Proposition 1 and Proposition 15 have the same $1/G$ structure but describe different things: the first is an exact population identity driven by subset-sampling noise, the second a first-order estimation coefficient driven by the shared training sample. Comparing magnitudes, $\mathbb{E}_S \Gamma_{SS} \asymp k$ while $\Gamma_{m,k} \asymp k^2/m$, so the additional diagonal contribution to the estimation coefficient is negligible when $m/(Gk) \rightarrow 0$. This is not the condition for the complete feasible losses: their leading difference is the population subset-sampling term $1/(Gk\bar{\tau}_m)$ in (86). Preserving the fourth-root optimizer therefore requires the*

stronger rate $mT^{1/4}/G \rightarrow 0$, equivalently $mk/G \rightarrow 0$ on $k \asymp T^{1/4}$. The diagonal/off-diagonal decomposition is consistent with Elliott et al. (2013, Sec. 2.5.1) and the finite-draw random-subspace analysis of Boot and Nibbering (2019, Sec. 3.2). The exact coefficient in (84), however, is specific to the common-training-sample forecast-combination estimator studied here.

F Additional Exogenous Theory Diagnostics

This appendix collects the numerical verification of the estimators used in Section 4.1.1, the comparative-static experiments, and the diagnostics that qualify the main-text claims.

F.1 Numerical verification of the estimators

Every quantity compared against the theory in the main text is itself a numerical object, so each is validated against an independently computed benchmark before use. Four checks are run at the start of every execution of the simulation driver, and the run aborts if any fails.

The closed-form expression for Γ_{SR} used by the integrator is compared with a direct matrix evaluation of $(b + v'_S D v_R) \text{tr}(DK_S DK_R) + v'_R DK_S DK_R D v_S$ built from explicit K_S and K_R matrices, over forty random subset pairs; the two agree to 2.2×10^{-16} , that is, to machine precision. The population infinite-bag U -statistic is compared with exact enumeration of all $\binom{9}{4} = 126$ subsets at $m = 9$, $k = 4$, and agrees to 5.0×10^{-6} against a Monte Carlo standard error of 7.1×10^{-6} . The stratified Γ integrator is compared with exact enumeration over all 126^2 ordered subset pairs and agrees to 4.1×10^{-5} against a standard error of 1.1×10^{-4} . Finally, the exact single-subset factor of Proposition 6 is verified directly on a design with a general, non-grouped precision vector ($T = 60$, $m = 12$, $k = 9$, $b = 0.8$): across 20,000 independent training samples the mean feasible loss is 1.0428 against the predicted $\{b+1/A_S\}(T-1)/(T-k) = 1.0419$, a discrepancy of 9.5×10^{-4} against a tolerance of four Monte Carlo standard errors, 2.2×10^{-3} . Each check passes with margin; the largest error observed is 43 percent of its tolerance, that is, within 1.8 Monte Carlo standard errors. All constrained subset solves complete without a single numerical failure, so no fallback weighting is ever invoked.

The strength of these checks should not be overstated. Only the first is an exact algebraic identity verified to machine precision; the remaining three compare Monte Carlo estimators against exact benchmarks at tolerances set to four times the relevant Monte Carlo standard error. Passing therefore means “consistent with the benchmark at the simulation’s own resolution”, and the tolerances would tighten if the replication counts were raised. They establish

that the estimators are unbiased for the quantities they target, not that they are accurate to any fixed absolute precision.

F.2 General precision profiles

Figure 8 reports the optimizer experiment on the two bounded continuous profiles. These are constructed from deterministic log-scores, one symmetric and one right-skewed, and are recalibrated at every pool size so that the finite-pool mean precision and the profile constant $\mathcal{D}_m$ match the balanced two-point baseline exactly. Because the theory asserts that only b and $\mathcal{D}$ enter the leading constant, the profiles are matched on precisely those two quantities and left free to differ elsewhere; their dispersion indices are $H = 1.25$ and $H = 1.13$ against $H = 2.78$ for the baseline. The optimizers nonetheless coincide to within one subset, which is the intended test.

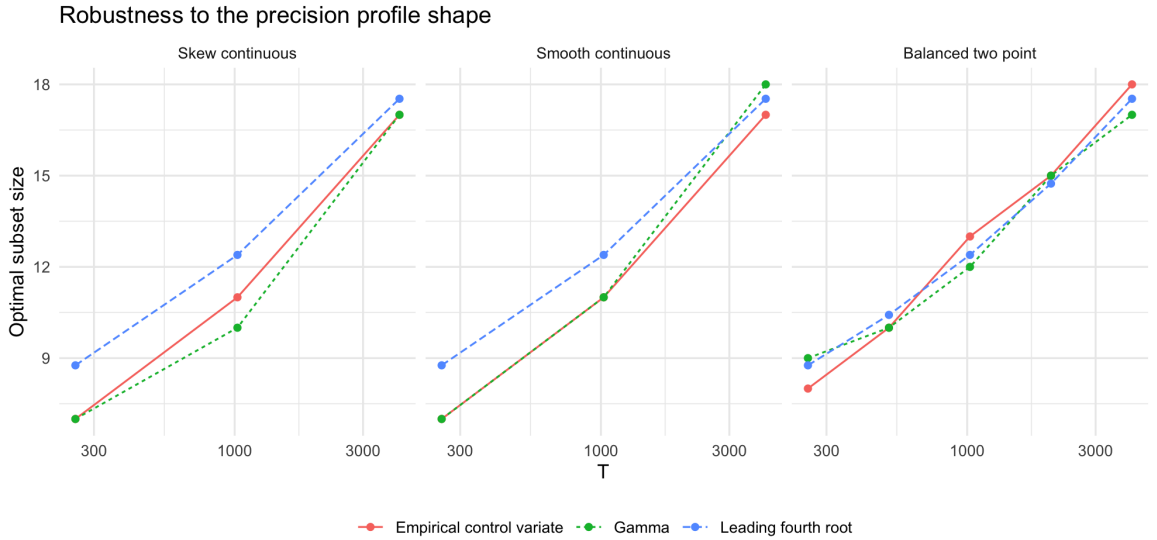


Figure 8: Optimizer under bounded continuous precision profiles matched to the baseline on $\bar{\tau}_m$ and $\mathcal{D}_m$.

F.3 Comparative statics in b , $\mathcal{D}$, and precision scale

The leading rule $k^* \sim (\mathcal{D}T/b)^{1/4}$ implies elasticities of $-1/4$ in b and $+1/4$ in $\mathcal{D}$. The elasticity with respect to the precision scale follows from the homogeneity of $\mathcal{D}_m$: under a common rescaling $\tau \mapsto c\tau$ the numerator of (4) scales as c^5 and $\bar{\tau}_m^6$ as c^6 , so $\mathcal{D}_m \mapsto \mathcal{D}_m/c$ and the implied elasticity of k^* with respect to mean precision is also $-1/4$.

Table 6 and Figure 9 report all three experiments at $T = 1024$ and $m = 160$. The empirical elasticities are -0.17 , 0.24 , and -0.09 ; the complete criterion predicts -0.17 , 0.20 , and -0.17 ;

the leading rule predicts -0.25 , 0.25 , and -0.25 . The dispersion experiment is the closest to its leading target and the two scale experiments the farthest. The useful observation is that the complete criterion misses the leading targets by about as much as the simulation does, so the gaps in this table reflect finite-design remainder in the criterion itself rather than error in its numerical implementation.

These elasticities carry substantially more uncertainty than the main-text experiments, and we do not use them to discriminate between the complete and leading predictions. Each is fitted on three to five integer optima by ordinary least squares with no replication bootstrap, and those optima are themselves uncertain by one to two subsets. Moving the largest precision-scale cell by a single subset, well inside its near-optimal set, changes the fitted elasticity from -0.09 to -0.14 . The experiments also vary one primitive at a time within a fixed bounded family, so they move higher moments of the precision distribution at the same time. A further caveat applies to the precision-scale experiment specifically: since $\mathcal{D}_m$ is homogeneous of degree -1 , rescaling all precisions is not an independent channel but a traversal of the dispersion channel along a known path, which is why the two experiments should agree and why the extent to which they do not (0.24 against -0.09 for integer optima) is itself a measure of the resolution available here. We read this table as confirming the sign and rough magnitude of each channel and nothing finer.

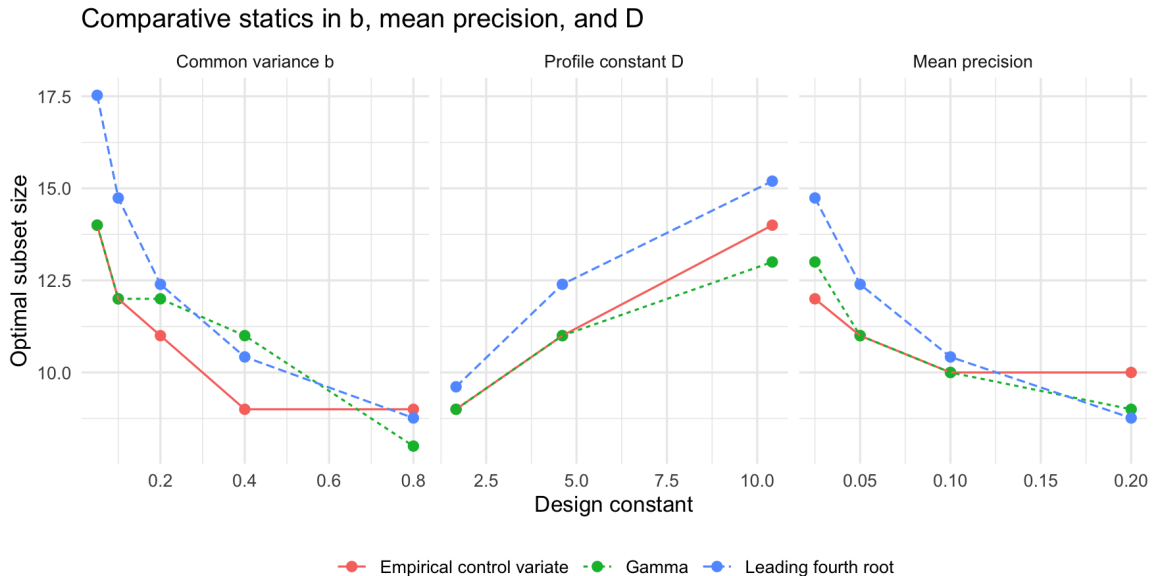


Figure 9: Comparative statics of the shared-sample optimizer at $T = 1024$, $m = 160$.

Experiment	Setting	$\mathcal{D}_m$	b	$\hat{k}^*$	k_Γ^*	$(\mathcal{D}_m T / b)^{1/4}$
Common variance b	0.05	4.61	0.05	14	14	17.53
	0.10	4.61	0.10	12	12	14.74
	0.20	4.61	0.20	11	12	12.39
	0.40	4.61	0.40	9	11	10.42
	0.80	4.61	0.80	9	8	8.76
Dispersion $\mathcal{D}_m$	1.67	1.67	0.20	9	9	9.61
	4.61	4.61	0.20	11	11	12.39
	10.42	10.42	0.20	14	13	15.20
Precision scale $\bar{\tau}_m$	0.025	9.22	0.20	12	13	14.74
	0.050	4.61	0.20	11	11	12.39
	0.100	2.30	0.20	10	10	10.42
	0.200	1.15	0.20	10	9	8.76
Fitted elasticity of $\hat{k}^*$			b : -0.17 $\mathcal{D}_m$: 0.24 $\bar{\tau}_m$: -0.09			
complete criterion / leading rule			-0.17 / -0.25 0.20 / 0.25 -0.17 / -0.25			

Notes: Each row is a separate experiment with 120 training replications and $G = 800$. The precision-scale rows vary $\bar{\tau}_m$ holding the relative shape of the profile fixed; because $\mathcal{D}_m$ is homogeneous of degree -1 in a common rescaling of the precisions, $\mathcal{D}_m \bar{\tau}_m$ is constant along those rows and this experiment traverses the dispersion channel rather than an independent one. Elasticities are ordinary least squares slopes of $\log \hat{k}^*$ on the logarithm of the setting, fitted on three to five integer optima with no bootstrap. They are sensitive to the integer resolution documented in Table 7: moving the last precision-scale cell by a single subset changes that elasticity from -0.09 to -0.14 . They are reported as evidence on sign and rough magnitude only.

Table 6: Comparative statics of the shared-sample optimizer at $T = 1024$, $m = 160$

F.4 Pool size

Figure 10 reports the fixed- T experiment underlying the pool-size discussion in the main text, with 120 replications per cell. The absence of systematic drift over $m \in \{60, 100, 160, 260, 400\}$ is consistent with the cancellation of the leading $1/m$ factors in Proposition 9. As noted in the main text, this supports the absence of a leading pool-size effect over the evaluated grid and not exact finite- m invariance.

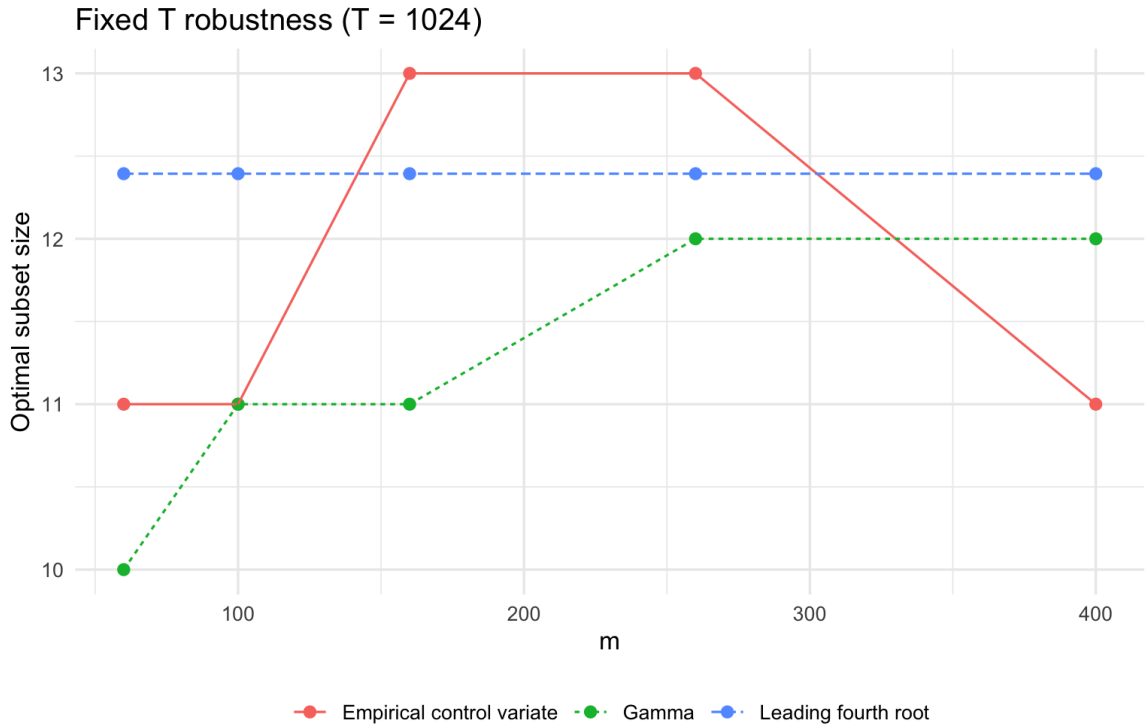


Figure 10: Pool size diagnostic for the shared sample infinite bag optimizer at fixed $T = 1024$.

F.5 Flat minima and the resolution of the integer optimizer

Table 7 quantifies how sharply the integer optimizer is identified. For each design it reports the set of subset sizes that cannot be separated from the complete-theory optimum at conventional Monte Carlo tolerance, together with the gap between the best and second-best integer expressed in units of a paired standard error. Within the two-point profile that ratio declines from 1.30 at $T = 256$ to 0.13 at $T = 4096$; it exceeds 1.96 in none of the eleven designs and exceeds unity in only three. The near-optimal sets widen correspondingly, from three subset sizes at $T = 256$ to ten at $T = 4096$.

The ratio should be read as a descriptive flatness diagnostic and not as a test statistic, and it errs in the conservative direction for the use made of it here. Both the pair being compared and the standard error are computed from the same replications: the two subset

sizes are selected as the empirical best and runner-up, which inflates the numerator even when the population curve is exactly flat, while the denominator conditions on that selection and so omits the variance contributed by which pair was chosen. Both effects push the ratio up, so an unbiased statistic would indicate even weaker resolution than the values reported, and the conclusion drawn from them is unaffected. At the largest sample sizes the inclusion rule is itself noise-driven, to the point that one near-optimal set is not contiguous: for the smooth profile at $T = 4096$ the set is $\{13, 15, \dots, 20\}$, with 14 excluded, which the minimum and maximum columns cannot represent. The integer argmin is thus progressively less identified as T grows, which is a property of the criterion rather than of the simulation: the curvature of the risk at its minimum falls faster than the Monte Carlo error. The same holds for the theoretical optimizer, whose own best-versus-runner-up gap exceeds twice its numerical integration error in only two of eleven designs and falls below one in five; the k_T^* column and the theoretical slopes computed from it should be read with that in mind at $T \geq 2048$.

Table 8 reports replication-bootstrap stability intervals for the descriptive rate slopes. All six intervals contain their corresponding point estimates. The two two-point intervals contain $1/4$; the four continuous-profile intervals lie above it, which is consistent with the exact criterion's own slopes of 0.32 and 0.34 on that grid rather than evidence against the asymptotic result. These are Monte Carlo stability diagnostics conditional on the numerical population integration; they are not confidence intervals for any empirical application, and the underlying five-point and three-point log-log slopes are descriptive summaries rather than formal inference.

F.6 Presentation cutoffs and finite-bag sensitivity

Cells in which the theoretical penalty is small relative to its own Monte Carlo standard error are uninformative about calibration and are shaded rather than dropped in Figure 2. Table 9 shows that this is a visibility convention with no bearing on the conclusion: raising the cutoff from two to five moves the median calibration ratio by less than 0.002 in every profile and removes at most three of 136 cells.

The main shared-sample experiment targets infinite-bag risk but computes only a finite number of subset draws. We therefore use the exact conditional bridge of Proposition 16 and check its sensitivity by varying $G \in \{50, 100, 200, 400, 800\}$ at $T = 1024$, $m = 160$, and $k = 12$, holding training seeds fixed across G so that the comparison is paired. The uncorrected G -bag risk falls from 0.357 at $G = 50$ to 0.328 at $G = 800$ and continues to display the expected $a + c/G$ drift. In contrast, the bridge-corrected estimate is 0.3253, 0.3255, 0.3257, 0.3257, and

Profile	T	m	$\hat{k}$	k_Γ	$k_{\min}$	$k_{\max}$	Gap/SE
two_type	256	80	8	9	8	10	1.30
two_type	512	114	10	10	10	12	1.11
two_type	1024	160	13	12	11	14	0.73
two_type	2048	226	15	15	13	17	0.16
two_type	4096	320	18	17	14	23	0.13
smooth	256	80	7	7	6	8	0.82
smooth	1024	160	11	11	10	12	0.52
smooth	4096	320	17	18	13	20	0.24
skew	256	80	7	7	7	8	1.40
skew	1024	160	11	10	9	13	0.32
skew	4096	320	17	17	15	18	0.93

Table 7: Monte Carlo resolution of the integer optimizer.

statistic	median	q025	q975	profile
raw	0.292	0.235	0.326	two_type
smooth	0.265	0.228	0.290	two_type
raw	0.320	0.272	0.416	smooth
smooth	0.320	0.292	0.357	smooth
raw	0.320	0.272	0.341	skew
smooth	0.301	0.266	0.320	skew

Table 8: Monte Carlo bootstrap stability intervals for descriptive rate slopes.

0.3257 across the same grid. It is stable to four significant digits from $G = 200$ onward, and the extrapolated intercept of the uncorrected sequence agrees with the corrected value. The paired difference against the largest bag is -9.3×10^{-6} with a standard error of 1.6×10^{-5} by $G = 400$. Thus subset-integration noise is not a binding constraint for the risk levels reported in the main text.

The estimation-penalty component is much smaller than the total risk and is therefore harder to resolve numerically. It still drifts monotonically by about fourteen percent over the same G grid. Accordingly, the penalty results should be read as showing no G -dependence detectable at the replication counts used, rather than as a separate demonstration of numerical convergence of that small component.

profile	cutoff	cells	median_ratio	q25	q75	median_abs_z
two_type	2.000	136	1.005	0.979	1.027	0.642
smooth	2.000	84	1.016	0.978	1.045	0.998
skew	2.000	84	1.009	0.977	1.032	0.790
two_type	3.000	133	1.006	0.982	1.028	0.643
smooth	3.000	83	1.016	0.977	1.045	1.000
skew	3.000	84	1.009	0.977	1.032	0.790
two_type	5.000	126	1.005	0.983	1.026	0.642
smooth	5.000	81	1.017	0.982	1.045	1.017
skew	5.000	81	1.010	0.979	1.031	0.739

Table 9: Sensitivity of penalty calibration to the presentation SNR cutoff.

F.7 The average-single-subset criterion

Figure 11 reports the full risk curves for the criterion that evaluates each fitted subset before averaging, on both the balanced two-point and the matched continuous profile. The exact Gaussian factor $(T - 1)/(T - k)$ of Proposition 6 tracks the empirical curves over the whole grid, and the resulting optimizer coincides with the theoretical minimizer in every cell.

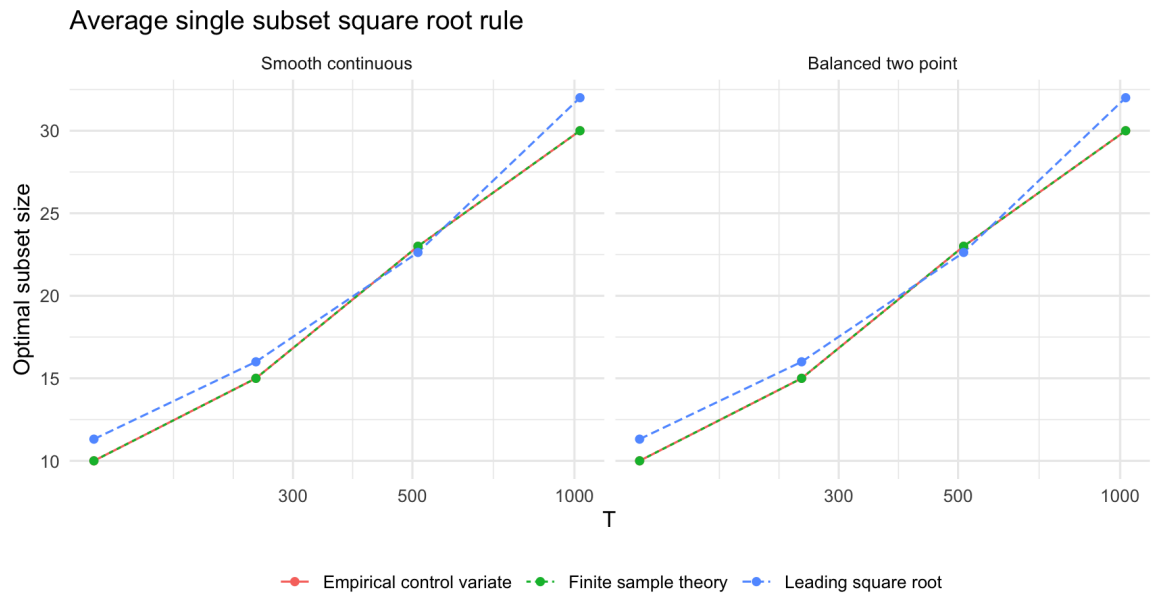


Figure 11: Average-single-subset criterion: empirical and exact-factor risk curves and their optimizers.

G U.S. SPF Design and Robustness

G.1 Screened versus Unscreened Equal Weighting

Section 5.3 reports that RSM improves on the screened equal-weight mean but not on the unscreened average of all respondents. Table 10 shows that this holds across every information set we consider, so it is a property of the panel rather than of a particular vintage design.

Design	Sample	RMSFE			RSM gain (%)	
		All-resp.	Screened	RSM	vs. all-resp.	vs. screened
Final vintage	Full	1.294	1.306	1.291	+0.24	+1.15
Final vintage	Ex-COVID	1.254	1.269	1.254	−0.06	+1.16
Final vintage	Non-crisis	1.181	1.199	1.183	−0.20	+1.34
Real-time weights, final outcome	Full	1.301	1.313	1.296	+0.32	+1.24
	Ex-COVID	1.260	1.275	1.259	+0.01	+1.24
	Non-crisis	1.187	1.205	1.183	+0.30	+1.84
Fully real time	Full	1.259	1.275	1.258	+0.11	+1.33
	Ex-COVID	1.209	1.229	1.212	−0.22	+1.36
	Non-crisis	1.154	1.176	1.156	−0.12	+1.74

Notes: The screened mean averages the eligible panel used to estimate weights; the all-respondent mean averages every forecaster replying in the round. The last two design blocks use the 146 origins common to all designs. No one-sided HAC(4) p -value for RSM against the all-respondent mean falls below 0.38 in any cell, whereas the all-respondent mean improves on the screened mean with p -values as low as 0.023. RSM is more accurate than the all-respondent mean in 49 percent of quarters.

Table 10: RSM Against Two Equal-Weight Benchmarks

The mechanism is transparent. The screen removes a median of 13 of 35 respondents at each origin, and the discarded forecasters are, on average, not worse than those retained; they are simply less regular respondents. Averaging is robust to their irregularity, whereas weight estimation is not, so the screen is a cost that RSM must pay and equal weighting need not. In the euro-area panel of Conflitti et al. (2015) the analogous comparison is immaterial—their balanced-panel and all-respondent benchmarks differ from the optimal combination by 0.003 and 0.002 in MSFE for HICP inflation—which is why the distinction has not received attention. It is material here, and we report it rather than selecting the benchmark that flatters the method.

G.2 Real-Time Information and Outcome Vintages

Table 11 separates the role of the information used to estimate weights from the release against which forecasts are scored. The first row is a conventional pseudo-out-of-sample exercise: both historical errors and evaluation outcomes use final-vintage data. The second row replaces the training outcomes with the vintages available at each origin but continues to score against final outcomes. The final row is fully real time and scores the vintage-correct combinations against first-release outcomes. To hold the evaluation dates fixed, the table uses the 134 non-crisis origins common to all three designs; this sample excludes 2008Q3–2009Q4 and 2020Q1–2021Q2.

Training outcomes	Evaluation outcome	Mean	RSM	Gain (%)	HAC(4) p
Final vintage	Final vintage	1.205	1.189	1.35	0.104
Real-time vintage	Final vintage	1.205	1.183	1.84	0.138
Real-time vintage	First release	1.176	1.156	1.74	0.153

Notes: Entries under Mean and RSM are RMSFEs in annualized percentage points, with Mean the screened equal-weight benchmark. Gain is $100(1 - \text{RMSFE}_{\text{RSM}}/\text{RMSFE}_{\text{Mean}})$. The p -values are from one-sided HAC(4) tests based on squared-error loss. Circular-block-bootstrap p -values are 0.105, 0.132, and 0.144, respectively.

Table 11: SPF Results Across Real-Time Information Sets

The similarity of the three point estimates suggests that the RSM gain is not driven by revisions. It does not, however, strengthen the statistical evidence against equal weighting: sampling uncertainty remains substantial in each design. When the real-time-produced forecasts are scored against successive releases the gain over the screened mean remains positive and in fact rises—1.36, 1.71 and 1.57 percent against the first, second and third releases on the ex-COVID sample—but the ranking relative to simplex weighting changes sign across releases. This release sensitivity is another reason to emphasize regularization relative to unrestricted weights rather than uniform dominance over every constrained benchmark.

G.3 Pseudo-OOS and Real-Time Loss Curves

Figure 12 compares the Ex-COVID and Non-crisis fixed- k diagnostics in the original two-series style; the Full-sample calculations remain in the underlying output and are reported below. The pseudo-OOS design estimates weights using final-vintage historical outcomes and also scores forecasts against final outcomes. The fully real-time design uses the target vintage available at each origin and scores forecasts against first releases. The displayed pseudo-OOS

samples contain 146 and 136 origins in the Ex-COVID and Non-crisis panels; the corresponding real-time counts are 144 and 134. Consequently, vertical differences between the curves mix outcome-vintage and sample-composition effects and should not be read as a causal accuracy comparison.

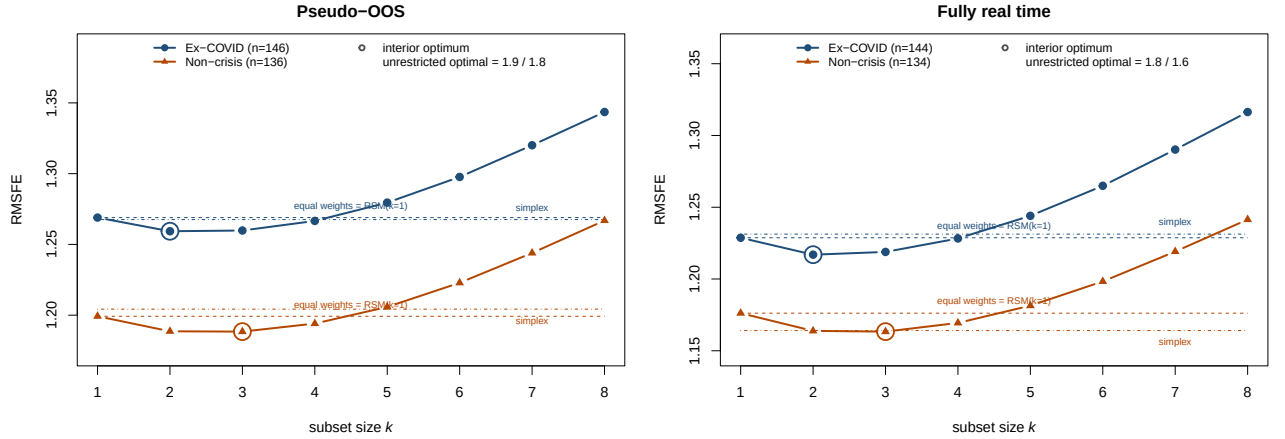


Figure 12: Ex-COVID and Non-crisis fixed- k loss curves in the original style: pseudo-OOS and fully real time

The shape and optimizer are nevertheless highly stable. In both designs the best fixed size is $k = 2$ for the Full and Ex-COVID samples and $k = 3$ for the Calm sample. The median theory-selected size is three in all pseudo-OOS panels and remains three in the real-time Full and Ex-COVID panels; it moves only to 2.5 in the real-time Calm panel. Over $k = 1, \dots, 8$, real-time RMSFE is lower by roughly 1.5–2.6 percent in the Full sample, 2.0–3.4 percent Ex-COVID, and about 2 percent in the Calm sample. More importantly for the paper’s mechanism, the fully estimated endpoint remains far above the interior minimum: 49.8 versus 44.5 percent in the Full pseudo-OOS and real-time designs, respectively. Thus real-time data modestly reduce the vertical contrast but do not weaken the U-shape or the selector validation.

G.4 Limits of the Structural Approximation

The empirical selector is theory-informed rather than a literal structural estimator. The first principal component explains a large share of the forecast-error correlation matrix and precision heterogeneity is pronounced, supporting the common-component approximation and a nontrivial regularization path. At the same time, exact individual unbiasedness and Gaussianity are not supported, some implied idiosyncratic variances require flooring, and the median ratio m/T is about 0.45. These departures motivate the finite- m safeguard and the external comparison with the realized loss curve. They also limit the claim: the application validates the selector’s

average level and the instability of unrestricted weights, not every distributional assumption or quarter-specific movement in k .

H Global and Conditional Weights

It is possible to establish a correspondence between a global weight and a conditional weight. The latter is conditional on being select in a subset S . More formally, when subsets of size k are selected according to Assumption 4, we can write the RSM global weight by definition as follows:

$$\bar{v}_{k,i} = \mathbb{E}_S[v_{S,i}] = P(i \in S) \cdot \mathbb{E}_S[w_{S,i}|i \in S] = \frac{k}{m} \mathbb{E}_S[w_{S,i}|i \in S] \quad (89)$$

Equation 89 illustrates how the global and conditional weights are related to each other. Suppose after using RSM, a forecast i has a weight greater than the equal weight. In this case, forecasts earn above-average weight in the global bagged combination if and only if they earn above-average weight within the subsets they appear in:

$$\bar{v}_{k,i} > \frac{1}{m} \iff \mathbb{E}_S[w_{S,i}|i \in S] > \frac{1}{k}$$

This relationship between the weights has a natural interpretation. A forecast receives more than the equal global weight if and only if its mean weight, across the subsets in which it appears, exceeds $1/k$. Therefore, the global weight is dictated by its subset counterpart. This equivalence holds because the two benchmarks $\frac{1}{m}$ and $\frac{1}{k}$ differ by the inclusion probability under uniform rule $\frac{k}{m}$. The same quantity links global and conditional weights. Hence global importance is achieved locally. Alternatively, a forecast that is useful across different subsets is assigned a higher value at the global level.

The correspondence makes the SPF weight profiles interpretable: a reported global weight can be read directly as a statement about how much weight a forecaster earns inside the subsets that contain that forecaster.